EXAM PREPARATION

HCIA-Datacom V2.0
Study Guide

Simple notes, pictures and practice questions for Chapters 1 to 23

23
chapters, from network basics to campus design

69
diagrams, each with a "How to read it" note

115
exam-style practice questions with answers

INSIDE EVERY CHAPTER

- Pictures that show the big idea first
- Plain-English notes for each topic
- Key facts to learn by heart
- Huawei commands with examples
- Exam traps and common mistakes
- 5 questions to check yourself

Name _____ Class _____

Study notes based on the Huawei HCIA-Datacom V2.0 training material. This is not an official Huawei publication.

How to use this guide

Each chapter follows the same pattern, so you always know where to look. Spend about 30 to 45 minutes on a chapter.

1. **Look at the pictures first.** Read the "How to read it" note under each one. Pictures give you the big idea before the details.
2. **Read the notes.** Each topic starts with a plain explanation, then the boxes below.
3. **Test yourself.** Cover the boxes with a piece of paper and say the key facts out loud. Recalling from memory works much better than reading again.
4. **Do the 5 "Check yourself" questions** without looking back. Then mark them with the answer key at the back.
5. **Go back to anything you got wrong.** Before the exam, revise the Quick reference pages.

What the boxes mean

A plain-English explanation of the idea comes first.

> **REMEMBER**
> - The key facts and numbers to learn by heart.

COMMAND / EXAMPLE

```
[HUAWEI] what you type on the device
```

> **EXAM TRAP** A common wrong answer, or a detail the exam likes to test.

Two kinds of commands

The course slides use **YunShan OS (V600)** commands, with port names like `GE1/0/1`. The eNSP labs use **VRP (V200)**, with names like `GigabitEthernet0/0/1`. Where the two differ, the notes show both. There is a full list at the back: Slides vs eNSP commands.

The practice questions

The questions use the same styles as the HCIA exam: choose one answer, choose all the correct answers, and true or false. When a question says "Choose TWO", every correct option must be picked to get the mark.

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Answer key 114

Quick reference and slides vs eNSP commands 118

CHAPTER 1

Data Communication Network Basics

EXAM FOCUS
Topologies, LAN/MAN/WAN, enterprise vs carrier, device roles (switch = MAC, router = IP, firewall = egress), MPU/SFU/LPU, key terms

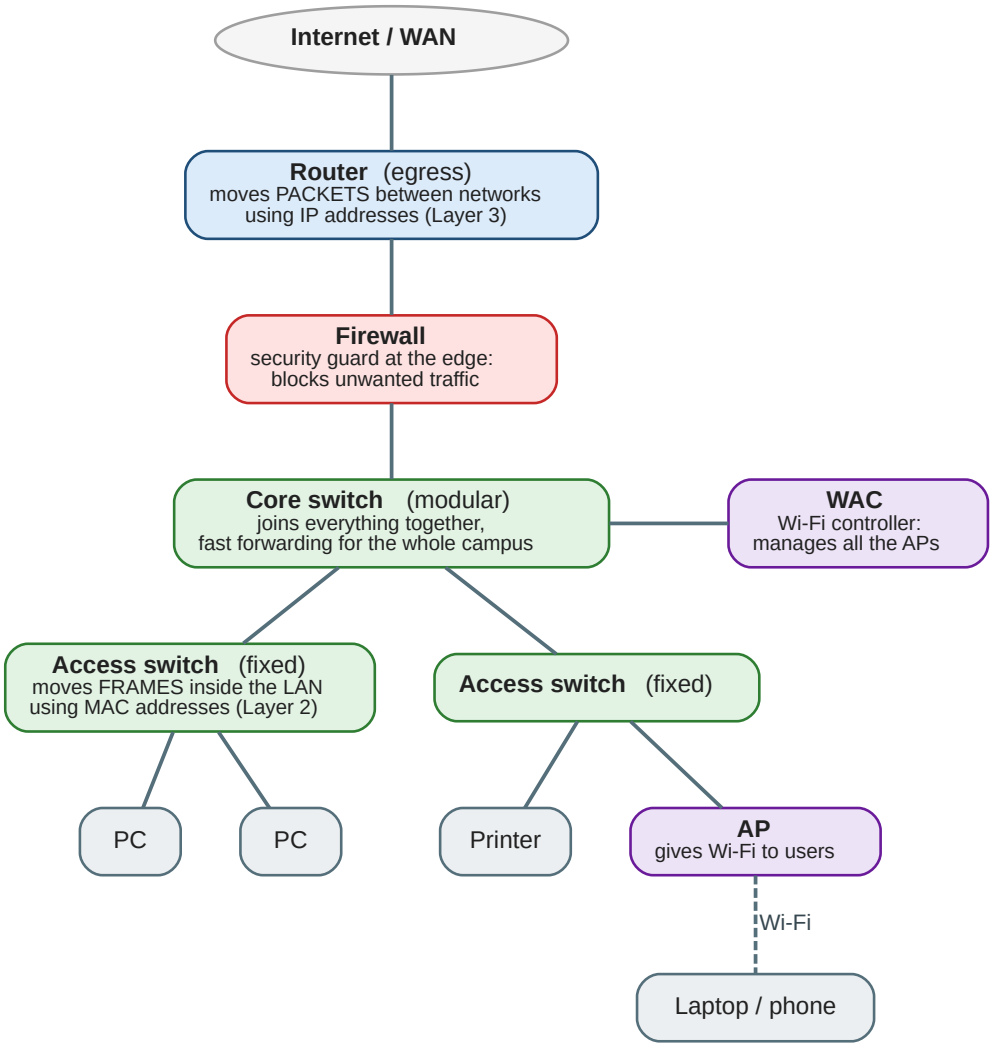
IN THIS CHAPTER

1.1 Communication basics · 1.2 Network topologies · 1.3 Network size: LAN, MAN, WAN · 1.4 Who runs the network · 1.5 Main network devices · 1.6 Inside a big (modular) device · 1.7 Key terms · 1.8 Huawei certification path

Pictures Look at these first

PICTURE 1.1

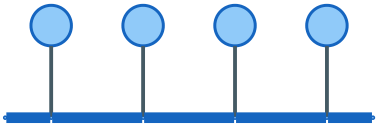
A typical campus network: what each device does



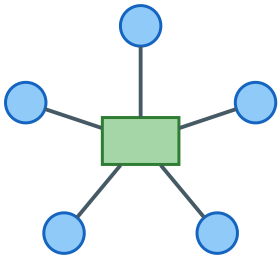
HOW TO READ IT
Read from the top. Traffic from the Internet passes the router, then the firewall, then the core switch, then an access switch, and finally reaches a PC or an AP. Each box says the device's one main job.

PICTURE 1.2

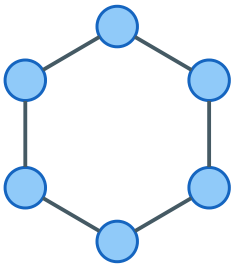
Network topology shapes



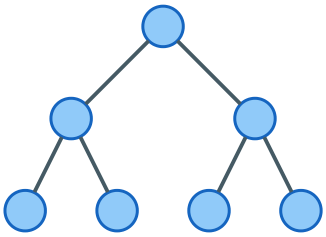
Bus
one shared cable



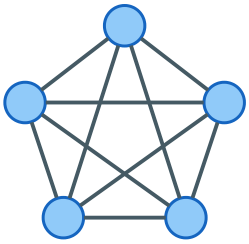
Star
all to one centre switch



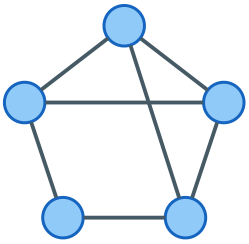
Ring
each device to two neighbours



Tree
layers, like a family tree



Full-mesh
everyone to everyone



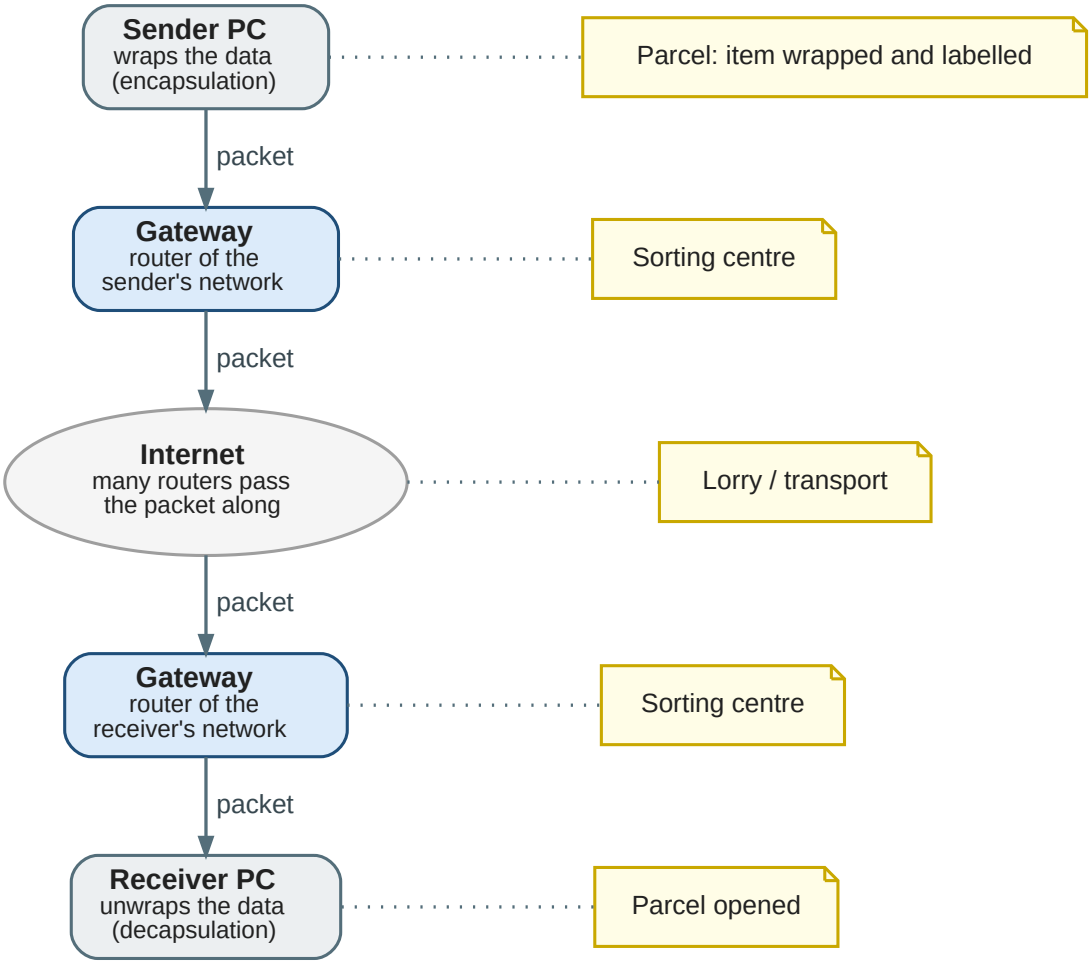
Partial-mesh
some extra backup links

HOW TO READ IT

Topology = the shape of the connections. Star is what you see in most offices. Full-mesh gives the most backup paths but needs the most cables: n(n-1)/2 links.

PICTURE 1.3

How data travels: like sending a parcel



HOW TO READ IT

Sending data works like posting a parcel. The sender wraps it (encapsulation), routers pass it along like sorting centres, and the receiver unwraps it (decapsulation).

Notes

Read, then cover the boxes and test yourself

1.1 Communication basics

Communication = moving information from a sender to a receiver through something in between. Like a phone call: you talk, the line carries it, your friend hears it.

REMEMBER

- Source = the sender (PC, phone)
- Channel = the path (cable, fibre, radio waves)
- Destination = the receiver

EXAM TRAP Drum signals and pigeon post are communication, but **NOT** network communication. Email and file transfer over a cable **ARE**.

1.2 Network topologies

Topology = the shape of how devices are connected.

REMEMBER

- Bus, Star, Ring, Tree, Full-mesh, Partial-mesh
- Star = most common in offices (everything plugs into one switch)
- Real networks mix several shapes

COMMAND / EXAMPLE

```
Full-mesh links = n(n−1)/2
Example: 5 routers = 5×4/2 = 10 links
```

EXAM TRAP If the centre switch of a star fails, everything connected to it goes down.

1.3 Network size: LAN, MAN, WAN

Networks are grouped by how big an area they cover.

REMEMBER

- LAN – a building or campus (a few km). Ethernet, Wi-Fi
- MAN – a city. Ethernet 10G/100G
- WAN – between cities or countries. PPP, MPLS, SRv6
- The Internet is the biggest WAN

EXAM TRAP Match the technology to the type: Wi-Fi/Ethernet → LAN; PPP/MPLS → WAN.

1.4 Who runs the network

Networks are also grouped by who owns them.

REMEMBER

- Enterprise network – built by one organisation for itself (school, company, data centre)
- Carrier network – built by a telco to sell service (Internet backbone, 4G/5G, MPLS VPN)

EXAM TRAP Two ways to classify networks: by area covered and by owner.

1.5 Main network devices

Each device has one main job.

REMEMBER

- Switch – moves **FRAMES** inside a LAN using MAC addresses
- Router – moves **PACKETS** between networks using IP addresses
- Firewall – security guard at the network edge
- AP + WAC – Wi-Fi access points and the controller that manages them

EXAM TRAP Switch = MAC (Layer 2). Router = IP (Layer 3). Don't mix them up.

1.6 Inside a big (modular) device

A modular switch or router is a chassis you fill with cards. Each card does one job.

REMEMBER

- MPU – Main Processing Unit (the brain)
- SFU – Switch Fabric Unit (connects the cards together)
- LPU – Line Processing Unit (the cards with ports)
- Fixed device = all-in-one box with a set number of ports (used at the access layer)

COMMAND / EXAMPLE

```
Modular: CloudEngine S12700H
Fixed: CloudEngine S5755
```

EXAM TRAP Know what MPU / SFU / LPU stand for. LPUs talk to each other **THROUGH** the SFU.

1.7 Key terms

Sending data is like sending a parcel: wrap it, label it, deliver it, unwrap it.

REMEMBER

- Payload – the actual data
- Header / trailer – labels added before / after the data
- Encapsulation – wrapping (adding headers)
- Decapsulation – unwrapping
- Gateway – device that connects two different networks

1.8 Huawei certification path

Huawei certificates go from beginner to expert.

REMEMBER

- HCIA (Associate) → HCIP (Professional) → HCIE (Expert)

EXAM TRAP Order: A → P → E.

Check yourself

Answers: page 114

1. CHOOSE ONE

   Which device forwards frames inside a LAN by looking at MAC addresses?

   A Router
   B Firewall
   C Access point controller (WAC)
   D Switch

2. CHOOSE ONE

   A full-mesh network connects 6 routers. How many links are needed?

   A 12
   B 15
   C 30
   D 36

3. CHOOSE ONE

   In a modular (chassis) switch, which card connects all the other cards together?

   A MPU
   B LPU
   C Power module
   D SFU

4 TRUE OR FALSE?

PPP and MPLS are typical LAN technologies.

☐ True ☐ False

5 CHOOSE ONE

What is the correct order of Huawei certification levels, from beginner to expert?

A HCIP, HCIA, HCIE

B HCIA, HCIE, HCIP

C HCIE, HCIP, HCIA

D HCIA, HCIP, HCIE

CHAPTER 2

Network Reference Models

EXAM FOCUS
OSI 7 & TCP/IP layers, PDU names, protocol-to-layer mapping, well-known ports, TCP vs UDP, PPP LCP → auth → NCP, encapsulation

IN THIS CHAPTER

2.1 Why use layers · 2.2 Who makes the standards · 2.3 OSI model (7 layers) · 2.4 TCP/IP and the 5-layer model · 2.5 Data names at each layer (PDU) · 2.6 Protocols at each layer · 2.7 Common port numbers · 2.8 TCP vs UDP · 2.9 DNS and HTTP · 2.10 Network vs data link layer · 2.11 Physical layer · 2.12 PPP · 2.13 Encapsulation

Pictures Look at these first

PICTURE 2.1

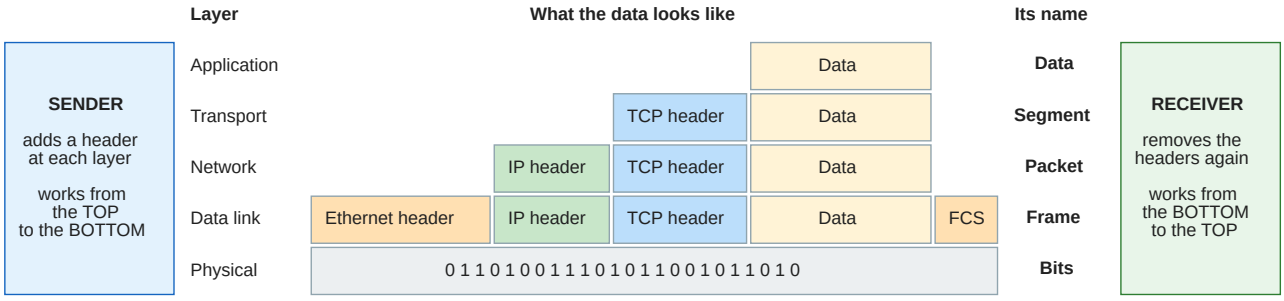
OSI model vs TCP/IP model (and what each layer does)

OSI model (7)	TCP/IP (4)	Course model (5)	Data name (PDU)	Example protocols	Device
7 Application	Application	Application	**Data**	HTTP, FTP, Telnet, SSH, DNS, DHCP, SMTP, SNMP	PC, server (the apps)
6 Presentation					
5 Session					
4 Transport	Transport	Transport	**Segment**	TCP, UDP (ports)	PC, server
3 Network	Internet	Network	**Packet**	IP, ICMP, IGMP	**Router**
2 Data link	Network interface	Data link	**Frame**	Ethernet, PPP (MAC)	**Switch**
1 Physical		Physical	**Bits**	Cable, fibre, radio	Cable, NIC

HOW TO READ IT
Read across each row. OSI layers 5, 6 and 7 all sit inside the TCP/IP Application layer. The data has a different name at each layer: Data, Segment, Packet, Frame, Bits.

PICTURE 2.2

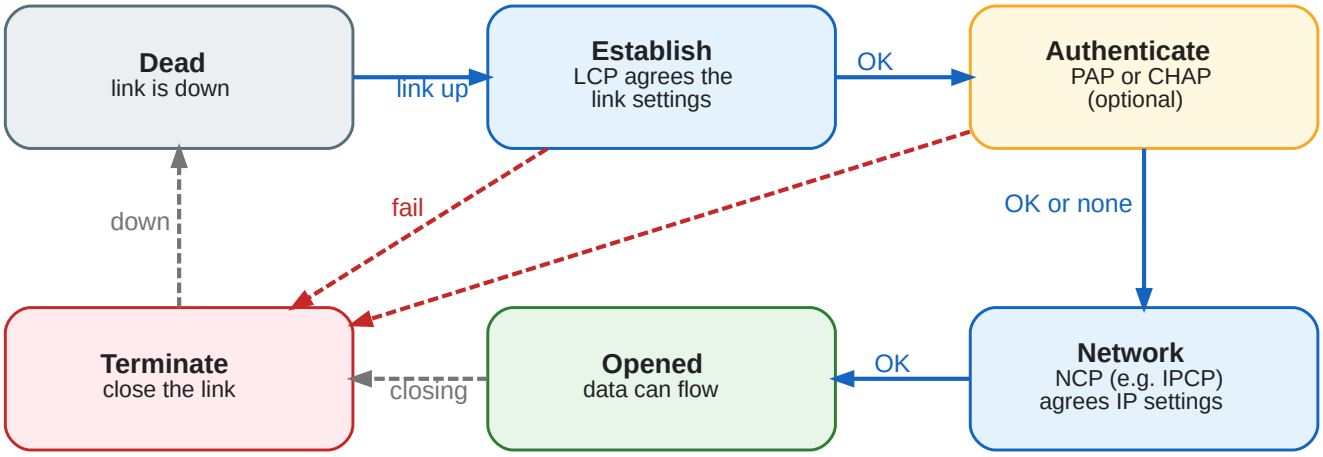
Encapsulation: the sender wraps, the receiver unwraps



HOW TO READ IT
Going DOWN the layers, the sender adds a header at each step (and an FCS trailer at the data link layer). Going UP, the receiver removes them in reverse order.

PICTURE 2.3

PPP: how the link comes up



HOW TO READ IT

PPP connects two routers over a WAN link. Follow the arrows: LCP agrees the link settings, then PAP or CHAP checks the password (optional), then NCP agrees the network settings such as the IP address.

Notes

Read, then cover the boxes and test yourself

2.1 Why use layers

A network model splits the job of sending data into layers. Each layer does one job, like departments in a company.

REMEMBER

- Easier to design, build and troubleshoot
- All vendors can follow one standard
- Layers work together through standard interfaces

EXAM TRAP Short-answer question: give 2–3 advantages of a layered model.

2.2 Who makes the standards

Standards bodies write the rules so devices from different makers can talk.

REMEMBER

- IETF – Internet protocols (TCP/IP), published as RFCs
- IEEE – Ethernet (802.3) and Wi-Fi (802.11)
- ISO – the OSI model

EXAM TRAP Ethernet & Wi-Fi = IEEE. RFC = IETF. OSI = ISO.

2.3 OSI model (7 layers)

The 7-layer 'textbook' model.

REMEMBER

7 Application – apps use the network
6 Presentation – data format
5 Session – manages the conversation
4 Transport – end-to-end delivery (TCP/UDP)
3 Network – IP addresses and routing
2 Data link – frames and MAC addresses
1 Physical – bits on the cable or air

COMMAND / EXAMPLE

```
Memory trick (7 → 1):
All People Seem To Need Data Processing
```

2.4 TCP/IP and the 5-layer model

The model the real Internet uses. This course uses a 5-layer version of it.

REMEMBER

- TCP/IP model (4): Application, Transport, Internet, Network Interface
- Course model (5): Application, Transport, Network, Data Link, Physical

EXAM TRAP OSI layers 5, 6 and 7 all fit inside the TCP/IP Application layer.

2.5 Data names at each layer (PDU)

The data has a different name at each layer.

REMEMBER

- Application → Data
- Transport → Segment
- Network → Packet
- Data link → Frame
- Physical → Bits

EXAM TRAP Network layer = **PACKET**. Data link = **FRAME**. Very common question.

2.6 Protocols at each layer

Each layer has its own protocols.

REMEMBER

- Application: HTTP, FTP, Telnet, DNS, DHCP, SMTP, SNMP, TFTP
- Transport: TCP, UDP
- Network: IP, ICMP, IGMP
- Data link: Ethernet, PPP

EXAM TRAP ICMP (used by ping) is a **NETWORK** layer protocol, not transport.

2.7 Common port numbers

A port number tells the computer which app the data is for, like a unit number in an apartment block.

REMEMBER

- HTTP 80 (TCP)
- SMTP 25 (TCP)
- Telnet 23 (TCP)
- FTP 20 + 21 (TCP)
- DNS 53 (UDP)

COMMAND / EXAMPLE

```
Full port list: see the Quick reference at the back.
```

2.8 TCP vs UDP

TCP = registered post (tracked, guaranteed). UDP = normal letter (fast, no guarantee).

REMEMBER

- TCP: connects first, reliable, in order – web, email, file transfer
- UDP: no connection, fast, may lose data – video calls, games, DNS

EXAM TRAP Needs reliability → TCP. Needs low delay → UDP.

2.9 DNS and HTTP

DNS is the Internet's phone book: it turns a name into an IP address. Then HTTP fetches the web page.

REMEMBER

- DNS: www.huawei.com → 1.2.3.4
- HTTP: browser asks, web server sends the page
- Both are client/server

EXAM TRAP When you open a website, DNS happens **FIRST**, then HTTP.

2.10 Network vs data link layer

Network layer = finds the route between networks (like the address on an envelope). Data link layer = delivers within the same local network.

REMEMBER

- Routers read the destination IP and check their routing table
- MAC address = 48 bits, built into the network card
- Ethernet (IEEE 802.3) = most common data link protocol

EXAM TRAP MAC = 48 bits, IPv4 = 32 bits, IPv6 = 128 bits.

2.11 Physical layer

Turns bits into signals: electrical, light or radio.

REMEMBER

- Media: twisted pair (RJ45), optical fibre, hybrid cable, wireless

2.12 PPP

PPP is used on a direct WAN link between two routers, e.g. branch office to HQ.

REMEMBER

Setup happens in this order:

1. LCP – agree on link settings
2. Authentication – PAP or CHAP (optional)
3. NCP – agree on network settings (e.g. IP address)

EXAM TRAP The order LCP → PAP/CHAP → NCP is often tested.

2.13 Encapsulation

Sending = wrapping a gift in more and more layers. Receiving = unwrapping it.

REMEMBER

- Sender adds: TCP header → IP header → Ethernet header + FCS trailer → bits
- Receiver removes them in reverse order
- A switch looks up to Layer 2; a router looks up to Layer 3

EXAM TRAP The FCS trailer is added only at the data link layer.

Check yourself

Answers: page 114

1 CHOOSE ONE

At which layer is the data unit called a PACKET?

A Transport layer
B Network layer
C Data link layer
D Physical layer

2 CHOOSE ONE

ping uses ICMP. Which layer does ICMP belong to?

A Application layer
B Transport layer
C Network layer
D Data link layer

3 CHOOSE ONE

In what order does a PPP link come up?

A NCP, LCP, authentication
B Authentication, LCP, NCP
C LCP, authentication, NCP
D LCP, NCP, authentication

4 CHOOSE TWO

Which TWO statements about UDP are true?

A It does not set up a connection before sending
B It guarantees that data arrives in order
C It suits video calls, games and DNS
D It uses a three-way handshake

5 CHOOSE ONE

Which organisation writes the Ethernet (802.3) and Wi-Fi (802.11) standards?

A IEEE
B IETF
C ISO
D Wi-Fi Alliance

CHAPTER 3

Network Device Operating Systems

EXAM FOCUS

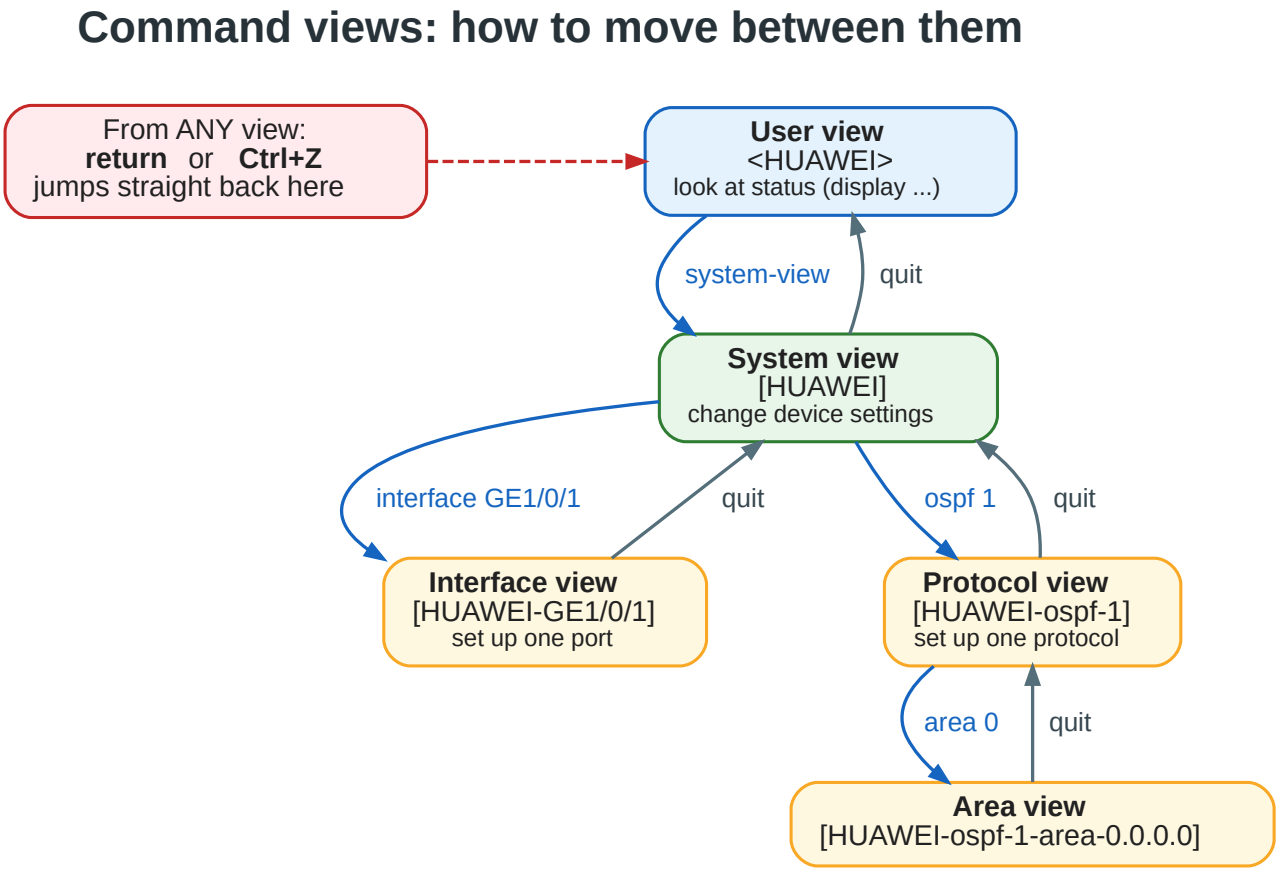
YunShan OS vs VRP, login methods, command views, privilege levels 0–3, hotkeys, undo, save/startup/reset, file types & commands

IN THIS CHAPTER

3.1 YunShan OS vs VRP · 3.2 Ways to log in · 3.3 Command views · 3.4 Privilege levels · 3.5 CLI shortcuts · 3.6 Error messages · 3.7 undo and hotkeys · 3.8 First-time setup · 3.9 Files and storage · 3.10 Saving the configuration · 3.11 Software upgrade

Pictures Look at these first

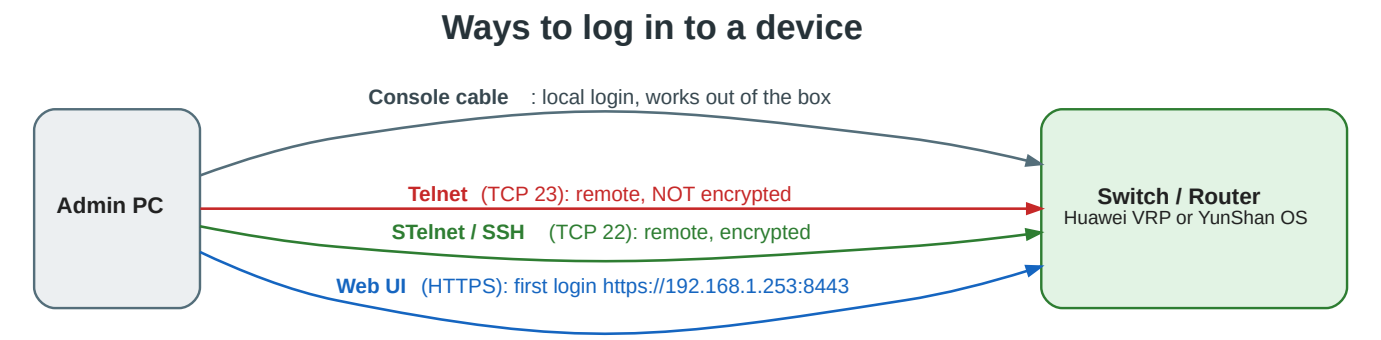
PICTURE 3.1



HOW TO READ IT

Each view is like a room. Type the command on an arrow to move. quit goes back one level; return or Ctrl+Z jumps straight back to the user view.

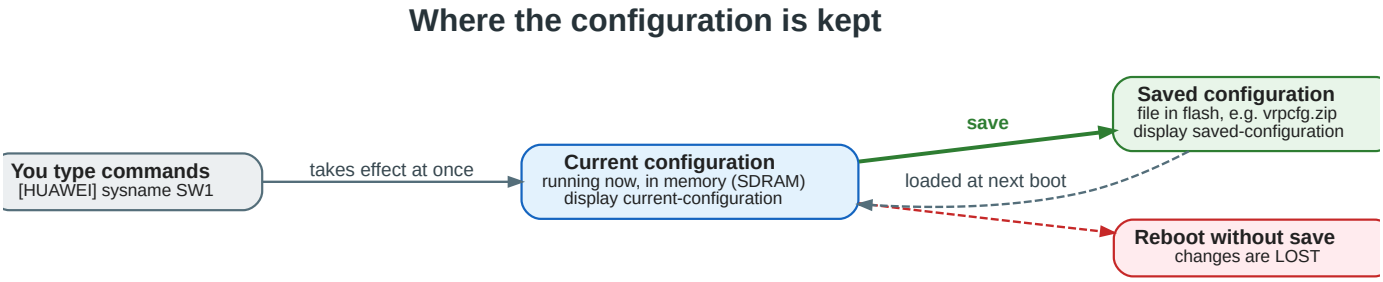
PICTURE 3.2



HOW TO READ IT

Console is local (a cable to the console port) and works on a brand-new device. Telnet sends everything as plain text; STelnet (SSH) is encrypted. The web UI uses HTTPS.

PICTURE 3.3



HOW TO READ IT

Commands you type change the current configuration in memory straight away. Only save copies it to flash. If you reboot without saving, your changes are lost.

Notes Read, then cover the boxes and test yourself

3.1 YunShan OS vs VRP

The 'Windows' of Huawei network devices. Newer devices run YunShan OS (V600); older ones run VRP (V200).

REMEMBER

- Both have a CLI (commands) and a Web UI
- Check which one with display version

COMMAND / EXAMPLE

```
<HUAWEI> display version
```

EXAM TRAP Slides use V600 port names (GE1/0/1). eNSP uses V200 names (GigabitEthernet0/0/1).

3.2 Ways to log in

You can manage a device from right beside it (console cable) or over the network.

REMEMBER

- Console – cable to the console port; local; works out of the box
- Telnet – remote, **NOT** encrypted
- STelnet (SSH) – remote, encrypted
- Web – HTTPS. First login: https://192.168.1.253:8443

EXAM TRAP First console login asks you to set a password (8–16 characters). STelnet is safer than Telnet.

3.3 Command views

Views are like rooms: you must be in the right room to run certain commands.

REMEMBER

- User view <HUAWEI> – check status
- System view [HUAWEI] – change settings
- Interface / protocol views – set up one port or one protocol
- quit = back one level; return or Ctrl+Z = back to user view

COMMAND / EXAMPLE

```
<HUAWEI> system-view
[HUAWEI] interface GE1/0/1
[HUAWEI-GE1/0/1] quit
[HUAWEI] return
```

EXAM TRAP < > = user view. [ ] = system view or deeper.

3.4 Privilege levels

Users can only run commands at or below their own level.

REMEMBER

- 0 Visit – ping, tracert
- 1 Monitoring – display commands
- 2 Configuration – service settings
- 3 Management – files, upgrades, debugging

EXAM TRAP ping = level 0. File operations = level 3.

3.5 CLI shortcuts

The CLI helps you type less.

REMEMBER

- Short forms work if they're unique: dis cu = display current-configuration
- Tab = auto-complete
- ? = show the options
- Up arrow / Ctrl+P = previous command

COMMAND / EXAMPLE

```
<HUAWEI> dis cu
[HUAWEI-ui-vty0-4] authentication-mode ?
```

EXAM TRAP Typing 'd c' matches more than one command → Ambiguous error.

3.6 Error messages

The error tells you what went wrong.

REMEMBER

- Unrecognized – no such command
- Wrong parameter – bad value
- Incomplete – something is missing
- Too many parameters
- Ambiguous – short form matches more than one command

EXAM TRAP Match each error message to its cause.

3.7 undo and hotkeys

undo = the reverse of a command.

REMEMBER

- undo restores a default, turns a feature off, or deletes a setting
- Ctrl+G = display current-configuration
- Ctrl+L = display ip routing-table
- Ctrl+O = undo debugging all
- Ctrl+Z = back to user view

COMMAND / EXAMPLE

```
[Server] undo sysname
[HUAWEI]
```

EXAM TRAP The default device name is HUAWEI.

3.8 First-time setup

First things to set on a new device: name, time, management IP.

REMEMBER

- sysname – device name (system view)
- clock timezone / clock datetime – time (user view)
- header login / header shell – login banners
- Management IP on MEth0/0/0 or a VLANIF

COMMAND / EXAMPLE

```
[HUAWEI] sysname SW1
<SW1> clock timezone MYT add 08:00:00
[SW1] interface meth 0/0/0
[SW1-MEth0/0/0] ip address 10.1.1.1 24
```

EXAM TRAP clock commands are run in **USER** view.

3.9 Files and storage

The device keeps software and settings as files, just like a PC.

REMEMBER

- .cc = system software
- .cfg / .zip = configuration file
- .pat = patch
- Flash = permanent storage; SDRAM = running memory

COMMAND / EXAMPLE

```
dir, cd, pwd, mkdir, copy, move, rename, delete, undelete
```

EXAM TRAP delete moves a file to the recycle bin (undelete brings it back). delete /unreserved = gone forever.

3.10 Saving the configuration

Changes are lost on reboot unless you save them.

REMEMBER

- Current config = what is running now (in memory)
- Saved config = what loads at next startup (in flash)
- save = save now
- startup saved-configuration = choose the file for next boot
- reset saved-configuration = wipe the next-startup config

COMMAND / EXAMPLE

```
<HUAWEI> save
<HUAWEI> display startup
<HUAWEI> display saved-configuration
<HUAWEI> reboot fast
```

EXAM TRAP reboot asks you to save first; reboot fast does not.

3.11 Software upgrade

Upgrade = copy the new file in, point the device to it, reboot.

REMEMBER

1. Upload the new .cc (and .pat)
2. save
3. startup system-software
4. startup saved-configuration
5. startup patch
6. display startup to check
7. reboot

COMMAND / EXAMPLE

```
<HUAWEI> startup system-software new.cc
<HUAWEI> startup patch new.PAT all
```

EXAM TRAP Know the **ORDER** of the upgrade steps.

Check yourself

Answers: page 114

1 CHOOSE ONE

Which prompt shows that you are in the SYSTEM view?

- A <HUAWEI>
- B [HUAWEI]
- C HUAWEI#
- D HUAWEI>

2 CHOOSE ONE

What is the LOWEST command level that can run ping and tracert?

- A Level 0 (Visit)
- B Level 1 (Monitoring)
- C Level 2 (Configuration)
- D Level 3 (Management)

3 CHOOSE ONE

What happens when you enter reboot fast?

- A The device asks you to save first
- B The device reboots without asking you to save
- C The device saves, then reboots
- D The command is rejected in user view

4 CHOOSE ONE

Which file extension is the device's system software?

- A .cfg
- B .pat
- C .zip
- D .cc

5 TRUE OR FALSE?

A file removed with delete (without /unreserved) can be brought back with undelete.

- ☐ True
- ☐ False

CHAPTER 4

Ethernet Switching Basics

EXAM FOCUS

Ethernet II vs 802.3 frame, MAC address types, learning from SOURCE MAC, flood/forward/discard, ARP request/reply

IN THIS CHAPTER

4.1 Ethernet frame · 4.2 MAC address · 4.3 How a switch learns · 4.4 What a switch does with a frame · 4.5 Collision vs broadcast domain · 4.6 ARP

Pictures Look at these first

PICTURE 4.1

The Ethernet II frame

Destination MAC	Source MAC	Type	Data	FCS
6 bytes	6 bytes	2 bytes	46 to 1500 bytes	4 bytes
who it is FOR	who SENT it (switch learns this)	what is inside: 0x0800 IPv4 0x0806 ARP 0x86DD IPv6	the payload, e.g. an IP packet	error check

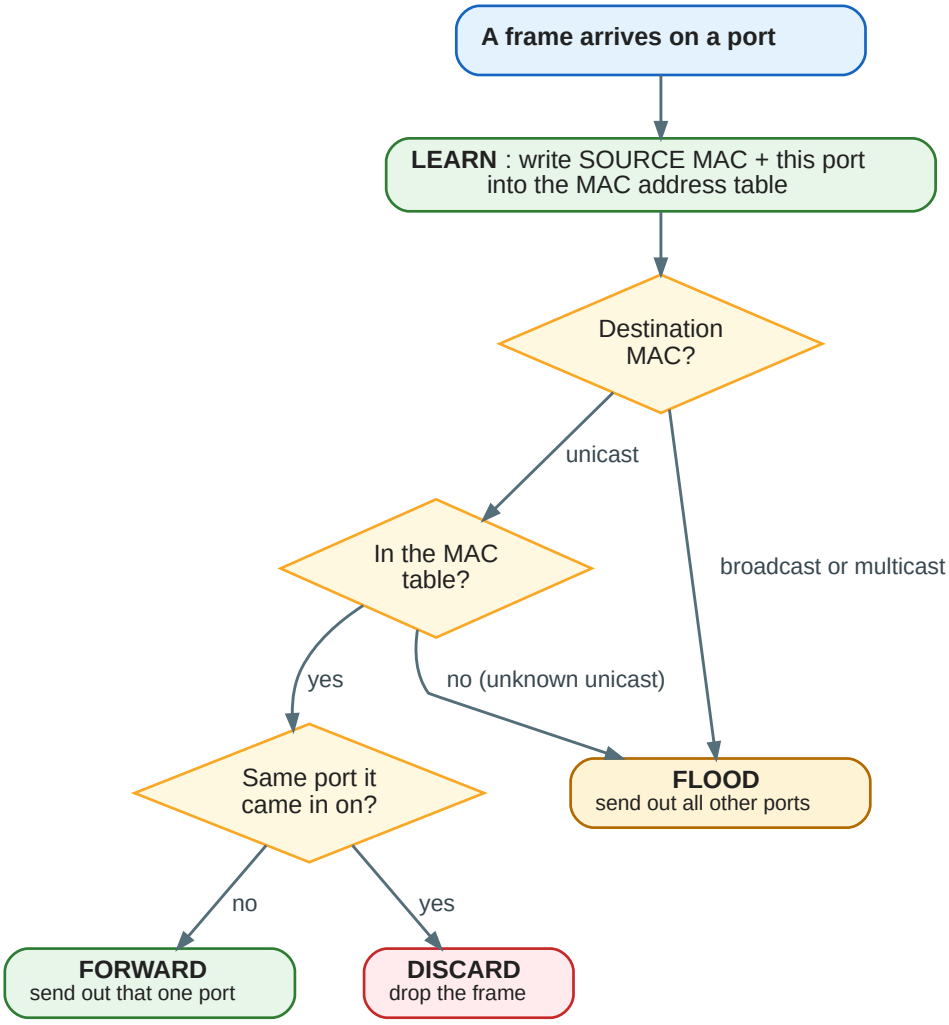
Whole frame: 64 to 1518 bytes. IEEE 802.3 frames put a **Length** (up to 1500) where Ethernet II has **Type** .

HOW TO READ IT

A frame is the envelope used inside a LAN. The switch reads the destination MAC to decide where to send it, and learns the source MAC. The Type field says what is inside.

PICTURE 4.2

What a switch does with every frame

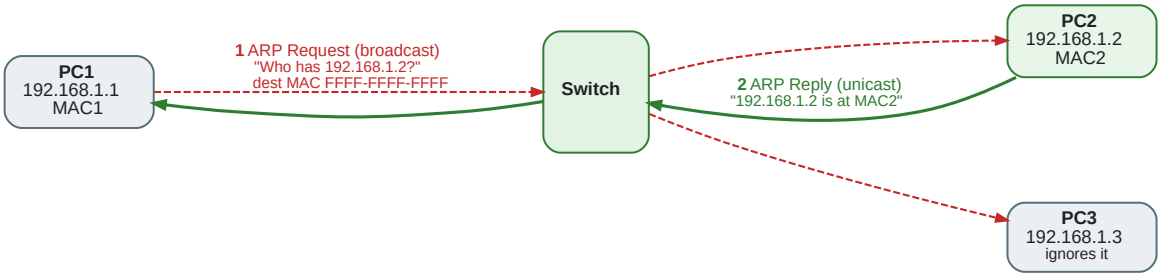


HOW TO READ IT

Follow the arrows. The switch always LEARNS the source MAC first. Then it FLOODS, FORWARDS or DISCARDS depending on the destination MAC.

PICTURE 4.3

ARP: finding the MAC address for an IP address



HOW TO READ IT

PC1 knows PC2's IP but not its MAC. Step 1: PC1 BROADCASTS an ARP request, so every PC receives it. Step 2: only PC2 answers, with a UNICAST ARP reply. PC1 saves the answer in its ARP table.

Notes Read, then cover the boxes and test yourself

4.1 Ethernet frame

A frame is the 'envelope' used to carry data inside a LAN.

REMEMBER

- Destination MAC | Source MAC | Type | Data | FCS
- Type says what's inside: 0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6
- Frame size 64–1518 bytes; MTU 1500

EXAM TRAP Ethernet II has a **TYPE** field; IEEE 802.3 has a **LENGTH** field (≤ 1500).

4.2 MAC address

A MAC address is like the serial number of a network card.

REMEMBER

- 48 bits, e.g. 0050-5600-0001
- First 24 bits = vendor code (OUI)
- Unicast = one device; Multicast = a group; Broadcast = everyone (FFFF-FFFF-FFFF)

EXAM TRAP A **SOURCE** MAC is always unicast.

4.3 How a switch learns

The switch learns 'who is on which port' by looking at the **SENDER** of each frame.

REMEMBER

- Learns the **SOURCE** MAC + the port it came in on
- Uses the **DESTINATION** MAC to decide where to send
- Unused entries expire after 300 s

COMMAND / EXAMPLE

```
display mac-address
```

EXAM TRAP The MAC table is built from the **SOURCE** MAC (slide quiz).

4.4 What a switch does with a frame

There are three possible actions.

REMEMBER

- Flood – send out all other ports (broadcast, multicast, unknown destination)
- Forward – send out the one known port
- Discard – drop it (destination is on the same port it came in on)

EXAM TRAP Unknown unicast is **FLOODED**, not dropped (true/false trap).

4.5 Collision vs broadcast domain

Collision domain = devices that can talk over each other. Broadcast domain = devices that all hear the same broadcasts.

REMEMBER

- Hub = 1 collision domain
- Switch = each port is its own collision domain, but one broadcast domain (per VLAN)
- Router / VLAN = splits broadcast domains

EXAM TRAP Switches split collision domains; routers and VLANs split broadcast domains.

4.6 ARP

ARP asks: 'Who has this IP address? Tell me your MAC.'

REMEMBER

- ARP Request = **BROADCAST** to everyone
- ARP Reply = **UNICAST** back to the asker
- The answer is saved in the ARP table

COMMAND / EXAMPLE

```
PC: arp -a
Huawei: display arp
```

EXAM TRAP Request = broadcast, Reply = unicast. To reach a remote host, the PC ARPs for its **GATEWAY**.

Check yourself

Answers: page 114

1 CHOOSE ONE

How does a switch build its MAC address table?

- A From the destination MAC address of each frame
- B From ARP replies only
- C From the IP addresses inside each packet
- D From the source MAC address and the port the frame came in on

2 TRUE OR FALSE?

A switch drops a unicast frame if the destination MAC is not in its MAC table.

- ☐ True
- ☐ False

3 CHOOSE TWO

Which TWO statements about ARP are true?

- A The ARP request is sent as a broadcast
- B The ARP reply is sent as a broadcast
- C The ARP reply is sent as a unicast
- D ARP finds the IP address for a known MAC address

4 CHOOSE ONE

An Ethernet frame has Type = 0x0806. What does it carry?

- A IPv4
- B ARP
- C IPv6
- D An 802.1Q VLAN tag

5

CHOOSE ONE

Which device splits BROADCAST domains?

A Hub

B Repeater

C Switch with all ports in one VLAN

D Router

CHAPTER 5

VLAN Fundamentals and Configuration

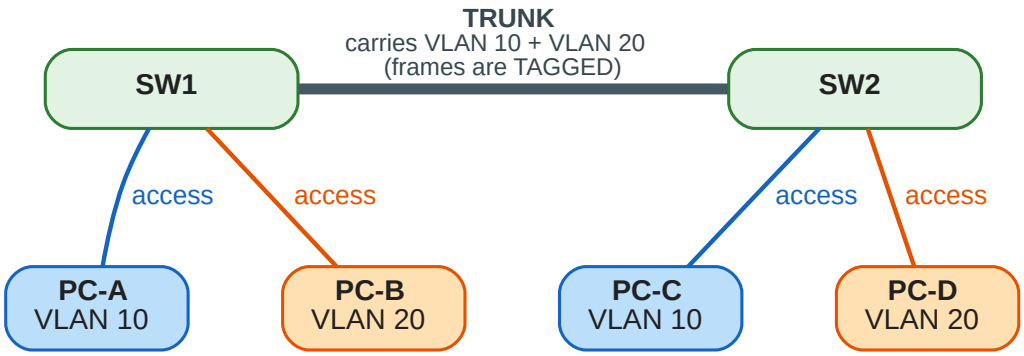
EXAM FOCUS
802.1Q tag fields, VID range, PVID, access/trunk/hybrid send-receive rules, VLAN commands

IN THIS CHAPTER
5.1 Why VLANs · 5.2 VLAN tag (802.1Q) · 5.3 How ports join a VLAN · 5.4 Port types · 5.5 Configure VLANs · 5.6 Check VLANs

Pictures Look at these first

PICTURE 5.1

VLANs: one network split into separate groups

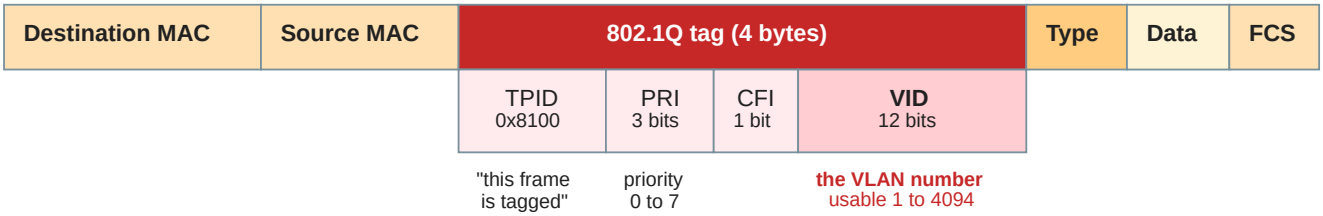


PC-A **can** reach PC-C (both VLAN 10).
PC-A **cannot** reach PC-B or PC-D without a router or VLANIF (Chapter 9).

HOW TO READ IT
Blue PCs are in VLAN 10 and orange PCs are in VLAN 20. PCs reach each other only inside the same VLAN. The trunk link between the switches carries both VLANs, with a tag on each frame saying which VLAN it belongs to.

PICTURE 5.2

The 802.1Q VLAN tag inside a frame



HOW TO READ IT
The switch adds a 4-byte tag between the Source MAC and Type fields. The 12-bit VID is the VLAN number, so VLANs 1 to 4094 can be used (0 and 4095 are reserved).

PICTURE 5.3

Access, trunk and hybrid ports

Port type	Usually connects to	How many VLANs	Frames LEAVING the port	Key command
Access	PC, printer, phone	1	always **untagged**	`port default vlan 10`
Trunk	another switch, router	many	**tagged** , except the PVID VLAN (that one leaves untagged)	`port trunk allow-pass vlan 10 2`
Hybrid	either (default on many Huawei S switches)	many	you choose per VLAN: **tagged** or **untagged**	`port hybrid untagged vlan 10 20`

HOW TO READ IT

Every port gives untagged frames its PVID (home VLAN). The difference is how frames LEAVE the port: access always removes the tag, trunk keeps it (except for the PVID VLAN), hybrid lets you choose per VLAN.

Notes Read, then cover the boxes and test yourself

5.1 Why VLANs

A VLAN splits one physical switch into several separate virtual switches – like putting walls in an open-plan office.

REMEMBER

- Each VLAN = its own broadcast domain
- Less broadcast traffic, better security
- Group users by department, not by where they sit

EXAM TRAP Different VLANs cannot talk without a Layer 3 device (router or VLANIF) – see Ch 9.

5.2 VLAN tag (802.1Q)

A tag is a small label added to the frame saying which VLAN it belongs to.

REMEMBER

- Tag = 4 bytes, inserted after the source MAC
- TPID = 0x8100 (means 'this frame is tagged')
- PRI = 3 bits of priority
- VID = 12-bit VLAN number
- Usable VLANs 1–4094; default VLAN = 1

EXAM TRAP VID is 12 bits → 4096 values, but 0 and 4095 are reserved.

5.3 How ports join a VLAN

There are several ways to decide which VLAN a device belongs to.

REMEMBER

- Port-based (most common, the default)
- MAC-based
- IP subnet-based
- Protocol-based
- Policy-based

EXAM TRAP By default every port is in VLAN 1.

5.4 Port types

The port type decides whether frames keep their VLAN tag when they leave.

REMEMBER

- Access – to PCs; one VLAN; frames leave **UNTAGGED**
- Trunk – between switches; many VLANs; frames leave **TAGGED** (except the PVID VLAN)
- Hybrid – flexible; you choose tagged or untagged per VLAN
- PVID = the port's 'home' VLAN for untagged frames

EXAM TRAP On a trunk, frames in the PVID VLAN leave **UNTAGGED**.

5.5 Configure VLANs

Three steps: create the VLANs, set the port type, put the port in the VLAN(s).

REMEMBER

- Access port → port default vlan
- Trunk port → port trunk allow-pass vlan
- Hybrid port → port hybrid tagged / untagged vlan

COMMAND / EXAMPLE

```
[SW] vlan batch 10 20
[SW-GE1/0/1] port link-type access
[SW-GE1/0/1] port default vlan 10
[SW-GE1/0/24] port link-type trunk
[SW-GE1/0/24] port trunk allow-pass vlan 10 20
[SW-GE1/0/3] port link-type hybrid
[SW-GE1/0/3] port hybrid untagged vlan 10 20
```

EXAM TRAP vlan batch 10 20 = two VLANs. vlan batch 10 to 20 = eleven VLANs.

5.6 Check VLANs

Use display commands to confirm ports are in the right VLAN.

REMEMBER

- display vlan – VLANs and their ports
- display port vlan – port type, PVID, allowed VLANs

COMMAND / EXAMPLE

```
display vlan
display port vlan
```

EXAM TRAP UT = untagged, TG = tagged.

Check yourself

Answers: page 114

1 CHOOSE ONE

How many bits long is the VID field in an 802.1Q tag?

A 8
B 10
C 12
D 16

2 CHOOSE ONE

How many VLANs does vlan batch 10 to 20 create?

A 2
B 10
C 11
D 20

3 CHOOSE ONE

On a trunk port, how do frames from the PVID VLAN leave the port?

A Tagged
B Dropped
C Untagged
D Tagged with VLAN 1

4 CHOOSE ONE

Which command allows VLANs 10 and 20 on a trunk port?

A port default vlan 10 20
B port hybrid pvid vlan 10 20
C vlan batch 10 20 trunk
D port trunk allow-pass vlan 10 20

5 TRUE OR FALSE?

PCs in VLAN 10 and VLAN 20 on the same switch can talk to each other without a Layer 3 device.

☐ True ☐ False

CHAPTER 6

STP Fundamentals and Configuration

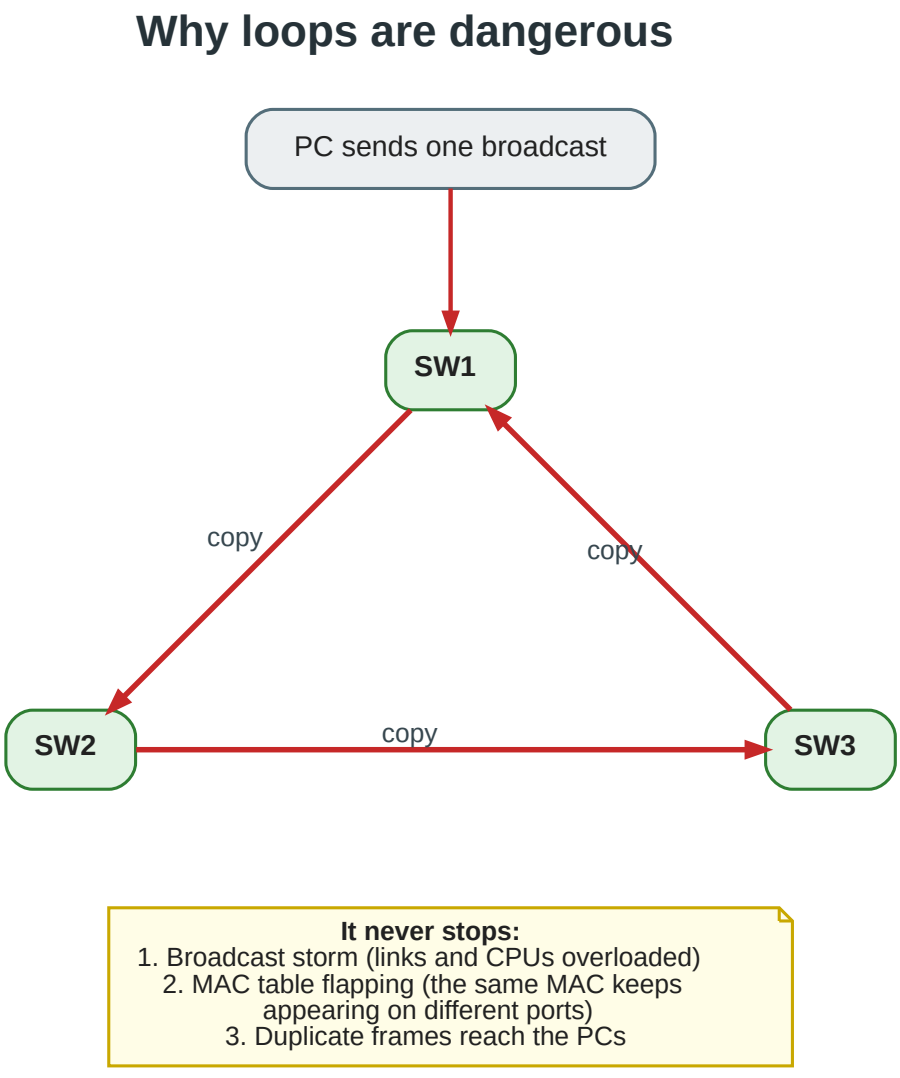
EXAM FOCUS
Root bridge/port/designated port election, BID & costs, timers, port states, RSTP roles & edge port, MSTP/VBST

IN THIS CHAPTER

6.1 Why STP · 6.2 Bridge ID and cost · 6.3 STP election (4 steps) · 6.4 Tie-breakers · 6.5 Timers and port states · 6.6 RSTP (802.1w) · 6.7 MSTP and VBST · 6.8 STP commands

Pictures Look at these first

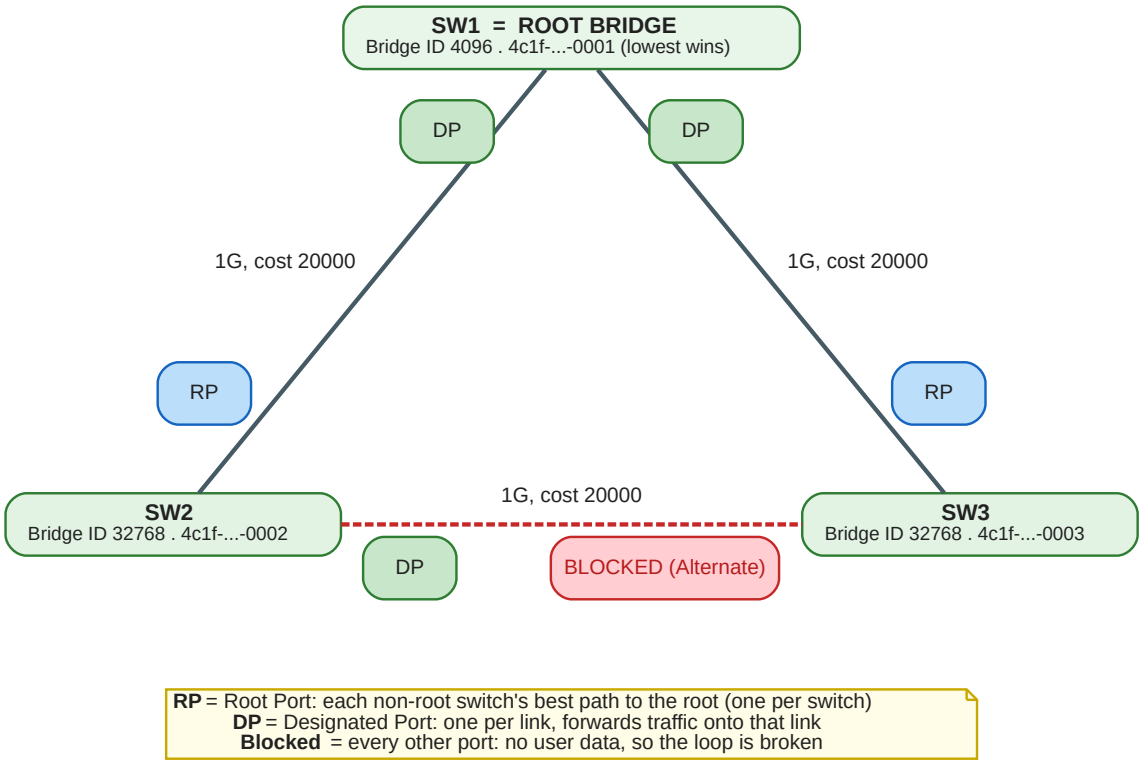
PICTURE 6.1



HOW TO READ IT
With backup links and no STP, a broadcast frame goes round the triangle forever and multiplies. The result is a broadcast storm, MAC table flapping and duplicate frames.

PICTURE 6.2

STP result: root bridge, root ports, designated ports, blocked port



HOW TO READ IT

SW1 has the lowest Bridge ID, so it is the root bridge and all its ports are designated (DP). SW2 and SW3 each pick a root port (RP) towards SW1. On the SW2-SW3 link SW2 wins (lower Bridge ID), so SW3's port is blocked.

PICTURE 6.3

STP port states (and the RSTP version)

STP state	Disabled	Blocking	Listening	Learning	Forwarding
Receives BPDUs	No	Yes	Yes	Yes	Yes
Sends BPDUs	No	No	Yes	Yes	Yes
Learns MAC addresses	No	No	No	Yes	Yes
Forwards user data	No	No	No	No	**Yes**
How long	port down	until elected	**15 s** Forward Delay	**15 s** Forward Delay	stays
RSTP name	**Discarding**			**Learning**	**Forwarding**

HOW TO READ IT

A port must wait 15 s in Listening and 15 s in Learning before it forwards, about 30 s in total. RSTP keeps only three states and moves much faster.

Notes Read, then cover the boxes and test yourself

6.1 Why STP

Backup links are good, but they create loops, and in a loop frames go round forever. STP blocks some ports so there is only **ONE** active path.

REMEMBER

Problems a loop causes:

- Broadcast storm
- MAC address flapping (the table keeps changing)
- Duplicate frames

EXAM TRAP Know these 3 loop problems.

6.2 Bridge ID and cost

STP picks a 'boss' switch and the cheapest paths to it. The lowest number always wins.

REMEMBER

- Bridge ID = priority (default 32768, in steps of 4096) + MAC
- Lowest Bridge ID = Root Bridge
- Port cost by speed: 100M = 200000, 1G = 20000, 10G = 2000

EXAM TRAP In STP, **LOWER** is better for everything.

6.3 STP election (4 steps)

STP always follows these four steps in order.

REMEMBER

1. Elect the Root Bridge (lowest Bridge ID)
2. Every other switch picks **ONE** Root Port (cheapest path to the root)
3. Every link picks **ONE** Designated Port
4. All other ports are **BLOCKED**

EXAM TRAP The root bridge has no root port; all of its ports are designated.

6.4 Tie-breakers

If the costs are equal, compare these in order.

REMEMBER

Root port: lowest path cost → lowest neighbour Bridge ID → lowest neighbour Port ID → lowest own Port ID
Designated port: lowest path cost → lowest Bridge ID → lowest Port ID

EXAM TRAP Typical exam question: find the root bridge and port roles from a diagram.

6.5 Timers and port states

STP is slow because ports wait before forwarding, to be safe.

REMEMBER

- Hello 2 s, Forward Delay 15 s, Max Age 20 s
- States: Disabled → Blocking → Listening → Learning → Forwarding
- Blocking → Forwarding ≈ 30 s; root failure recovery ≈ 50 s

EXAM TRAP Only root ports and designated ports reach Forwarding.

6.6 RSTP (802.1w)

Rapid STP = the faster upgrade of STP.

REMEMBER

- Recovers in seconds
- New roles: Alternate (backup for the root port) and Backup (backup for a designated port)
- Only 3 states: Discarding, Learning, Forwarding
- Edge port = port to a PC; goes straight to Forwarding

COMMAND / EXAMPLE

```
[SW] stp mode rstp
[SW-GE1/0/1] stp edged-port enable
```

EXAM TRAP Alternate = backs up the **ROOT** port. Backup = backs up a **DESIGNATED** port.

6.7 MSTP and VBST

STP/RSTP use one tree for all VLANs, so the blocked link sits idle. MSTP and VBST use several trees to share the load.

REMEMBER

- MSTP (802.1s) – groups of VLANs share an instance (tree)
- VBST (Huawei) – one tree per VLAN
- Huawei default mode = MSTP

COMMAND / EXAMPLE

```
[SW] stp mode mstp
```

EXAM TRAP Load-sharing across VLANs needs MSTP or VBST, not STP/RSTP.

6.8 STP commands

Make your core switch the root so traffic takes sensible paths.

REMEMBER

- stp root primary = make this switch root (priority 0)
- stp root secondary = backup root (priority 4096)
- Or set the priority yourself (multiple of 4096)

COMMAND / EXAMPLE

```
[SW] stp root primary
[SW] stp priority 4096
[SW-GE1/0/1] stp cost 2000
[SW] stp bpdu-protection
display stp brief
```

EXAM TRAP display stp brief shows each port's role (**ROOT**/DESI/ALTE) and state.

Check yourself

Answers: page 114

1 CHOOSE ONE

Four switches run STP with default settings except priority. Which one becomes the root bridge?

A Priority 32768, MAC 00e0-fc00-0000
B Priority 4096, MAC 00e0-fc00-0009
C Priority 8192, MAC 00e0-fc00-0002
D Priority 32768, MAC 00e0-fc00-0001

2 CHOOSE THREE

Which THREE problems can a Layer 2 loop cause?

A Broadcast storm
B MAC address flapping
C Duplicate frames
D IP address conflict

3 CHOOSE ONE

In RSTP, which port role is the backup for the ROOT port?

A Backup port
B Alternate port
C Edge port
D Designated port

4 CHOOSE ONE

What is the default STP Forward Delay timer?

A 2 seconds
B 15 seconds
C 20 seconds
D 50 seconds

5 TRUE OR FALSE?

The root bridge has exactly one root port.

☐ True ☐ False

CHAPTER 7

Ethernet Link Aggregation

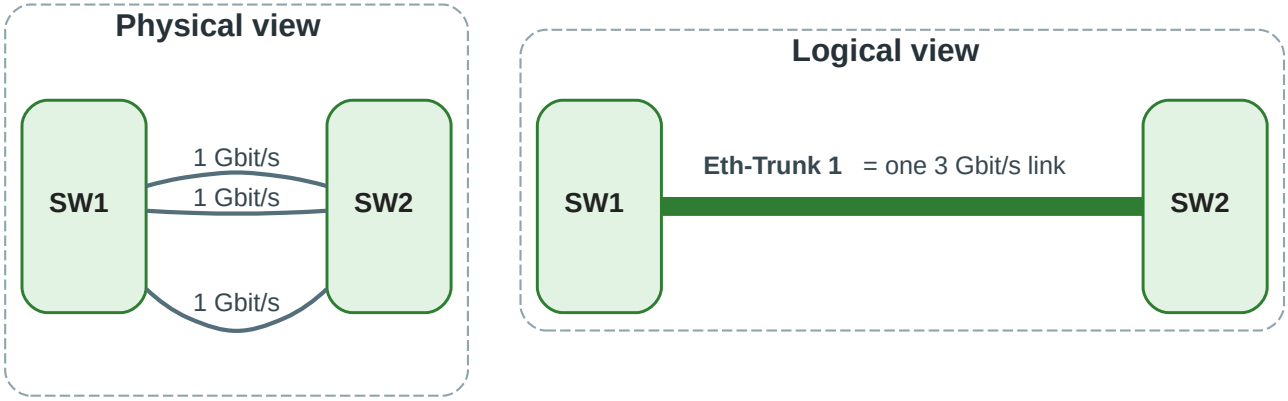
EXAM FOCUS
Manual vs LACP, system priority (Actor) vs interface priority (active links), preemption, load balancing, Eth-Trunk config

IN THIS CHAPTER
7.1 What Eth-Trunk is · 7.2 Manual mode · 7.3 LACP mode · 7.4 Load balancing · 7.5 Check Eth-Trunk

Pictures Look at these first

PICTURE 7.1

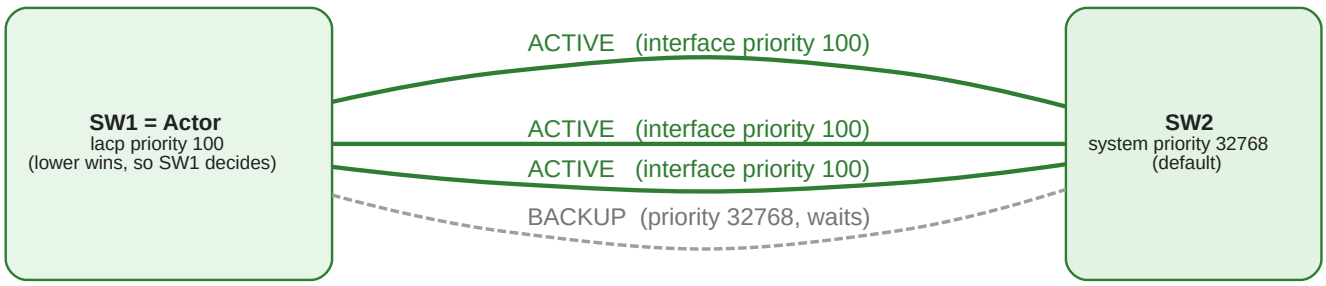
Eth-Trunk: many cables, one logical link



HOW TO READ IT
Left: what is really plugged in, three 1G cables. Right: what the switches and STP see, one 3G link called Eth-Trunk 1. If one cable breaks, the other two keep working.

PICTURE 7.2

LACP mode: who decides, and which links are active



HOW TO READ IT
Lower priority numbers win. SW1 has system priority 100, so it is the Actor and decides. Its three ports with interface priority 100 become active; the fourth waits as a backup. max active-linknumber 3 keeps only 3 links active. If one fails, the backup link takes over.

PICTURE 7.3

Manual mode vs LACP mode

	Manual mode	LACP mode (IEEE 802.3ad)
Negotiation	none (no LACPDUs)	LACPDUs between the switches
Active links	all member links	chosen by priority; others are backups
Spots wiring mistakes	no	yes
Command (V600)	`mode manual load-balance`	`mode lacp`
Command (eNSP V200)	`mode manual load-balance`	`mode lacp-static`
Check it	`display eth-trunk 1` (Selected = active, Unselected = backup)	

HOW TO READ IT

Manual mode just bundles the ports with no negotiation, so every link is active. LACP mode lets the two switches talk, choose the active links and keep spare links as backups.

Notes

Read, then cover the boxes and test yourself

7.1 What Eth-Trunk is

Bundle several cables into one big logical link. You get more speed, and if one cable breaks the others keep working.

REMEMBER

- More bandwidth
- Link backup
- Load sharing
- STP sees it as **ONE** link, so nothing gets blocked

EXAM TRAP All member ports must have the same speed and duplex.

7.2 Manual mode

You bundle the ports yourself. The switches don't negotiate.

REMEMBER

- All member links are active
- Can't detect wiring mistakes on the other side

COMMAND / EXAMPLE

```
[SW] interface Eth-Trunk 1
[SW-Eth-Trunk1] mode manual load-balance
[SW-Eth-Trunk1] trunkport GE 1/0/1 to 1/0/3
```

EXAM TRAP Manual mode sends no LACP packets.

7.3 LACP mode

The two switches talk to each other (LACP, IEEE 802.3ad) to agree which links to use. Spare links can wait on standby.

REMEMBER

- System priority (default 32768, lower wins) → picks the 'boss' side (Actor)
- Interface priority (default 32768, lower wins) → picks which links are active
- max active-linknumber → limits active links
- lacp preempt enable → a better link takes over again when it comes back

COMMAND / EXAMPLE

```
[SW] lacp priority 100
[SW] interface Eth-Trunk 1
[SW-Eth-Trunk1] mode lacp   (eNSP: mode lacp-static)
[SW-Eth-Trunk1] max active-linknumber 2
[SW-GE1/0/1] eth-trunk 1
```

EXAM TRAP System priority = which **DEVICE** decides. Interface priority = which **LINKS** are active.

7.4 Load balancing

Traffic is shared per conversation (flow), so packets from one conversation stay on one cable and arrive in order.

REMEMBER

- Based on MAC or IP addresses (source/destination)
- Set on the sending side

COMMAND / EXAMPLE

```
[SW-Eth-Trunk1] load-balance src-dst-ip
```

EXAM TRAP It only affects traffic going **OUT** of that switch.

7.5 Check Eth-Trunk

Confirm which links are working.

REMEMBER

- Selected = active link
- Unselected = standby or not working

COMMAND / EXAMPLE

```
display eth-trunk 1
display interface brief
```

Check yourself

Answers: page 115

1 CHOOSE ONE

In LACP mode, what decides which switch becomes the Actor (the side that decides)?

A Interface priority
B System priority
C max active-linknumber
D The fastest port speed

2 TRUE OR FALSE?

In manual mode all member links are active and no LACP packets are sent.

☐ True ☐ False

3 CHOOSE ONE

Which is a requirement for Eth-Trunk member ports?

A They must have the same speed and duplex
B They must be in different VLANs
C They must use optical fibre
D They must be on different switches

4 CHOOSE ONE

Why does STP not block the extra cables inside an Eth-Trunk?

A STP sees the Eth-Trunk as one logical link
B STP is turned off on Eth-Trunk ports
C LACP blocks BPDUs
D The member ports are in different VLANs

5 CHOOSE ONE

How does Eth-Trunk share traffic across its member links?

A Per packet, round robin
B Per flow (conversation)
C Per byte
D Only by VLAN number

CHAPTER 8

IP Address and Configuration

EXAM FOCUS

IPv4 header, classes, private/special ranges, subnetting & VLSM, ICMP/ping/tracert, DHCP DORA, renewal 50%/87.5%, DHCP config

IN THIS CHAPTER

8.1 IPv4 address · 8.2 IPv4 header · 8.3 Address classes · 8.4 Private and special addresses · 8.5 Subnetting · 8.6 ICMP, ping and tracert · 8.7 Setting an interface IP · 8.8 DHCP – how it works · 8.9 DHCP lease · 8.10 DHCP configuration

Pictures Look at these first

PICTURE 8.1

An IPv4 address = network part + host part

IP address	**192**	**168**	**1**	**10**
in binary	11000000	10101000	00000001	00001010
Mask /24	11111111	11111111	11111111	00000000
in decimal	255	255	255	0
	NETWORK part (24 bits) = 192.168.1			**HOST part (8 bits)**

Network address	**First usable**	**Last usable**	**Broadcast**	**Usable hosts**
192.168.1. **0** host bits all 0	192.168.1. **1**	192.168.1. **254**	192.168.1. **255** host bits all 1	$2^8 - 2 =$ **254**

HOW TO READ IT

The mask /24 means the first 24 bits are the network (the 'street') and the last 8 bits are the host (the 'house number'). Host bits all 0 = network address; host bits all 1 = broadcast address.

PICTURE 8.2

Subnetting: one /24 cut into four /26 subnets

192.168.20.0 /24 (256 addresses)				
Subnet (/26)	**192.168.20.0/26**	**192.168.20.64/26**	**192.168.20.128/26**	**192.168.20.192/26**
Network address	.0	.64	.128	.192
Usable hosts	.1 to .62	.65 to .126	.129 to .190	.193 to .254
Broadcast	.63	.127	.191	.255
How many hosts	$2^6 - 2 =$ **62** each (block size = 256 - 192 = 64)			

HOW TO READ IT

Borrowing 2 host bits turns 192.168.20.0/24 into four /26 subnets of 64 addresses each. Each subnet loses 2 addresses (network and broadcast), so 62 hosts fit in each. A department of 40 hosts fits in any /26.

PICTURE 8.3

DHCP: how a PC gets its IP address (D-O-R-A)



HOW TO READ IT

Read from top to bottom. The new PC has no IP yet, so it broadcasts DISCOVER and REQUEST. The server answers with OFFER and ACK. The address is a loan (lease), renewed at 50% and 87.5% of the lease time.

Notes Read, then cover the boxes and test yourself

8.1 IPv4 address

An IP address is like a home address: the network part is the street, the host part is the house number.

REMEMBER

- 32 bits, written like 192.168.1.10
- Subnet mask shows where the network part ends (/24 = 255.255.255.0)
- Network address = host bits all 0
- Broadcast address = host bits all 1
- Usable hosts = 2^n − 2

COMMAND / EXAMPLE

```
192.168.1.0/24 → usable hosts .1 to .254
```

EXAM TRAP Always subtract 2 (network + broadcast address).

8.2 IPv4 header

The information written on the outside of every IP packet.

REMEMBER

- 20 bytes minimum
- TTL – drops by 1 at each router; at 0 the packet is thrown away (stops loops)
- Protocol – what's inside: 1 ICMP, 6 TCP, 17 UDP, 89 OSPF
- Source and destination IP

EXAM TRAP TTL stops packets looping forever.

8.3 Address classes

An old system that grouped addresses by their first number.

REMEMBER

- A: 1–126 (/8)
- B: 128–191 (/16)
- C: 192–223 (/24)
- D: 224–239 multicast
- E: 240–255 reserved

COMMAND / EXAMPLE

```
201.222.5.64 → Class C
```

EXAM TRAP 127.x.x.x is loopback, not a normal Class A address.

8.4 Private and special addresses

Private addresses are for inside your network only – like extension numbers in an office.

REMEMBER

- Private: 10.0.0.0/8, 172.16.0.0/12 (172.16–172.31), 192.168.0.0/16
- 127.0.0.1 = loopback (the device itself)
- 169.254.x.x = PC found no DHCP server
- 255.255.255.255 = broadcast

EXAM TRAP 172.32.x.x is **PUBLIC**, not private.

8.5 Subnetting

Cutting one big network into smaller ones by borrowing host bits.

REMEMBER

- Hosts per subnet = 2^(host bits) − 2
- Pick the smallest subnet that fits
- /26 = 62 hosts, /27 = 30 hosts, /28 = 14 hosts

COMMAND / EXAMPLE

```
Need 40 hosts → /26 (62 hosts)
Valid /26 subnets of 192.168.20.0/24: .0, .64, .128, .192
```

EXAM TRAP Slide quiz: 40 hosts → 192.168.20.64/26 and 192.168.20.128/26. /27 is too small.

8.6 ICMP, ping and tracert

ICMP is the network's messaging system. ping asks 'are you there?'; tracert shows every router along the way.

REMEMBER

- Echo Request (type 8) / Echo Reply (type 0) = ping
- Time Exceeded (type 11) = used by tracert
- Destination Unreachable (type 3)

COMMAND / EXAMPLE

```
ping 10.1.1.2
ping -c 10 -s 1000 10.1.1.2
tracert 10.1.1.2
```

EXAM TRAP ICMP works at the **NETWORK** layer.

8.7 Setting an interface IP

Give the interface an address and mask.

REMEMBER

- Router ports take an IP directly
- V600 switch ports must first change to Layer 3 (undo portswitch)

COMMAND / EXAMPLE

```
[HUAWEI-GE1/0/1] undo portswitch
[HUAWEI-GE1/0/1] ip address 192.168.1.1 24
display ip interface brief
```

8.8 DHCP – how it works

DHCP hands out IP addresses automatically, like a hotel front desk handing out room keys.

REMEMBER

4 steps (DORA):

1. DISCOVER – client shouts 'any DHCP server out there?'
2. OFFER – server offers an IP
3. REQUEST – client says 'I'll take it'
4. ACK – server confirms

- Server port UDP 67, client port UDP 68

EXAM TRAP Order: DISCOVER – OFFER – REQUEST – ACK.

8.9 DHCP lease

The address is borrowed, not owned. The default lease is 1 day.

REMEMBER

- At 50% of the lease → client renews by **UNICAST** to its server
- At 87.5% → client **BROADCASTS** to any server
- DHCP relay = passes DHCP messages to a server in another subnet

EXAM TRAP Client sends REQUEST at first join, 50% and 87.5%. Giving the address back is RELEASE, not REQUEST.

8.10 DHCP configuration

Turn DHCP on, then use an interface pool (simple) or a global pool (flexible).

REMEMBER

- dhcp enable – must be done first!
- Interface pool: dhcp select interface
- Global pool: ip pool + dhcp select global
- Relay: dhcp select relay + dhcp relay server-ip

COMMAND / EXAMPLE

```
[R2] dhcp enable
[R2] ip pool POOL10
[R2-ip-pool-POOL10] network 192.168.10.0 mask 24
[R2-ip-pool-POOL10] gateway-list 192.168.10.254
[R2-GE0/0/1] dhcp select global
[R1-GE0/0/1] ip address dhcp-alloc
display ip pool
```

EXAM TRAP Forgot dhcp enable = nothing works.

Check yourself

Answers: page 115

1. CHOOSE ONE

How many usable host addresses does a /27 subnet have?

A 14
B 30
C 32
D 62

2. CHOOSE ONE

Which address is a PRIVATE address?

A 172.32.1.1
B 11.1.1.1
C 192.169.1.1
D 172.20.5.1

3. CHOOSE ONE

What is the correct order of DHCP messages when a PC first gets an address?

A DISCOVER, OFFER, REQUEST, ACK
B REQUEST, OFFER, DISCOVER, ACK
C DISCOVER, REQUEST, OFFER, ACK
D OFFER, DISCOVER, ACK, REQUEST

4. CHOOSE ONE

When 50% of the DHCP lease has passed, what does the client do?

A Unicasts a REQUEST to its own DHCP server
B Broadcasts a DISCOVER
C Sends a RELEASE
D Broadcasts a REQUEST to any server

5. CHOOSE ONE

A department needs 40 hosts, taken from 192.168.20.0/24. Which subnet is valid and big enough?

A 192.168.20.32/27
B 192.168.20.96/26
C 192.168.20.48/28
D 192.168.20.64/26

CHAPTER 9

IP Routing Basics

EXAM FOCUS

Routing table fields, longest match → preference → cost, preference values, inter-VLAN (sub-interface/VLANIF), static/default/floating routes

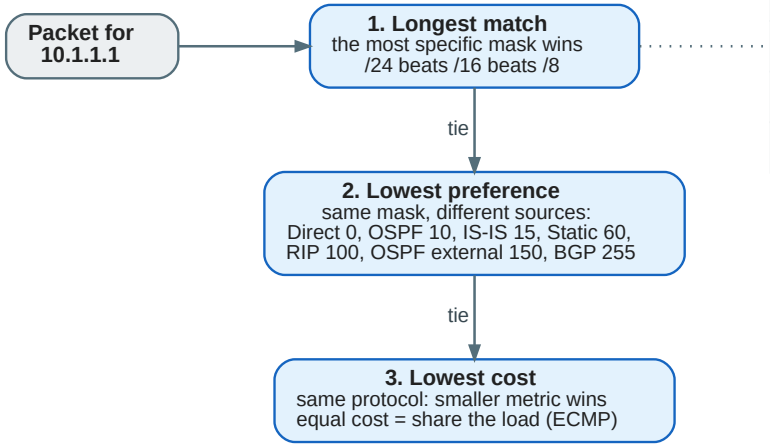
IN THIS CHAPTER

9.1 Routing table · 9.2 Choosing the best route · 9.3 Inter-VLAN routing: router · 9.4 Inter-VLAN routing: Layer 3 switch · 9.5 Static routes · 9.6 Default and floating routes · 9.7 Testing routing

Pictures Look at these first

PICTURE 9.1

How a router chooses the best route



Example routing table

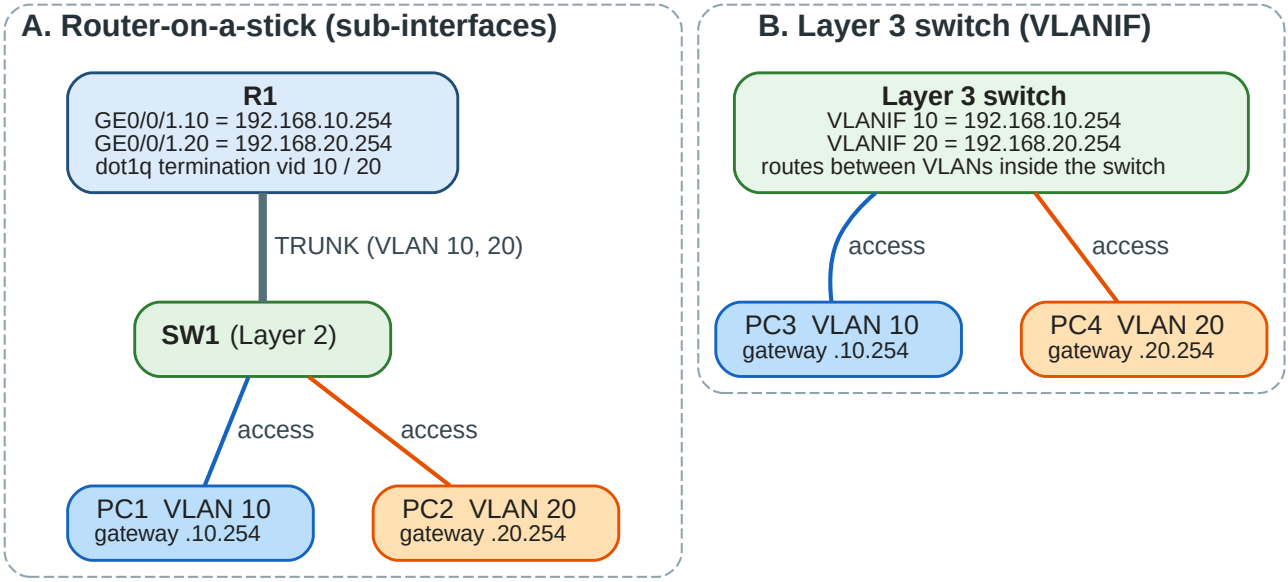
Destination	Proto	Pre	Result
10.0.0.0/8	Static	60	matches
10.1.0.0/16	OSPF	10	matches
10.1.1.0/24	Static	60	**WINS (longest)**

HOW TO READ IT

The router checks these three rules in order and stops at the first one that decides. In the example, three routes match 10.1.1.1 but the /24 wins at step 1, even though OSPF has a better preference.

PICTURE 9.2

Inter-VLAN routing: two ways to do it

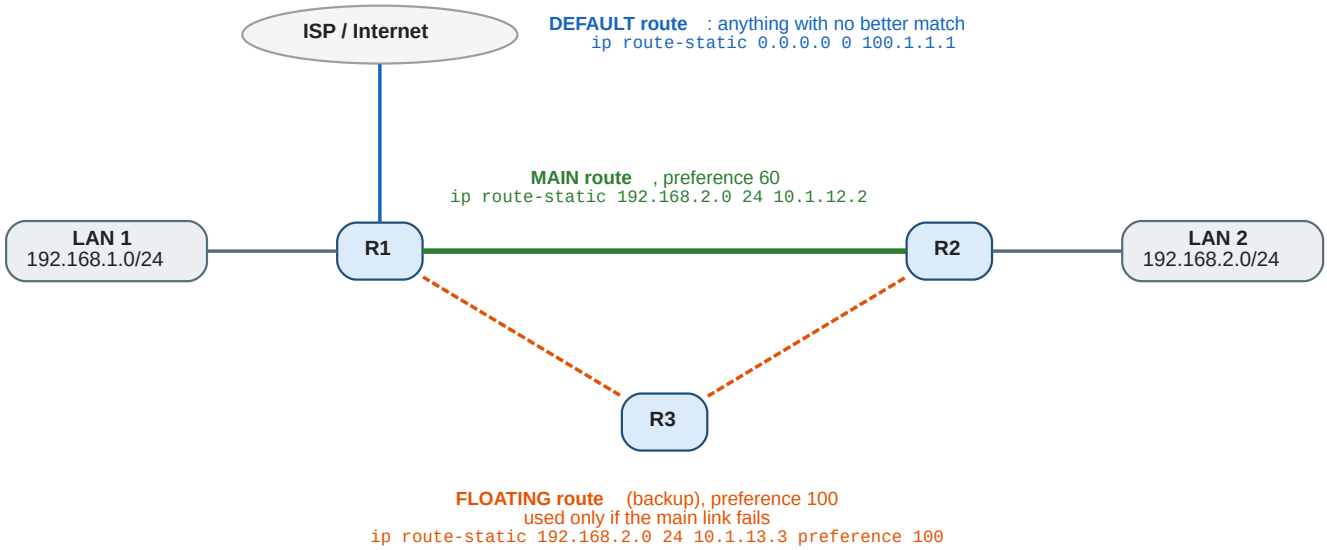


HOW TO READ IT

VLANs cannot talk to each other without a Layer 3 device. Left: one router port split into sub-interfaces over a trunk. Right: a Layer 3 switch with a VLANIF for each VLAN. Each PC uses the .254 address of its own VLAN as its gateway.

PICTURE 9.3

Static, default and floating routes



HOW TO READ IT

The main static route (preference 60) is used while its link works. The floating route has a higher preference (100), so it waits as a backup. The default route 0.0.0.0/0 catches everything else and sends it to the ISP.

Notes Read, then cover the boxes and test yourself

9.1 Routing table

The routing table is the router's map: 'to reach this network, go this way'.

REMEMBER

- Shows: Destination/Mask, Proto (where the route came from), Pre (preference), Cost, NextHop, Interface
- Routes come from: Direct, Static, Dynamic (OSPF etc.)

COMMAND / EXAMPLE

```
display ip routing-table
```

EXAM TRAP Direct routes only appear when the interface is UP.

9.2 Choosing the best route

If several routes match, the router decides in this order.

REMEMBER

1. Longest match (most specific mask) wins
2. Same mask → lowest Preference wins
3. Same protocol → lowest Cost wins (equal cost = share the load)

COMMAND / EXAMPLE

```
Preferences: Direct 0, OSPF 10, Static 60, RIP 100, OSPF external 150, BGP 255
```

EXAM TRAP Longest match beats preference: a /24 static route beats a /16 OSPF route.

9.3 Inter-VLAN routing: router

VLANs need a Layer 3 device to talk to each other.

REMEMBER

- One router port per VLAN – simple but wastes ports
- Router-on-a-stick – one trunk cable, one sub-interface per VLAN

COMMAND / EXAMPLE

```
[R1] interface GE0/0/1.10
[R1-GE0/0/1.10] dot1q termination vid 10
[R1-GE0/0/1.10] ip address 192.168.10.254 24
[R1-GE0/0/1.10] arp broadcast enable   (eNSP)
```

EXAM TRAP The switch port facing a router-on-a-stick must be a **TRUNK**.

9.4 Inter-VLAN routing: Layer 3 switch

A Layer 3 switch has a router built in. Each VLAN gets a VLANIF interface as its gateway.

REMEMBER

- VLANIF 10 = gateway for VLAN 10
- Fastest and most common option
- PC sends to the gateway MAC → switch routes it to the other VLAN

COMMAND / EXAMPLE

```
[SW1] vlan batch 10 20
[SW1] interface Vlanif 10
[SW1-Vlanif10] ip address 192.168.10.254 24
```

EXAM TRAP A VLANIF is UP only if the VLAN exists **AND** at least one port in it is UP.

9.5 Static routes

You type the route in by hand. Simple, but it doesn't adapt if a link fails.

REMEMBER

- Give destination + mask + next hop (or exit interface)
- The other router also needs a route **BACK**

COMMAND / EXAMPLE

```
[R1] ip route-static 192.168.2.0 24 10.1.12.2
```

EXAM TRAP Ping fails if the return route is missing!

9.6 Default and floating routes

Default route = 'if you don't know where to send it, send it here' (usually towards the Internet). Floating route = a spare route that waits until the main one fails.

REMEMBER

- Default route: 0.0.0.0 0
- Floating route: same destination, **HIGHER** preference number

COMMAND / EXAMPLE

```
[R1] ip route-static 0.0.0.0 0 100.1.1.1
[R1] ip route-static 192.168.2.0 24 10.1.13.3 preference 100
```

EXAM TRAP Bigger preference number = less preferred = the backup.

9.7 Testing routing

Check reachability, then find where it breaks.

REMEMBER

- ping – can I reach it?
- ping -a – choose which source IP to test from
- tracert – where does the path stop?

COMMAND / EXAMPLE

```
ping -a 192.168.1.1 192.168.2.1
tracert 192.168.2.1
```

Check yourself

Answers: page 115

1 CHOOSE ONE

R1 has two routes that match 10.1.1.1: OSPF 10.1.0.0/16 (preference 10) and static 10.1.1.0/24 (preference 60). Which route is used?

A The OSPF route, because preference 10 is lower
B Both routes, sharing the load
C Neither; the packet is dropped
D The static route, because /24 is the longer match

2 CHOOSE ONE

What is the default preference of a static route on a Huawei device?

A 0
B 10
C 60
D 100

3 CHOOSE ONE

How do you make a static route a FLOATING (backup) route?

A Give it a lower preference value than the main route
B Give it a higher cost
C Use a longer mask
D Give it a higher preference value than the main route

4 TRUE OR FALSE?

A VLANIF interface comes UP as soon as it is given an IP address.

☐ True ☐ False

5 CHOOSE ONE

PC1 (192.168.1.0/24) cannot ping PC2 (192.168.2.0/24). R1 has a route to 192.168.2.0/24, but R2 has no route to 192.168.1.0/24. What is the problem?

A ARP is disabled
B STP is blocking the link
C DNS is not configured
D The return route is missing

CHAPTER 10

OSPF Fundamentals and Configuration

EXAM FOCUS

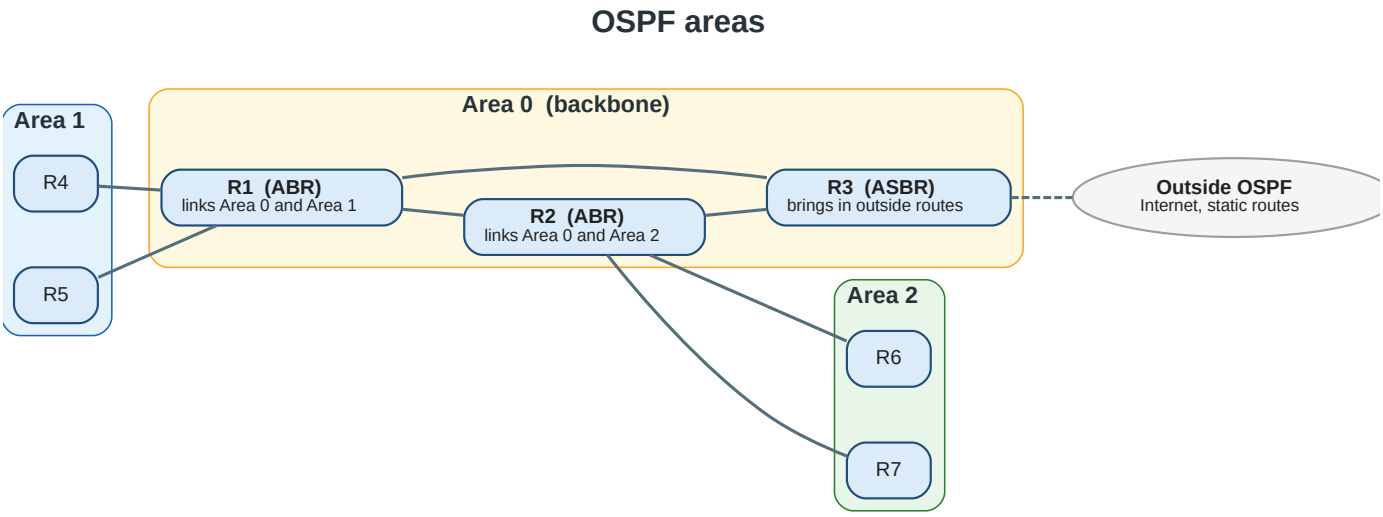
Router ID, areas, 5 packet types, neighbour states, DR/BDR election, Hello parameters, cost, config & display commands

IN THIS CHAPTER

10.1 What OSPF is · 10.2 Router ID and areas · 10.3 OSPF packets · 10.4 Neighbour states · 10.5 Must match to become neighbours · 10.6 DR and BDR · 10.7 OSPF cost · 10.8 Configure OSPF · 10.9 Check OSPF

Pictures Look at these first

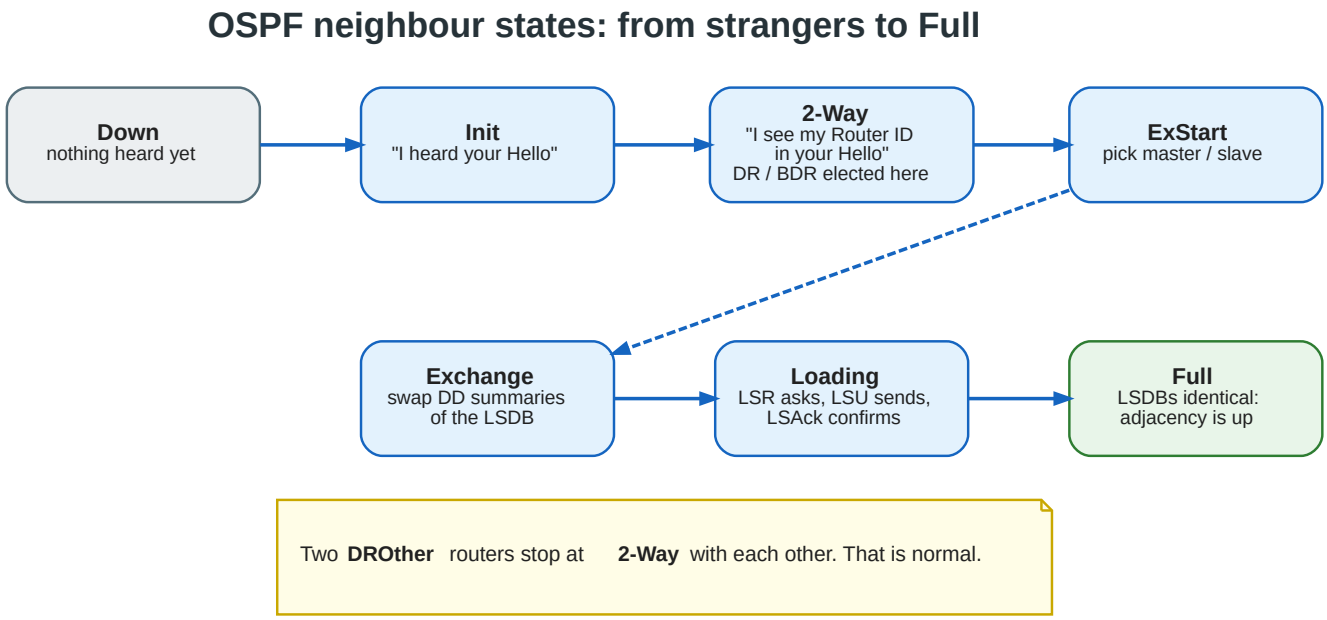
PICTURE 10.1



HOW TO READ IT

Area 0 is the backbone and every other area must connect to it. An ABR sits between two areas. An ASBR brings in routes from outside OSPF.

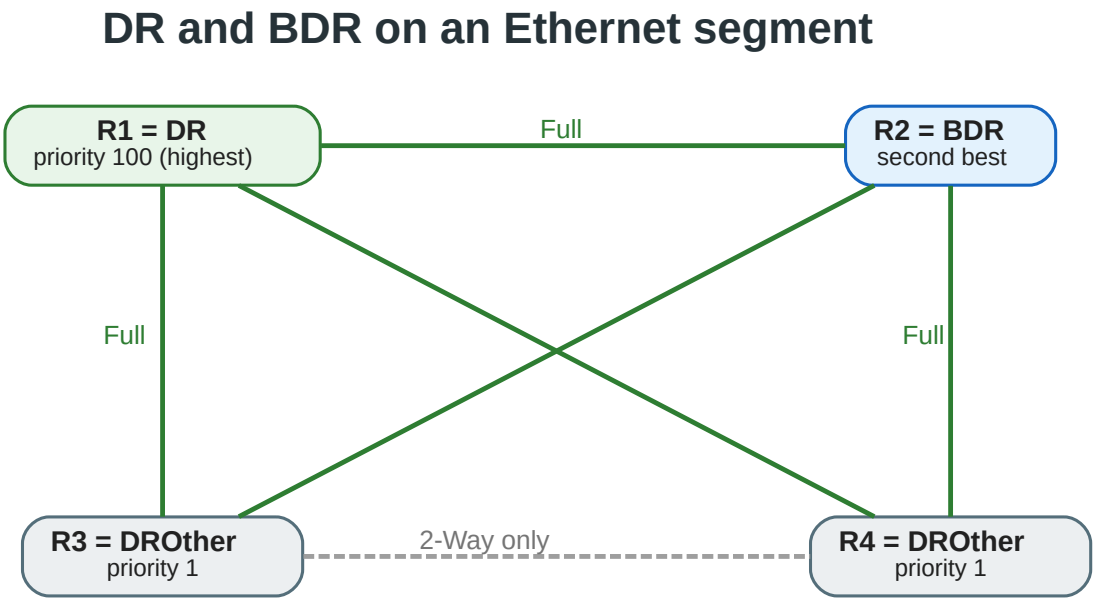
PICTURE 10.2



HOW TO READ IT

Two routers pass through these states in order. At 2-Way they can see each other and the DR/BDR is elected. Full means their link-state maps (LSDBs) are identical.

PICTURE 10.3



Election is **not preemptive** : a new router with a higher priority does NOT take over until OSPF restarts. Point-to-point links have no DR.

HOW TO READ IT

On Ethernet, every router becomes Full only with the DR and the BDR, not with every other router. The DR is the router with the highest priority (default 1), then the highest router ID. 0 means never DR.

Notes Read, then cover the boxes and test yourself

10.1 What OSPF is

OSPF is a routing protocol where every router builds the same full map of the network, then works out the shortest path by itself.

REMEMBER

- Link-state protocol; uses the SPF (Dijkstra) calculation
- Runs directly on IP (protocol 89)
- Multicast 224.0.0.5 (all OSPF routers) and 224.0.0.6 (DR/BDR)
- Preference 10

EXAM TRAP OSPF does **NOT** use TCP or UDP.

10.2 Router ID and areas

Router ID = the router's name tag in OSPF. Areas = splitting a big network into regions.

REMEMBER

- Router ID: 32 bits, must be unique, best set by hand
- Area 0 = backbone; every other area must connect to it
- ABR = router that sits between areas

COMMAND / EXAMPLE

```
[R1] ospf 1 router-id 1.1.1.1
[R1-ospf-1] area 0
```

EXAM TRAP A new router ID only takes effect after reset ospf process.

10.3 OSPF packets

OSPF uses five types of message.

REMEMBER

1. Hello – find and keep neighbours
2. DD – summary of the map
3. LSR – 'please send me this part'
4. LSU – 'here it is' (carries LSAs)
5. LSAck – 'got it'

EXAM TRAP LSU carries the LSAs.

10.4 Neighbour states

The steps two routers go through to become full neighbours.

REMEMBER

Down → Init → 2-Way → ExStart → Exchange → Loading → Full

- Full = maps fully synced

COMMAND / EXAMPLE

```
display ospf peer brief
```

EXAM TRAP Two DROther routers stay at 2-Way – that is **NORMAL**.

10.5 Must match to become neighbours

If any of these don't match, the routers won't become neighbours.

REMEMBER

- Area ID
- Hello and Dead timers (default 10 s / 40 s)
- Authentication
- Subnet mask (on Ethernet)
- Router IDs must be **DIFFERENT**

EXAM TRAP Most common lab problem: wrong area, wrong timers or duplicate router ID.

10.6 DR and BDR

On Ethernet, one router (the DR) acts as team leader so every router doesn't have to sync with every other router.

REMEMBER

- Highest priority wins (default 1; 0 = never DR)
- Tie → highest router ID
- No takeover after the election (non-preemptive)
- Point-to-point links have no DR

COMMAND / EXAMPLE

```
[R1-GE0/0/1] ospf dr-priority 100
```

EXAM TRAP Raising the priority later does **NOT** make it DR until OSPF restarts.

10.7 OSPF cost

Cost = how 'expensive' a link is. A faster link has a lower cost.

REMEMBER

- Cost = 100 Mbit/s ÷ interface speed (minimum 1)
- Route cost = add up the costs along the path

COMMAND / EXAMPLE

```
[R1-ospf-1] bandwidth-reference 10000
[R1-GE0/0/1] ospf cost 10
```

EXAM TRAP By default 100M and 1G links both have cost 1. Change the reference on **ALL** routers.

10.8 Configure OSPF

Start OSPF, set the router ID, join an area, list your networks.

REMEMBER

- The network command uses a **WILDCARD** mask (the opposite of a subnet mask)
- silent-interface = don't send Hellos towards PCs

COMMAND / EXAMPLE

```
[R1] ospf 1 router-id 1.1.1.1
[R1-ospf-1] area 0
[R1-ospf-1-area-0.0.0.0] network 10.1.12.0 0.0.0.255
[R1-ospf-1] silent-interface GE0/0/2
[R1-ospf-1] default-route-advertise always
```

EXAM TRAP 0.0.0.255 is a wildcard, **NOT** 255.255.255.0.

10.9 Check OSPF

Full in the peer list means a healthy neighbour.

REMEMBER

- Neighbours, interfaces, the map (LSDB) and learned routes

COMMAND / EXAMPLE

```
display ospf peer brief
display ospf interface
display ospf lsdb
display ip routing-table protocol ospf
```

Check yourself

Answers: page 115

1. CHOOSE ONE

How are OSPF packets carried?

A Directly over IP, protocol number 89
B TCP port 89
C UDP port 89
D UDP port 520

2. CHOOSE ONE

Which OSPF packet type carries LSAs?

A Hello
B DD
C LSU
D LSR

3. CHOOSE TWO

Which TWO must MATCH for two routers to become OSPF neighbours?

A Area ID
B Router ID
C Hello and Dead timers
D Interface cost

4. CHOOSE ONE

On an Ethernet segment, two DROther routers normally stay in which neighbour state with each other?

A Init
B ExStart
C 2-Way
D Full

5 CHOOSE ONE

With the default reference bandwidth (100 Mbit/s), what is the OSPF cost of a 1 Gbit/s interface?

A 1
B 10
C 100
D 1000

CHAPTER 11

IPv6 Address and Configuration

EXAM FOCUS
Format & compression, header, address types & prefixes, EUI-64, NDP (RS/RA/NS/NA), SLAAC vs DHCPv6, config

IN THIS CHAPTER
11.1 Why IPv6 and how it's written · 11.2 IPv6 header · 11.3 IPv6 address types · 11.4 EUI-64 · 11.5 NDP · 11.6 Getting an IPv6 address · 11.7 Configure IPv6

Pictures Look at these first

PICTURE 11.1

Writing an IPv6 address the short way

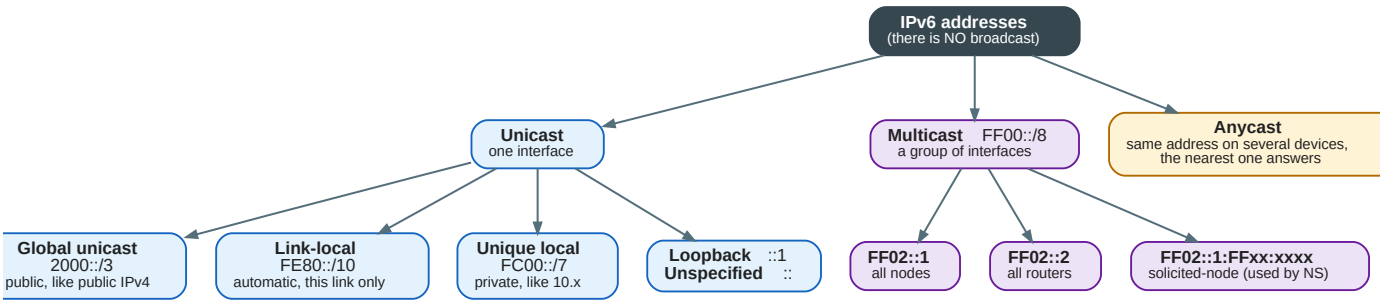
Full address 8 groups x 16 bits	2001:0DB8:0000:0000:0008:0800:200C:417A
Step 1 drop leading zeros	2001:DB8:0:0:8:800:200C:417A
Step 2 zero groups become ::	**2001:DB8::8:800:200C:417A**

Red = the zeros that get removed. **:: only once** , otherwise nobody knows how many zero groups each :: hides.

HOW TO READ IT
An IPv6 address is 128 bits written as 8 groups of 4 hex digits. Step 1: drop the leading zeros in each group. Step 2: replace ONE run of all-zero groups with ::. You may use :: only once.

PICTURE 11.2

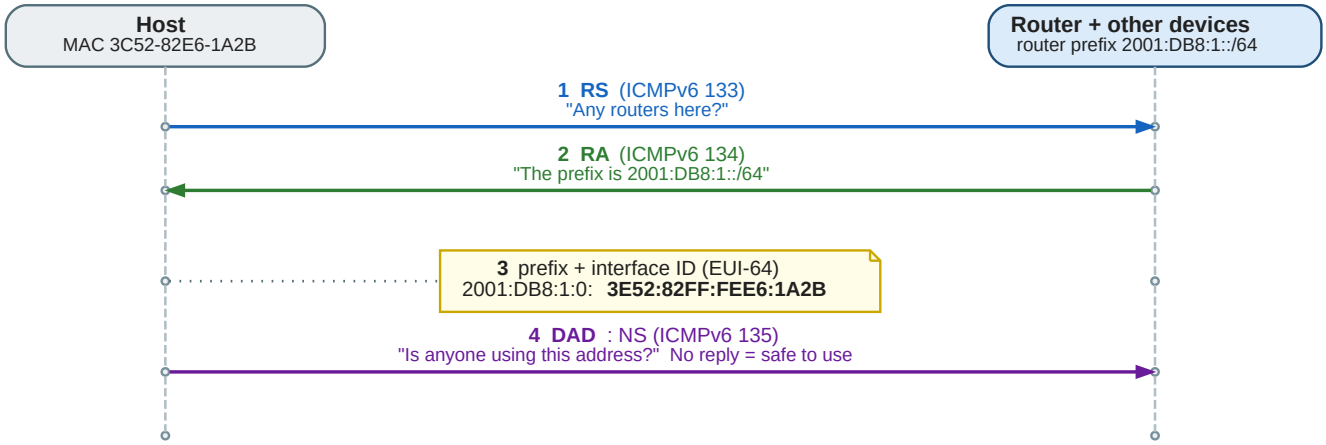
IPv6 address types



HOW TO READ IT
IPv6 has no broadcast. Unicast addresses identify one interface, multicast addresses reach a group, and anycast addresses reach the nearest of several devices. Remember the prefixes: 2000::/3, FE80::/10, FC00::/7, FF00::/8.

PICTURE 11.3

SLAAC: a host builds its own IPv6 address



HOW TO READ IT

1: the host asks for a router (RS). 2: the router replies with the prefix (RA). 3: the host adds its own 64-bit interface ID to the prefix. 4: it checks nobody else uses the address (DAD) before using it.

Notes

Read, then cover the boxes and test yourself

11.1 Why IPv6 and how it's written

IPv4 ran out of addresses. IPv6 is far bigger: 128 bits.

REMEMBER

- 8 groups of 4 hex digits, separated by :
- Drop leading zeros in each group
- Replace **ONE** run of all-zero groups with ::
- There is no broadcast in IPv6

COMMAND / EXAMPLE

```
2001:0DB8:0000:0000:0008:0800:200C:417A
→ 2001:DB8::8:800:200C:417A
```

EXAM TRAP :: can be used only **ONCE** in an address.

11.2 IPv6 header

Simpler than IPv4, and always the same size.

REMEMBER

- 40 bytes, fixed
- Hop Limit = IPv4's TTL
- Next Header = IPv4's Protocol field
- Extra options go in extension headers

11.3 IPv6 address types

Different prefixes mean different jobs.

REMEMBER

- Global unicast 2000::/3 – public (like public IPv4)
- Link-local FE80::/10 – automatic, only works on the local link
- Unique local FC00::/7 – private
- Multicast FF00::/8 (FF02::1 all nodes, FF02::2 all routers)
- ::1 = loopback, :: = unspecified
- Anycast = same address on many devices; the nearest one answers

EXAM TRAP FE80 = link-local. FF = multicast.

11.4 EUI-64

A way to build the last 64 bits of the address from the MAC address.

REMEMBER

1. Split the MAC in half
2. Put FFFE in the middle
3. Flip the 7th bit

COMMAND / EXAMPLE

```
MAC 3C52-82E6-1A2B
→ 3E52:82FF:FEE6:1A2B
```

EXAM TRAP Don't forget to flip the 7th bit (3C → 3E).

11.5 NDP

IPv6's replacement for ARP (and more), using ICMPv6 messages.

REMEMBER

- RS 133 – host asks for a router
- RA 134 – router advertises the prefix
- NS 135 / NA 136 – find a neighbour's MAC
- DAD – check nobody else is using my address

COMMAND / EXAMPLE

```
display ipv6 neighbors
```

EXAM TRAP IPv6 has no ARP: it uses NS/NA instead.

11.6 Getting an IPv6 address

There are three ways a host can get an IPv6 address.

REMEMBER

- Manual
- SLAAC – host builds it from the router's prefix (stateless)
- DHCPv6 – a server gives it (stateful)
- RA flags: M = use DHCPv6 for the address; O = use DHCPv6 for other info (e.g. DNS)

EXAM TRAP M = Managed (stateful). O = Other (stateless).

11.7 Configure IPv6

Turn IPv6 on globally **AND** on the interface, then add the address.

REMEMBER

- undo ipv6 nd ra halt = let the router send RAs (needed for SLAAC)

COMMAND / EXAMPLE

```
[R1] ipv6
[R1-GE0/0/1] ipv6 enable
[R1-GE0/0/1] ipv6 address 2001:DB8:1::1 64
[R1-GE0/0/1] undo ipv6 nd ra halt
[R2-GE0/0/1] ipv6 address auto global
display ipv6 interface brief
```

EXAM TRAP Forgot 'ipv6' in system view = nothing works.

Check yourself

Answers: page 115

1. CHOOSE ONE

   What is the correct short form of 2001:0DB8:0000:0000:0008:0800:200C:417A?

   A 2001:DB8::8::200C:417A
   B 2001:DB8:8:800:200C:417A
   C 2001:DB8::8:800:200C:417A
   D 2001:DB8:0::8:8:200C:417A

2. CHOOSE ONE

   Which prefix is used by IPv6 link-local addresses?

   A 2000::/3
   B FC00::/7
   C FE80::/10
   D FF00::/8

3. CHOOSE ONE

   Which messages does IPv6 use instead of ARP to find a neighbour's MAC address?

   A RS and RA
   B DHCPv6 Solicit and Advertise
   C ICMPv6 Echo Request and Reply
   D NS and NA

4. TRUE OR FALSE?

   IPv6 uses broadcast to find other devices on the link.

   ☐ True ☐ False

5. CHOOSE ONE

   Using EUI-64, what interface ID is built from MAC 3C52-82E6-1A2B?

   A 3C52:82FF:FEE6:1A2B
   B 3E52:82E6:FFFE:1A2B
   C 3E52:82FF:FEE6:1A2B
   D 3C52:82E6:1A2B:FFFE

CHAPTER 12

ACL Fundamentals and Configuration

EXAM FOCUS

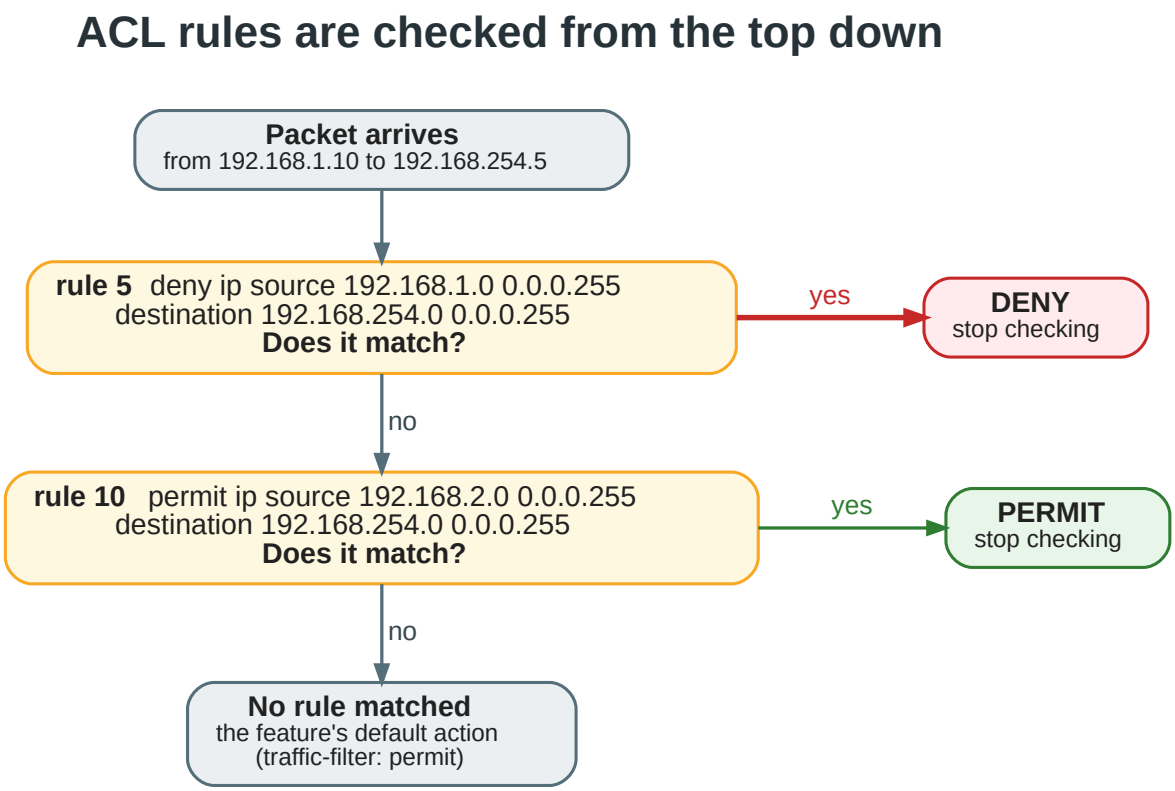
Basic vs advanced vs L2 numbers, wildcard masks, rule order & step, MQC (classifier → behavior → policy), placement

IN THIS CHAPTER

12.1 What an ACL is · 12.2 ACL types · 12.3 Rule order · 12.4 Wildcard mask · 12.5 Basic and advanced examples · 12.6 Applying an ACL with MQC · 12.7 Where to put the ACL

Pictures Look at these first

PICTURE 12.1



HOW TO READ IT

A packet from 192.168.1.10 to the server matches rule 5 first, so it is denied and the other rules are never checked. Order matters: the first rule that matches decides.

PICTURE 12.2

Wildcard masks: 0 = must match, 1 = don't care

Address	192	168	1	0
Wildcard	00000000	00000000	00000000	11111111
in decimal	0	0	0	255
Meaning	**MUST be 192.168.1**			**anything**

You write	It matches
`source 192.168.1.10 0`	exactly one host
`source 192.168.1.0 0.0.0.255`	the whole 192.168.1.0/24
`source any`	everything

HOW TO READ IT

A wildcard mask works like an upside-down subnet mask. Each 0 bit must match the address; each 1 bit can be anything. So 192.168.1.0 0.0.0.255 matches every address from 192.168.1.0 to 192.168.1.255.

PICTURE 12.3

MQC: four steps to put an ACL to work

Step	What it does	Command
0. ACL 3000	picks out WHICH traffic	`acl 3000` `rule deny ip source 192.168.1.0 0.0.0.255 ...`
1. Classifier	matches the ACL	`traffic classifier C1` `if-match acl 3000`
2. Behavior	says WHAT TO DO (permit or deny)	`traffic behavior B1` `deny`
3. Policy	joins classifier + behavior	`traffic policy P1` `classifier C1 behavior B1`
4. Interface	applies it, IN or OUT	`interface GE0/0/1` `traffic-policy P1 inbound`

HOW TO READ IT

An ACL only picks out traffic. MQC decides what to do with it: the classifier matches the ACL, the behavior says permit or deny, the policy joins them, and the policy is applied on an interface in one direction.

Notes Read, then cover the boxes and test yourself

12.1 What an ACL is

An ACL is a list of rules that picks out traffic. Think of a guest list: it only identifies people – the bouncer (another feature) decides what to do with them.

REMEMBER

- Each rule = permit or deny
- Used by: traffic filtering, NAT, routing policy, login control

EXAM TRAP An ACL on its own does nothing until a feature uses it.

12.2 ACL types

The ACL number tells you what it can check.

REMEMBER

- Basic 2000–2999 – checks **SOURCE** IP only
- Advanced 3000–3999 – source & destination IP, protocol, ports
- Layer 2 4000–4999 – MAC addresses, VLAN

EXAM TRAP Need to match a destination or a port → must use **ADVANCED**.

12.3 Rule order

Rules are checked top to bottom. The first match wins; the rest are ignored.

REMEMBER

- Rule numbers go up in steps of 5 (5, 10, 15...)
- The gaps let you insert rules later
- Put specific rules before general ones

EXAM TRAP Order matters – a broad permit at the top hides a deny further down.

12.4 Wildcard mask

Tells the ACL which bits must match. 0 = must match, 1 = don't care.

REMEMBER

- 0.0.0.0 (or 0) = one exact host
- 0.0.0.255 = a whole /24
- any = everything

COMMAND / EXAMPLE

```
rule permit source 192.168.1.0 0.0.0.255
rule deny source 192.168.1.10 0
```

EXAM TRAP A wildcard looks like an upside-down subnet mask.

12.5 Basic and advanced examples

Basic = source only. Advanced = add protocol, destination and port.

REMEMBER

- Basic ACL view: acl4-basic
- Advanced ACL view: acl4-advance

COMMAND / EXAMPLE

```
[R1] acl 2000
[R1-acl4-basic-2000] rule 5 deny source 192.168.1.0 0.0.0.255
[R1] acl 3000
[R1-acl4-advance-3000] rule permit tcp source 192.168.2.0 0.0.0.255 destination 192.168.254.10 0
destination-port eq 80
```

EXAM TRAP Valid basic rule (slide quiz): rule permit source any.

12.6 Applying an ACL with MQC

MQC puts the ACL to work in steps: which traffic → what action → bundle them → put it on a port.

REMEMBER

1. Classifier – match the ACL
2. Behavior – permit or deny
3. Policy – join classifier + behavior
4. Apply on an interface, inbound or outbound

COMMAND / EXAMPLE

```
[R1] traffic classifier C1
[R1-classifier-C1] if-match acl 3000
[R1] traffic behavior B1
[R1-behavior-B1] deny
[R1] traffic policy P1
[R1-trafficpolicy-P1] classifier C1 behavior B1
[R1-GE0/0/1] traffic-policy P1 inbound
(eNSP shortcut: traffic-filter inbound acl 3000)
```

EXAM TRAP Remember the order: classifier → behavior → policy → interface.

12.7 Where to put the ACL

Place it where it blocks the right traffic without blocking too much.

REMEMBER

- Advanced ACL – near the **SOURCE** (drop early)
- Basic ACL – near the **DESTINATION** (so you don't block too much)

COMMAND / EXAMPLE

```
display acl 3000
```

EXAM TRAP display acl shows how many times each rule matched.

Check yourself Answers: page 115

1 CHOOSE ONE

Which number range is used for ADVANCED ACLs?

- A 2000-2999
- B 3000-3999
- C 4000-4999
- D 1-99

2 CHOOSE ONE

You need to block web traffic (TCP port 80) to one server. Which ACL type must you use?

- A Basic ACL
- B Layer 2 ACL
- C Any type will work
- D Advanced ACL

3 CHOOSE ONE

Which wildcard mask matches exactly ONE host, such as 192.168.1.10?

- A 0.0.0.255
- B 255.255.255.255
- C 255.255.255.0
- D 0 (0.0.0.0)

4 CHOOSE ONE

What is the correct order when applying an ACL with MQC?

- A Classifier, behavior, policy, apply to interface
- B Behavior, classifier, interface, policy
- C Policy, classifier, behavior, apply to interface
- D Interface, policy, behavior, classifier

5 TRUE OR FALSE?

ACL rules are checked from top to bottom, and the first matching rule decides.

- ☐ True
- ☐ False

CHAPTER 13

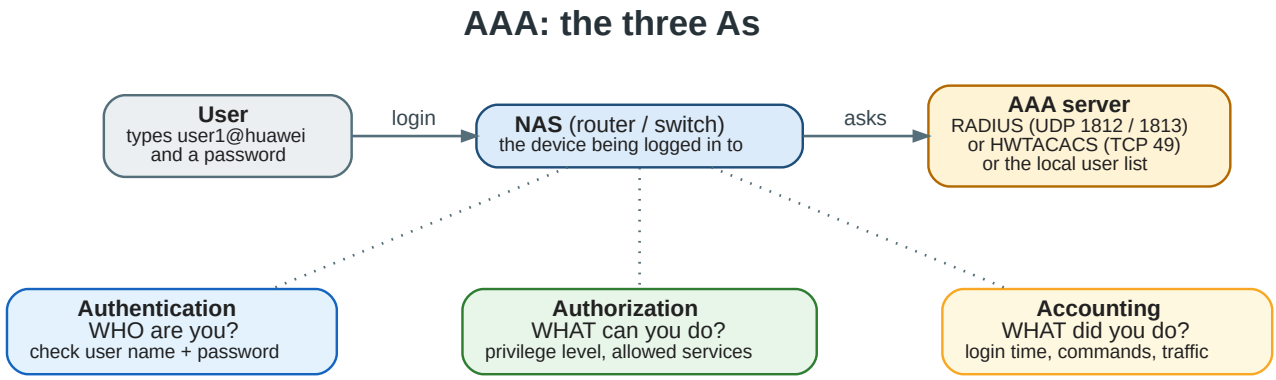
AAA Fundamentals and Configuration

EXAM FOCUS
3 As, authentication/authorization/accounting modes, RADIUS vs HWTACACS, domains, local-user config

IN THIS CHAPTER
13.1 AAA = the 3 As · 13.2 AAA methods · 13.3 RADIUS vs HWTACACS · 13.4 Domains · 13.5 Local AAA setup · 13.6 Check AAA

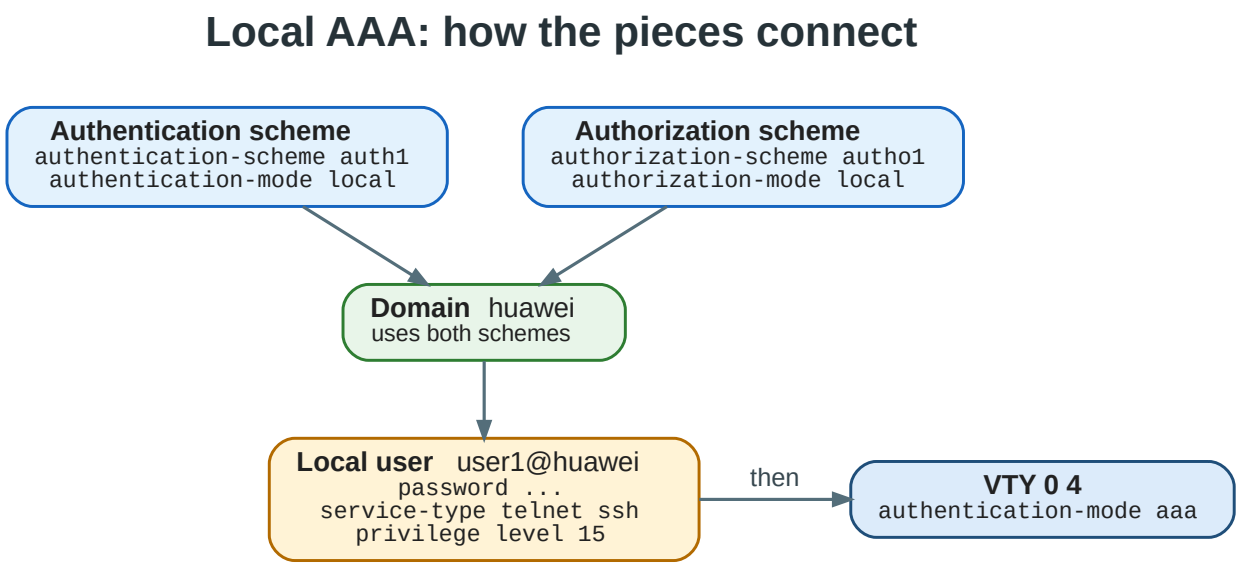
Pictures Look at these first

PICTURE 13.1



HOW TO READ IT
The user logs in through a network device (the NAS). The NAS checks with an AAA server, or with its own local user list. Authentication = who are you; Authorization = what may you do; Accounting = what did you do.

PICTURE 13.2



HOW TO READ IT
Schemes say HOW to check (local or a server). The domain groups the schemes. The local user belongs to the domain. Finally the VTY lines must use authentication-mode aaa, or the AAA settings are never used.

PICTURE 13.3

RADIUS vs HWTACACS

	RADIUS	HWTACACS
Transport	**UDP** 1812 (auth), 1813 (accounting)	**TCP 49**
Encryption	password only	the whole packet body
The three As	authentication + authorization together	all three separate
Best for	users joining the network	admins logging in to devices (per-command authorization)

HOW TO READ IT

Both are central AAA servers. RADIUS is common for user network access. HWTACACS (Huawei's TACACS+) is better for administrators logging in to devices because it encrypts everything and can authorize each command.

Notes

Read, then cover the boxes and test yourself

13.1 AAA = the 3 As

AAA is like a gym: check the membership card (Authentication), decide which rooms you may use (Authorization), log how long you stayed (Accounting).

REMEMBER

- Authentication – **WHO** are you?
- Authorization – **WHAT** can you do?
- Accounting – **WHAT** did you do?

EXAM TRAP Match each A to its question.

13.2 AAA methods

Each A can be done locally on the device or by a central server.

REMEMBER

- Authentication: none, local, RADIUS, HWTACACS
- Authorization: none, local, remote, if-authenticated
- Accounting: none, remote (no local option)

EXAM TRAP Accounting cannot be done locally.

13.3 RADIUS vs HWTACACS

Two kinds of central AAA server.

REMEMBER

- RADIUS – UDP 1812/1813; encrypts only the password; common for user access
- HWTACACS – TCP 49; encrypts everything; separates the 3 As; good for admin logins

EXAM TRAP UDP = RADIUS. TCP 49 = HWTACACS.

13.4 Domains

A domain is a group of users who share the same AAA rules.

REMEMBER

- User name format: name@domain
- Default domains: default and default_admin
- Schemes are attached to the domain, not to the user

13.5 Local AAA setup

Four steps: scheme → domain → user → point the VTY lines to AAA.

REMEMBER

- service-type controls how the user may log in (telnet, ssh, ftp...)
- privilege level 15 = full admin

COMMAND / EXAMPLE

```
[R1] aaa
[R1-aaa] authentication-scheme auth1
[R1-aaa-authen-auth1] authentication-mode local
[R1-aaa] domain huawei
[R1-aaa-domain-huawei] authentication-scheme auth1
[R1-aaa] local-user user1@huawei password irreversible-cipher Huawei@123
[R1-aaa] local-user user1@huawei service-type telnet ssh
[R1-aaa] local-user user1@huawei privilege level 15
[R1] user-interface vty 0 4
[R1-ui-vty0-4] authentication-mode aaa
```

EXAM TRAP Login fails? Check the service-type and the VTY authentication-mode.

13.6 Check AAA

See users, domains and who is logged in.

REMEMBER

- Use these after setup to confirm it worked

COMMAND / EXAMPLE

```
display local-user
display domain name huawei
display users
```

Check yourself

Answers: page 115

1 CHOOSE ONE

Which 'A' of AAA answers the question 'What did you do?'

- A Authentication
- B Authorization
- C Accounting
- D Administration

2 CHOOSE ONE

Which AAA function CANNOT be done locally on the device?

- A Authentication
- B Authorization
- C Accounting
- D All of them can be done locally

3 CHOOSE TWO

Which TWO statements about HWTACACS are true?

- A It uses TCP port 49
- B It encrypts only the password
- C It encrypts the whole message body
- D It uses UDP ports 1812 and 1813

4 CHOOSE ONE

A local user types the correct password but cannot log in with Telnet. What is a likely cause?

- A The user's service-type does not include telnet
- B The user has privilege level 15
- C The user belongs to a domain
- D OSPF is not running

5 CHOOSE ONE

Which are the default AAA domains on a Huawei device?

- A huawei and admin
- B local and remote
- C system and user
- D default and default_admin

CHAPTER 14

Network Services and Applications

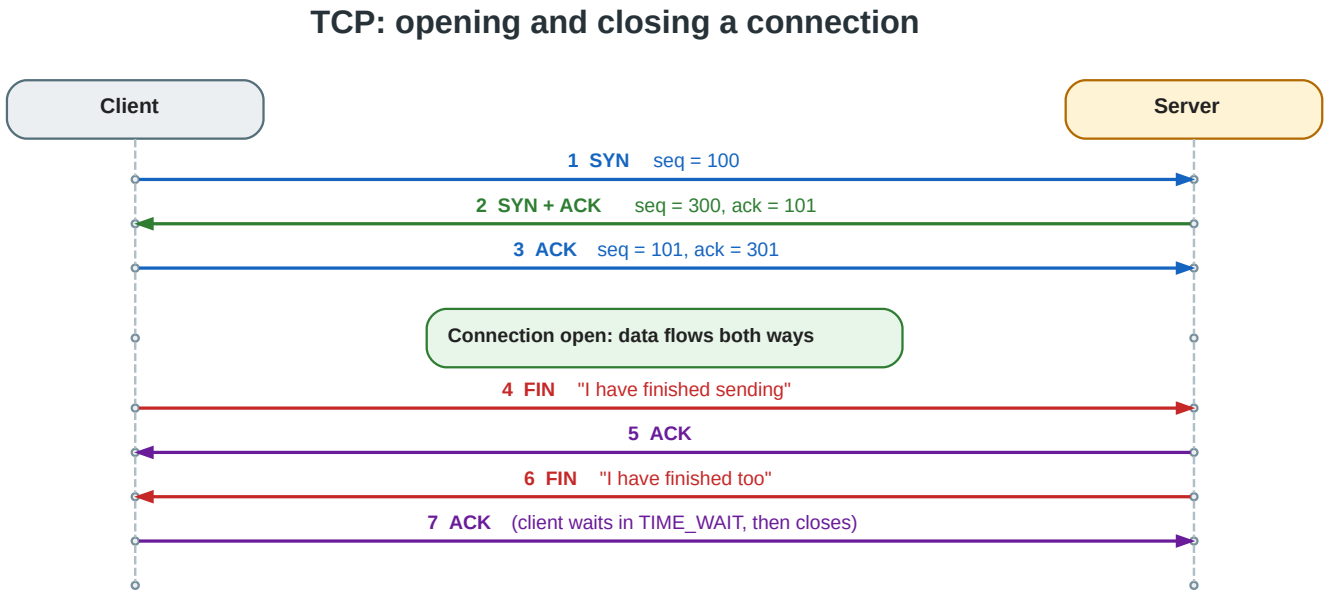
EXAM FOCUS
TCP header, 3-way handshake & 4-way close (seq/ack), port ranges, FTP active vs passive, TFTP, Telnet vs SSH config

IN THIS CHAPTER

14.1 Port numbers and headers · 14.2 TCP 3-way handshake · 14.3 TCP closing (4 steps) · 14.4 FTP · 14.5 TFTP · 14.6 Telnet · 14.7 SSH (STelnet and SFTP)

Pictures Look at these first

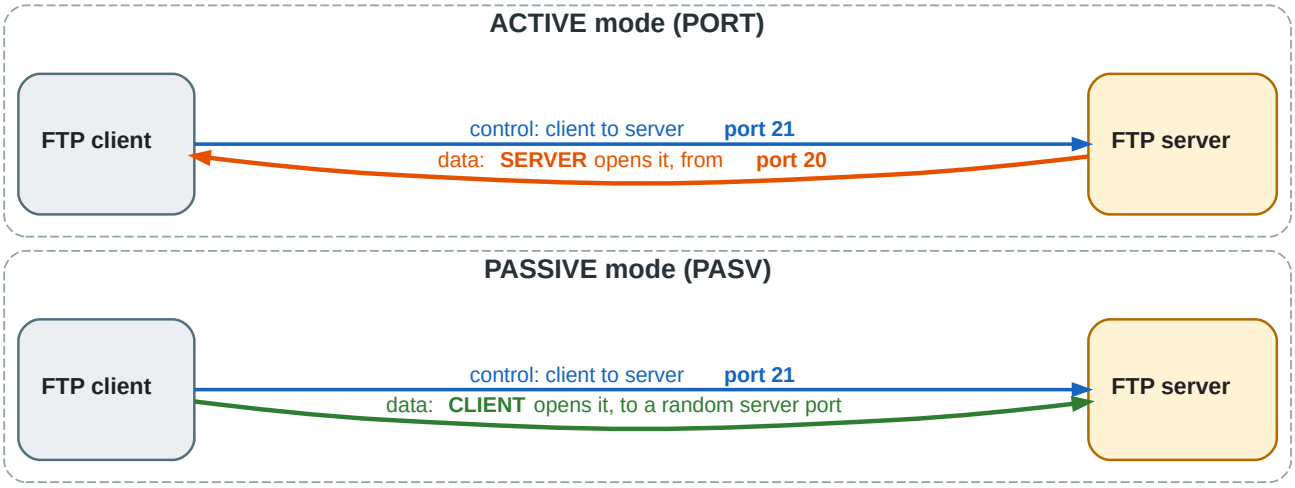
PICTURE 14.1



HOW TO READ IT
Opening takes 3 messages: SYN, SYN+ACK, ACK. Each ACK number is the other side's sequence number + 1. Closing takes 4 messages because each side closes its own direction: FIN, ACK, FIN, ACK.

PICTURE 14.2

FTP: active mode vs passive mode



HOW TO READ IT

FTP always uses two connections: control on TCP 21 and a separate data connection. In ACTIVE mode the server opens the data connection from port 20. In PASSIVE mode the client opens it, which works better through firewalls and NAT.

PICTURE 14.3

Remote access and file transfer: ports and security

Service	Used for	Port	Login?	Encrypted?
Telnet	remote CLI	TCP 23	yes	**No** (plain text)
STelnet (SSH)	remote CLI	TCP 22	yes	**Yes**
FTP	file transfer	TCP 20 + 21	yes	**No**
SFTP (SSH)	file transfer	TCP 22	yes	**Yes**
TFTP	simple file copy	UDP 69	**No**	**No**

HOW TO READ IT

Telnet sends passwords as plain text, so anyone with Wireshark can read them. SSH (STelnet and SFTP) encrypts everything. TFTP is the simplest: UDP, no login at all.

Notes Read, then cover the boxes and test yourself

14.1 Port numbers and headers

Ports tell the receiving computer which app gets the data.

REMEMBER

- 0–1023 well-known (servers)
- 1024–49151 registered
- 49152–65535 dynamic (clients)
- TCP header = 20 bytes; UDP header = 8 bytes

14.2 TCP 3-way handshake

Like starting a phone call: 'Hello?' – 'Hello, I can hear you' – 'Great, let's talk'.

REMEMBER

1. Client → SYN (seq = a)
2. Server → SYN+ACK (seq = b, ack = a+1)
3. Client → ACK (ack = b+1)

EXAM TRAP ack = the other side's seq + 1 (for SYN/FIN), or + the number of data bytes.

14.3 TCP closing (4 steps)

Each side says goodbye separately.

REMEMBER

FIN → ACK → FIN → ACK

- The side that closes first waits in TIME_WAIT

EXAM TRAP Closing takes 4 messages because each direction closes on its own.

14.4 FTP

FTP copies files using **TWO** connections: one for commands and one for the file itself.

REMEMBER

- Control connection = TCP 21
- Active mode: the **SERVER** opens the data connection from port 20
- Passive mode: the **CLIENT** opens the data connection (firewall-friendly)
- Use **BINARY** mode for software and zip files

COMMAND / EXAMPLE

```
[R2] ftp server enable
[R2-aaa] local-user huawei service-type ftp
[R2-aaa] local-user huawei ftp-directory flash:/
<R1> ftp 10.0.0.2
[ftp] binary
[ftp] get vrpcfg.zip
```

EXAM TRAP Active = server starts the data connection. Passive = client starts it.

14.5 TFTP

A very simple file-copy tool.

REMEMBER

- UDP 69
- No username or password
- Only get and put

COMMAND / EXAMPLE

```
<R1> tftp 10.0.0.2 get vrpcfg.zip
```

EXAM TRAP TFTP = UDP 69. FTP = TCP 20/21.

14.6 Telnet

Remote CLI access – but everything, including passwords, is sent as plain text.

REMEMBER

- TCP 23
- Uses VTY lines (vty 0 4)

COMMAND / EXAMPLE

```
[R2] telnet server enable
[R2] user-interface vty 0 4
[R2-ui-vty0-4] protocol inbound telnet
[R2-ui-vty0-4] authentication-mode aaa
<R1> telnet 10.0.0.2
```

EXAM TRAP Wireshark can read Telnet passwords – that's why SSH is preferred.

14.7 SSH (STelnet and SFTP)

Same idea as Telnet, but encrypted – like talking in a private code.

REMEMBER

- TCP 22
- STelnet = secure CLI; SFTP = secure file transfer
- The device needs a key pair first

COMMAND / EXAMPLE

```
[R2] rsa local-key-pair create
[R2] stelnet server enable
[R2] ssh user huawei authentication-type password
[R2] ssh user huawei service-type all
[R2-ui-vty0-4] protocol inbound ssh
[R1] ssh client first-time enable
[R1] stelnet 10.0.0.2
```

EXAM TRAP No key pair = SSH won't work.

Check yourself Answers: page 115

1 CHOOSE ONE

A client sends SYN with seq = 100. What ack number does the server put in its SYN+ACK?

- A 1
- B 100
- C 101
- D 200

2 CHOOSE ONE

In FTP ACTIVE mode, who opens the data connection?

- A The server, from port 20
- B The client, from port 21
- C The client, to a random server port
- D The server, from port 21

3 CHOOSE ONE

Which transport protocol and port does TFTP use?

- A TCP 21
- B TCP 69
- C UDP 21
- D UDP 69

4 TRUE OR FALSE?

Telnet encrypts the username and password before sending them.

- ☐ True
- ☐ False

5 CHOOSE ONE

Before STelnet (SSH) can work on a Huawei device, what must be created first?

- A An FTP directory
- B A VLANIF interface
- C A local RSA key pair
- D A DHCP pool

CHAPTER 15

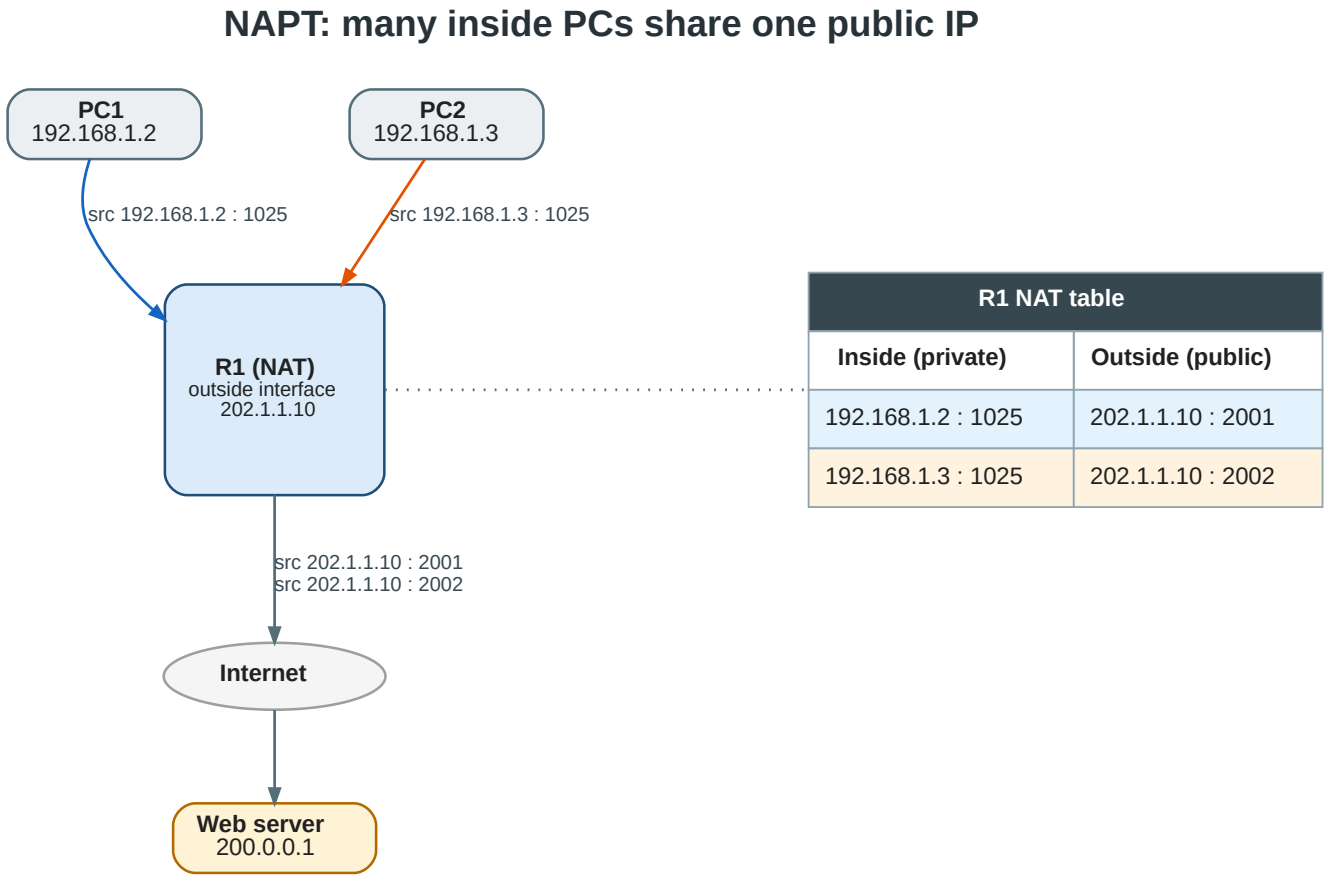
NAT Fundamentals and Configuration

EXAM FOCUS
Static NAT, No-PAT, NAPT, Easy IP, NAT Server, nat outbound/server commands, verification

IN THIS CHAPTER
15.1 Why NAT · 15.2 NAT types · 15.3 Source NAT setup (eNSP) · 15.4 NAT Server · 15.5 Check NAT

Pictures Look at these first

PICTURE 15.1



HOW TO READ IT
R1 swaps each private source address and port for its public address and a new port, and remembers the swap in its NAT table. Replies come back to 202.1.1.10, and R1 uses the table to send each reply to the right PC.

PICTURE 15.2

NAT types compared

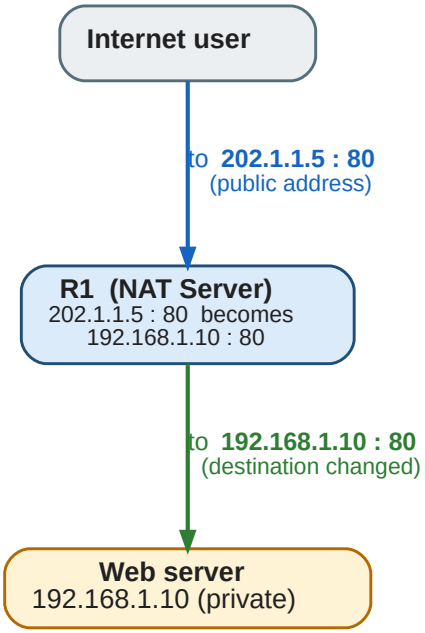
Type	What changes	Public addresses used	eNSP command (outside interface)
Static NAT	IP only, fixed 1 : 1	one per inside host	`nat static global 202.1.1.3 inside 192.168.1`
Dynamic NAT (No-PAT)	IP only, from a pool	pool, 1 : 1 while in use	`nat outbound 2000 address-group 1 no-pat`
NAPT	IP **and port**	pool shared by many users	`nat outbound 2000 address-group 1`
Easy IP	IP **and port**	the outside interface's own IP	`nat outbound 2000`
NAT Server	**destination** IP and port	one public IP:port per server	`nat server protocol tcp global 202.1.1.5 www inside 192.168.1.10 www`

HOW TO READ IT

The first four change the SOURCE address of traffic going out. NAT Server changes the DESTINATION address so people on the Internet can reach a server inside your network.

PICTURE 15.3

NAT Server: letting the Internet reach an inside server



HOW TO READ IT

An Internet user connects to the public address 202.1.1.5 port 80. R1 changes the destination to the private server 192.168.1.10 port 80. The user never sees the private address.

Notes Read, then cover the boxes and test yourself

15.1 Why NAT

NAT swaps private addresses for a public one at the exit to the Internet – like a receptionist putting everyone's calls through one main phone number.

REMEMBER

- Saves public IPv4 addresses
- Hides inside addresses
- Done on the router or firewall facing the Internet

EXAM TRAP Private ranges: 10/8, 172.16/12, 192.168/16.

15.2 NAT types

Four ways to translate addresses.

REMEMBER

- Static NAT – 1 private ↔ 1 public, fixed
- Dynamic NAT (No-PAT) – a pool of public IPs; changes the IP only
- NAPT – many users share public IPs using different **PORTS**
- Easy IP – NAPT using the router's own outside interface IP

EXAM TRAP No-PAT = IP only. NAPT = IP + port. Easy IP = interface IP + port.

15.3 Source NAT setup (eNSP)

The ACL picks who may go out; the address-group is the public pool; nat outbound goes on the outside port.

REMEMBER

- Easy IP is best when the public IP comes from DHCP or PPPoE

COMMAND / EXAMPLE

```
[R1] acl 2000
[R1-acl-basic-2000] rule permit source 192.168.1.0 0.0.0.255
[R1] nat address-group 1 202.1.1.10 202.1.1.20
[R1-GE0/0/1] nat outbound 2000 address-group 1 no-pat   (No-PAT)
[R1-GE0/0/1] nat outbound 2000 address-group 1   (NAPT)
[R1-GE0/0/1] nat outbound 2000   (Easy IP)
```

EXAM TRAP The slides show V600 syntax; these are the eNSP V200 commands.

15.4 NAT Server

Lets people on the Internet reach a server inside your network.

REMEMBER

- Public IP:port → private IP:port
- Outside users start the connection

COMMAND / EXAMPLE

```
[R1-GE0/0/1] nat server protocol tcp global 202.1.1.5 www inside 192.168.1.10 www
```

EXAM TRAP NAT Server changes the **DESTINATION** address.

15.5 Check NAT

Look for translation entries (sessions).

REMEMBER

- No sessions? Check the ACL and that NAT is on the **OUTSIDE** interface

COMMAND / EXAMPLE

```
display nat session all
display nat outbound
display nat server
```

Check yourself

Answers: page 116

1 CHOOSE ONE

Many inside users share the router's OWN outside interface IP, using different ports. Which NAT type is this?

A Easy IP
B Static NAT
C No-PAT
D NAT Server

2 CHOOSE ONE

Which NAT type translates the IP address ONLY, without changing ports?

A No-PAT (dynamic NAT)
B NAPT
C Easy IP
D NAT Server

3 CHOOSE ONE

For traffic coming in from the Internet, which address does NAT Server change?

A Source IP
B Destination IP
C Source MAC
D Nothing; it only logs the traffic

4 CHOOSE ONE

On which interface do you configure nat outbound?

A The inside (LAN) interface
B The loopback interface
C Every interface on the router
D The outside (Internet-facing) interface

5 TRUE OR FALSE?

Easy IP is a good choice when the public IP address is learned by DHCP or PPPoE.

☐ True ☐ False

CHAPTER 16

Data Center Network Basics

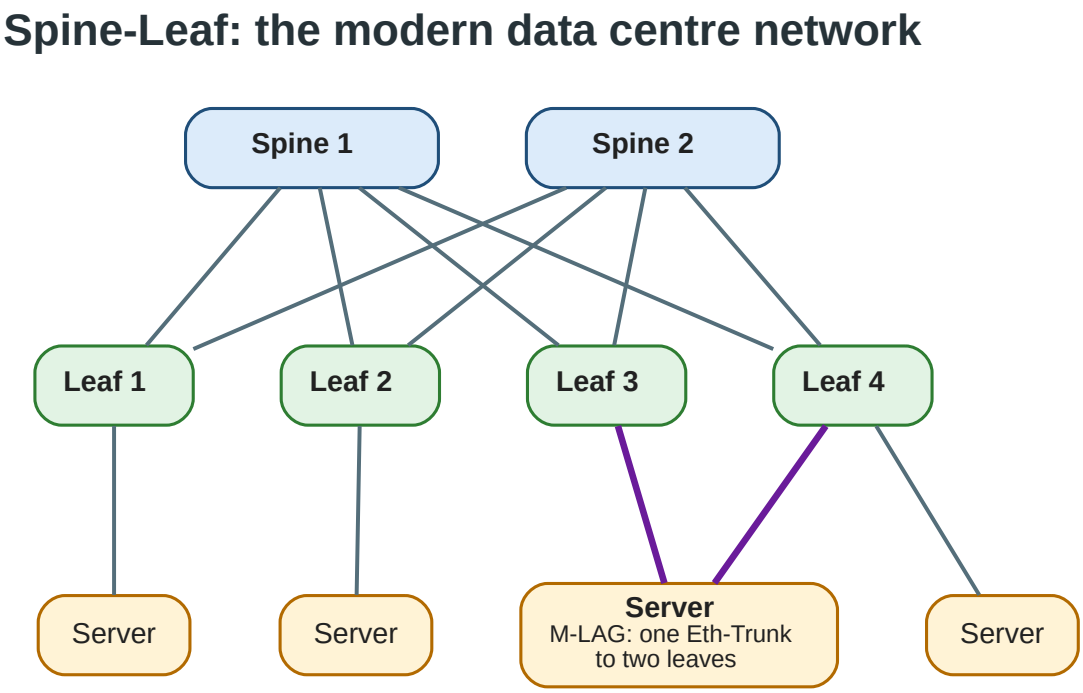
EXAM FOCUS
DC composition & evolution, general-purpose vs intelligent computing, Spine-Leaf, M-LAG, VXLAN/EVPN, PFC, ECN, RDMA/RoCE

IN THIS CHAPTER

16.1 What a data centre is · 16.2 How data centres grew · 16.3 Two kinds of computing · 16.4 General-purpose DCN versions · 16.5 Key DC technologies · 16.6 AI (intelligent computing) network · 16.7 PFC, ECN and RDMA · 16.8 Dual-track data centre

Pictures Look at these first

PICTURE 16.1



HOW TO READ IT
Every leaf connects to every spine, so any server reaches any other server in the same number of hops (leaf, spine, leaf). Need more ports? Add a leaf. Need more bandwidth? Add a spine. M-LAG lets one server connect to two leaves as if it were one link.

PICTURE 16.2

Data centre needs and the technology that answers each one

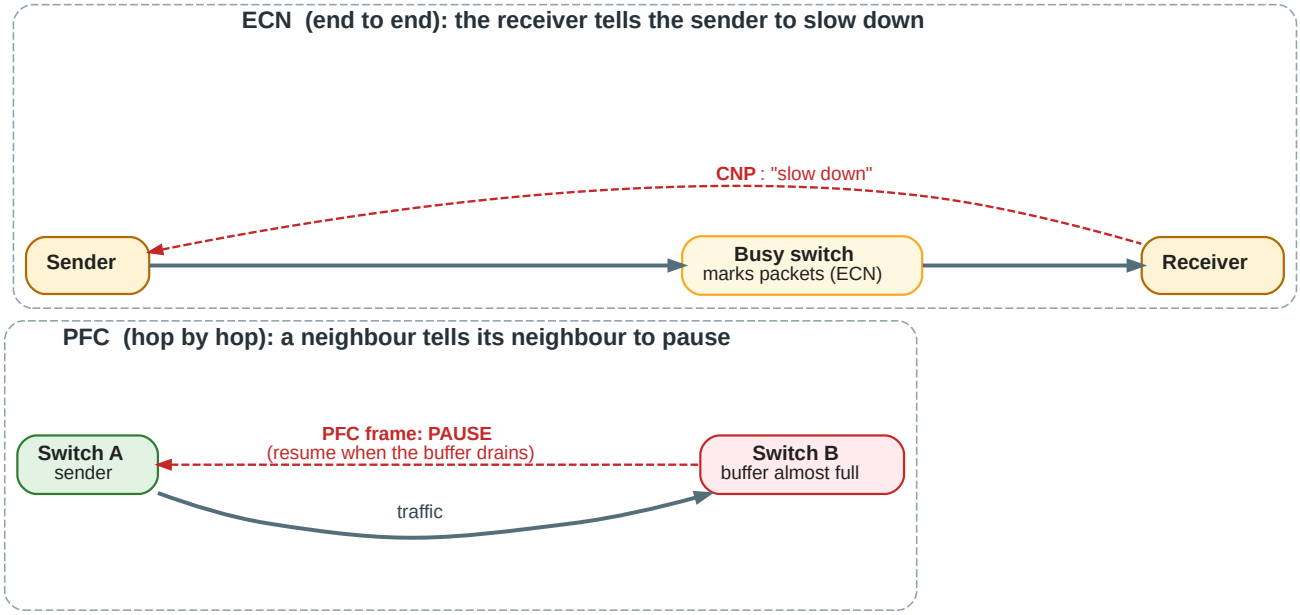
The need	Technology	The idea in one line
Grow easily	**Spine-Leaf**	add leaves for ports, spines for bandwidth
More bandwidth + no single point of failure	**M-LAG**	one link bundle across two switches
Keep customers apart, move VMs anywhere	**VXLAN**	24-bit VNI = about 16 million networks (VLAN only 4096)
Set up the network automatically	**EVPN + SDN controller**	tunnels and settings are created for you
AI: zero packet loss	**PFC + ECN**	pause or slow the sender before buffers overflow
AI: very low delay	**RDMA (RoCEv2)**	the NIC writes straight into the other server's memory

HOW TO READ IT

Each row is a problem a data centre has and the technology that solves it. The first four are for general-purpose computing (CPU servers). The last two are for AI clusters (GPU servers).

PICTURE 16.3

PFC and ECN: two ways to stop packet loss



HOW TO READ IT

PFC works between two neighbours: a full receiver tells the sender to pause. ECN works end to end: a busy switch marks packets, the receiver sends a CNP back, and the sender slows down.

Notes Read, then cover the boxes and test yourself

16.1 What a data centre is

A building full of servers that store and process a company's data.

REMEMBER

Three main parts:

- Computing – servers
- Storage – disks
- DCN – the network joining them

EXAM TRAP All data between computing and storage goes through the DCN.

16.2 How data centres grew

Data centres got more reliable and moved to the cloud over time.

REMEMBER

- Reliability: 1 DC → main + backup → two-site three-DC → many DCs
- Deployment: physical servers → VMs → private cloud → public cloud

EXAM TRAP Two-site three-DC = main DC + same-city DC (both active) + far-away backup.

16.3 Two kinds of computing

General-purpose = lots of different small jobs (CPU). Intelligent computing = one huge job split across many GPUs (AI).

REMEMBER

- General-purpose: CPU; lots of east-west traffic; needs low delay
- Intelligent: GPU; all-to-all traffic; needs huge bandwidth and **ZERO** packet loss

EXAM TRAP CPU = general-purpose. GPU = AI / intelligent computing.

16.4 General-purpose DCN versions

The design improved in three steps.

REMEMBER

- 1.0: STP + VRRP – STP blocks links (wasteful)
- 2.0: M-LAG – all links used, but limited size
- 3.0: Spine-Leaf + VXLAN – scales big, best paths

EXAM TRAP Know which version uses which technology.

16.5 Key DC technologies

Each need has a matching technology.

REMEMBER

- Grow easily → Spine-Leaf (spine = fast core, leaf = access)
- High bandwidth + backup → M-LAG (one link bundle across two switches)
- Separate customers + move VMs → VXLAN (24-bit ID ≈ 16 million networks)
- Automatic setup → VXLAN + EVPN + SDN controller

EXAM TRAP Slide quiz: separating tenants = VXLAN.

16.6 AI (intelligent computing) network

AI training sends huge bursts of data. Any lost packet slows the whole job down.

REMEMBER

- 400G/800G ports
- Lossless: PFC and ECN
- Low delay: RDMA
- Designs: Spine-Leaf or Dragonfly

EXAM TRAP Just adding bandwidth does **NOT** make a network lossless (true/false = False).

16.7 PFC, ECN and RDMA

Tools that stop packet loss and cut delay.

REMEMBER

- PFC – receiver tells the next hop 'pause!' when its buffer is full
- ECN – a busy switch marks packets; the receiver tells the sender to slow down
- RDMA – the network card writes straight into another server's memory, skipping the CPU (RoCEv2 = over Ethernet)

EXAM TRAP PFC = hop-by-hop pause. ECN = end-to-end slow-down.

16.8 Dual-track data centre

Future DCs run two networks side by side.

REMEMBER

- Shared building, power and cooling
- **SEPARATE** networks for general-purpose and AI

Check yourself

Answers: page 116

1 CHOOSE ONE

Which technology separates tenants (customers) in a data centre network?

A VXLAN
B STP
C VRRP
D PFC

2 CHOOSE ONE

How many bits long is the VXLAN network identifier (VNI)?

- A 12
- B 16
- C 20
- D 24

3 CHOOSE ONE

How does PFC prevent packet loss?

- A The switch marks packets and the sender slows down
- B The network card writes straight into another server's memory
- C The receiver tells the next hop to pause when its buffer is full
- D It adds more bandwidth to each link

4 TRUE OR FALSE?

Simply adding more bandwidth makes a data centre network lossless.

- ☐ True
- ☐ False

5 CHOOSE ONE

Which general-purpose DCN version uses Spine-Leaf with VXLAN?

- A DCN 1.0
- B DCN 2.0
- C DCN 3.0
- D All versions

CHAPTER 17

WLAN Technology Basics

EXAM FOCUS

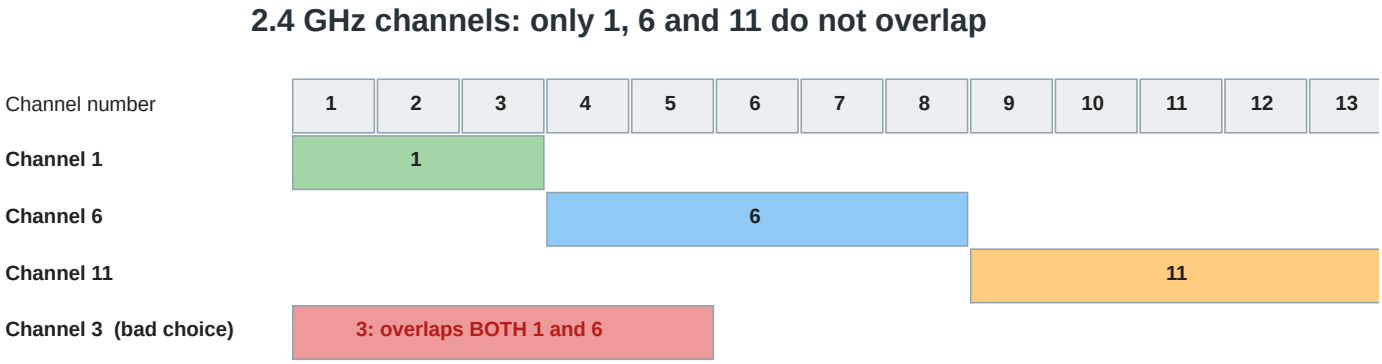
2.4/5/6 GHz bands & channels, modulation/speed levers, CSMA/CA, 802.11 standards ↔ Wi-Fi generations

IN THIS CHAPTER

17.1 What WLAN is · 17.2 Frequency bands · 17.3 What makes Wi-Fi faster · 17.4 CSMA/CA · 17.5 802.11 standards

Pictures Look at these first

PICTURE 17.1

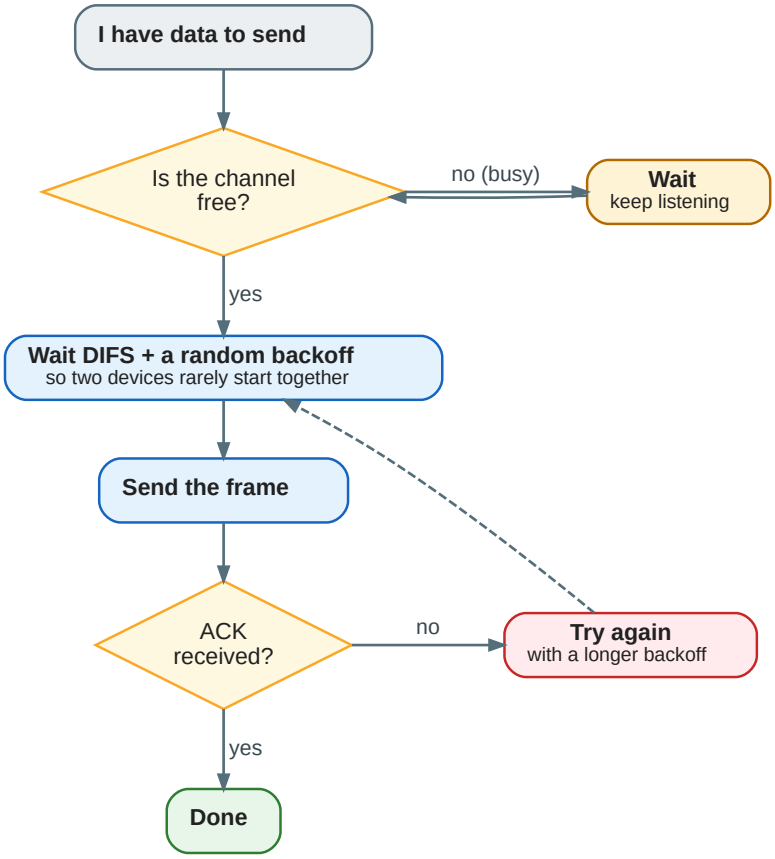


HOW TO READ IT

Each Wi-Fi channel is about 20 MHz wide, but channel numbers are only 5 MHz apart, so neighbouring channels overlap. Use 1, 6 and 11 for neighbouring APs. 5 GHz has many more channels that do not overlap.

PICTURE 17.2

CSMA/CA: Wi-Fi devices take turns



HOW TO READ IT

A Wi-Fi device cannot hear a collision while it is sending, so it tries to avoid one: listen first, wait a random time, send, then wait for an ACK. No ACK means the frame was lost, so it tries again.

PICTURE 17.3

802.11 standards and Wi-Fi generations

Name	Standard	Bands	Top speed (theory)	New idea
-	802.11b	2.4 GHz	11 Mbit/s	early Wi-Fi
-	802.11a	5 GHz	54 Mbit/s	OFDM
-	802.11g	2.4 GHz	54 Mbit/s	OFDM on 2.4 GHz
Wi-Fi 4	802.11n	2.4 + 5 GHz	600 Mbit/s	MIMO, 40 MHz channels
Wi-Fi 5	802.11ac	**5 GHz only**	6.9 Gbit/s	256-QAM, MU-MIMO, 160 MHz
Wi-Fi 6 / 6E	802.11ax	2.4 + 5 GHz (6E adds 6 GHz)	9.6 Gbit/s	OFDMA, 1024-QAM
Wi-Fi 7	802.11be	2.4 + 5 + 6 GHz	about 46 Gbit/s	320 MHz, 4096-QAM, multi-link

HOW TO READ IT

Each generation is faster than the one before. Remember which bands each one uses: Wi-Fi 5 is 5 GHz only, and Wi-Fi 6E and Wi-Fi 7 add the 6 GHz band.

Notes Read, then cover the boxes and test yourself

17.1 What WLAN is

Wi-Fi = a LAN without cables, using radio waves.

REMEMBER

- Standard: IEEE 802.11
- The Wi-Fi Alliance tests devices and names the generations (Wi-Fi 4, 5, 6, 7)

EXAM TRAP IEEE 802.11 = the standard. Wi-Fi = the brand name.

17.2 Frequency bands

Bands are like different roads: 2.4 GHz reaches further but is crowded; 5 GHz is faster but has shorter range.

REMEMBER

- 2.4 GHz – long range, crowded; only channels 1, 6, 11 don't overlap
- 5 GHz – more channels, faster, shorter range
- 6 GHz – newest and least crowded (Wi-Fi 6E/7)

EXAM TRAP Non-overlapping 2.4 GHz channels = 1, 6, 11.

17.3 What makes Wi-Fi faster

Four levers control the speed.

REMEMBER

- Wider channel (20 → 40 → 80 → 160 → 320 MHz)
- Higher modulation – more bits per signal (256 / 1024 / 4096-QAM)
- More antennas / spatial streams (MIMO)
- Better coding / shorter guard interval

EXAM TRAP 256-QAM = Wi-Fi 5, 1024-QAM = Wi-Fi 6, 4096-QAM = Wi-Fi 7.

17.4 CSMA/CA

Wi-Fi devices can't hear collisions, so they wait their turn – like people in a meeting waiting for silence before speaking.

REMEMBER

- Listen → wait → random backoff → send → wait for ACK
- RTS/CTS helps with hidden devices

EXAM TRAP Wi-Fi = CSMA/CA (avoid). Old Ethernet = CSMA/CD (detect).

17.5 802.11 standards

Each generation is faster than the last.

REMEMBER

- 802.11b – 2.4 GHz, 11 Mbps
- 802.11a – 5 GHz, 54 Mbps
- 802.11g – 2.4 GHz, 54 Mbps
- Wi-Fi 4 = 802.11n (2.4 / 5 GHz)
- Wi-Fi 5 = 802.11ac (5 GHz only)
- Wi-Fi 6 = 802.11ax (2.4 / 5 GHz; 6E adds 6 GHz)
- Wi-Fi 7 = 802.11be (2.4 / 5 / 6 GHz)

EXAM TRAP Wi-Fi 5 works on 5 GHz **ONLY**.

Check yourself

Answers: page 116

1 CHOOSE ONE

Which 2.4 GHz channels do NOT overlap with each other?

A 1, 5, 9
B 1, 6, 11
C 2, 7, 12
D 1, 7, 13

2 CHOOSE ONE

Which band(s) does Wi-Fi 5 (802.11ac) use?

A 2.4 GHz only
B 2.4 GHz and 5 GHz
C 5 GHz only
D 5 GHz and 6 GHz

3 CHOOSE ONE

Which channel access method does Wi-Fi use?

A CSMA/CA
B CSMA/CD
C Token passing
D TDMA

4 CHOOSE ONE

Wi-Fi 6 is the brand name for which IEEE standard?

A 802.11n
B 802.11ac
C 802.11ax
D 802.11be

5 CHOOSE ONE

What is the highest modulation used by Wi-Fi 7?

A 64-QAM
B 256-QAM
C 1024-QAM
D 4096-QAM

CHAPTER 18

WLAN Networking Architecture

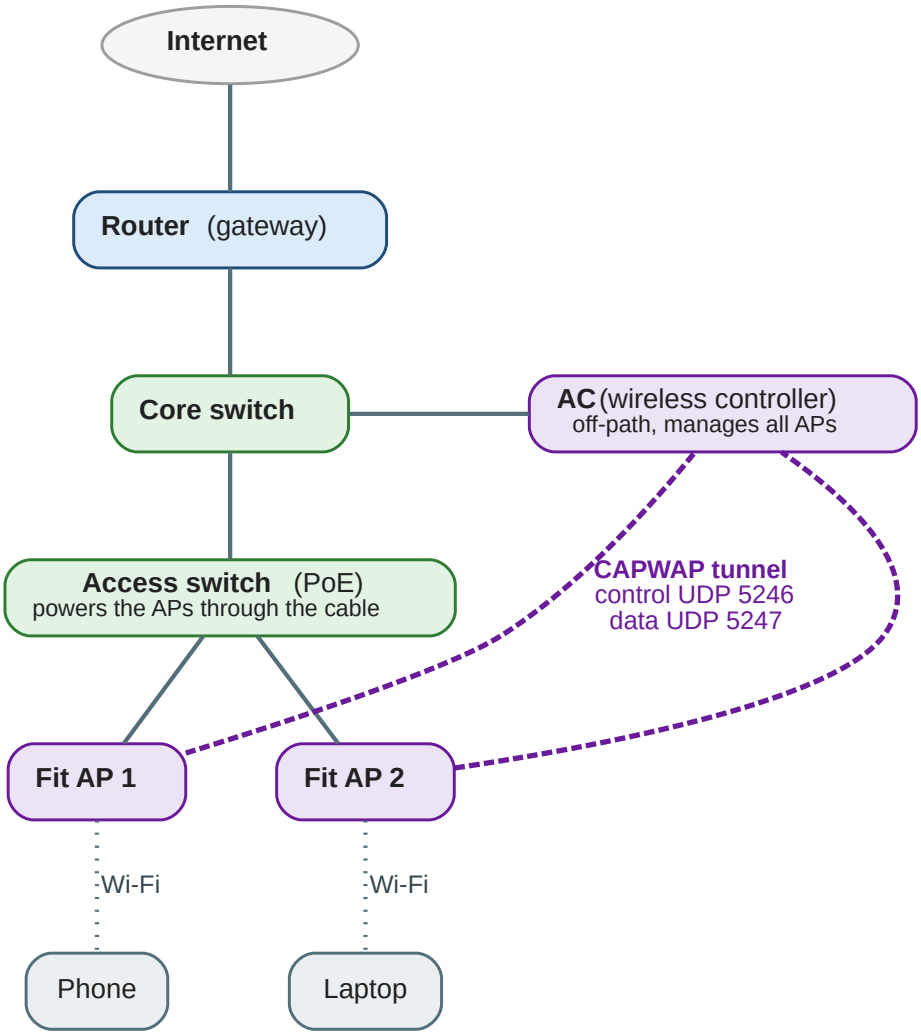
EXAM FOCUS
Fat/Fit/cloud/leader AP, CAPWAP 5246/5247, direct vs tunnel forwarding, in-path vs off-path, L2 vs L3 networking, PoE

IN THIS CHAPTER
18.1 Types of AP setup · 18.2 CAPWAP · 18.3 Forwarding modes · 18.4 Where the AC sits · 18.5 VLANs and PoE

Pictures Look at these first

PICTURE 18.1

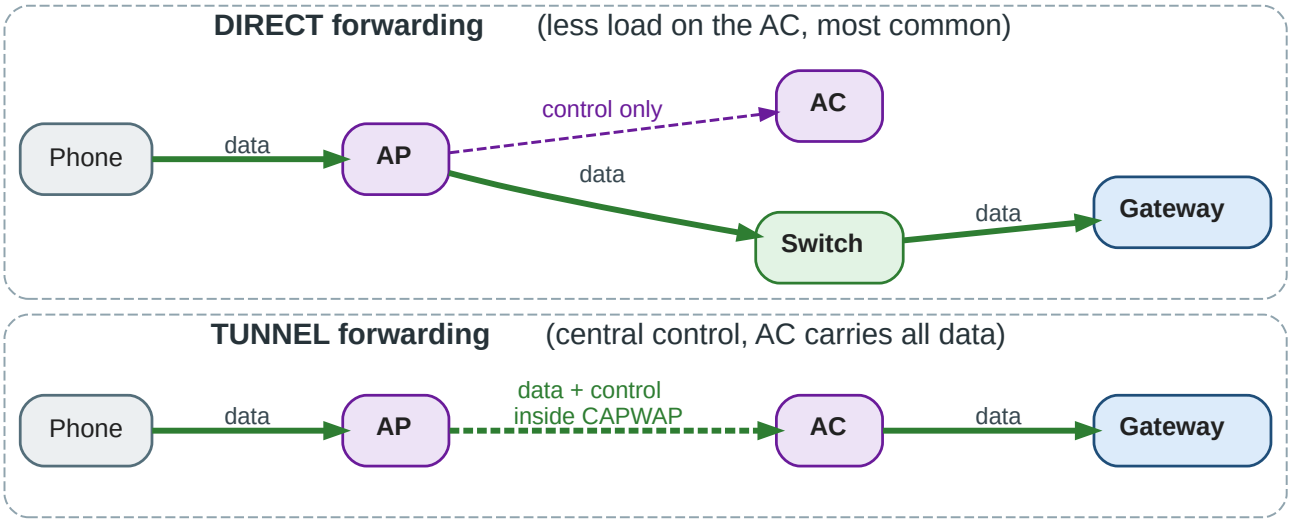
AC + Fit AP: one controller manages many APs



HOW TO READ IT
The AC (wireless controller) hangs off the core switch. Each Fit AP builds a CAPWAP tunnel to the AC: control messages use UDP 5246, and user data uses UDP 5247 when tunnel forwarding is used. PoE switches power the APs.

PICTURE 18.2

Direct forwarding vs tunnel forwarding



HOW TO READ IT

Control messages ALWAYS travel through the CAPWAP tunnel to the AC. The difference is the user data: direct forwarding sends it straight out from the AP, tunnel forwarding sends it through the AC first.

PICTURE 18.3

Four ways to run access points

Setup	Who manages the APs	Typical site
Fat AP	each AP manages itself	home, very small office
AC + Fit AP	the AC (wireless controller)	medium and large campus
Cloud AP	cloud platform (iMaster NCE-Campus)	many small branches
Leader AP	one AP leads the others	small or medium site, no AC

HOW TO READ IT

Pick the setup by size. A Fat AP works alone; Fit APs need an AC; cloud APs are managed online; a leader AP manages the other APs in a small site without an AC.

Notes Read, then cover the boxes and test yourself

18.1 Types of AP setup

Different ways to run access points.

REMEMBER

- Fat AP – works alone (home, tiny office)
- AC + Fit AP – a controller manages many APs (medium/large sites)
- Cloud AP – managed from the cloud (iMaster NCE-Campus); good for many branches
- Leader AP – one AP manages the others (small sites, no AC)

EXAM TRAP Large enterprise = AC + Fit AP.

18.2 CAPWAP

The tunnel between the AP and the AC (controller).

REMEMBER

- Control tunnel – UDP 5246 (management)
- Data tunnel – UDP 5247 (user traffic in tunnel mode)

EXAM TRAP 5246 = control, 5247 = data.

18.3 Forwarding modes

Where does user traffic go?

REMEMBER

- Direct forwarding – straight from the AP onto the network (the AC only handles control)
- Tunnel forwarding – all user traffic goes through the AC first

COMMAND / EXAMPLE

```
[AC-wlan-vap-prof-x] forward-mode direct-forward
[AC-wlan-vap-prof-x] forward-mode tunnel
```

EXAM TRAP Control traffic **ALWAYS** goes through CAPWAP. Only the **DATA** path changes.

18.4 Where the AC sits

The AC's position and subnet decide how APs find it.

REMEMBER

- In-path – AC sits on the traffic path
- Off-path – AC hangs off the core switch (easy to add)
- Layer 2 – AP and AC in the same subnet (AP finds the AC by broadcast)
- Layer 3 – different subnets (AP finds the AC via DHCP Option 43)

EXAM TRAP Common real setup: off-path + direct forwarding.

18.5 VLANs and PoE

Keep AP management and user traffic apart, and power APs through the cable.

REMEMBER

- Management VLAN – AP ↔ AC (CAPWAP)
- Service VLAN – user traffic. Keep them different!
- PoE: 802.3af 15.4 W, 802.3at 30 W, 802.3bt 60–90 W

EXAM TRAP PoE+ = 802.3at = 30 W.

Check yourself

Answers: page 116

1 CHOOSE ONE

Which UDP port does the CAPWAP CONTROL tunnel use?

A 5247
B 5246
C 1812
D 161

2 TRUE OR FALSE?

With direct forwarding, control traffic between the AP and AC no longer uses the CAPWAP tunnel.

☐ True ☐ False

3 CHOOSE ONE

The APs and the AC are in DIFFERENT subnets. How do the APs usually learn the AC's IP address?

A By broadcast
B From DHCP Option 43
C From DNS only
D From the AP's MAC address

4 CHOOSE ONE

How much power does PoE+ (802.3at) provide per port?

A 15.4 W
B 30 W
C 60 W
D 90 W

5 CHOOSE ONE

Which AP setup suits a large enterprise campus best?

A Fat AP
B Leader AP
C A home Wi-Fi router
D AC + Fit AP

CHAPTER 19

WLAN Fundamentals and Configuration

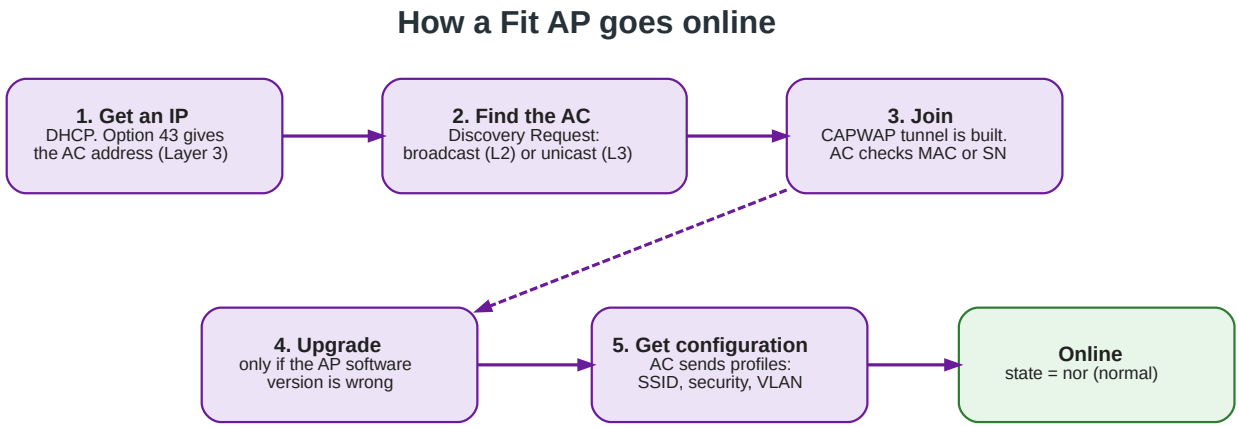
EXAM FOCUS
AP onboarding, AP auth modes, profiles (regulatory, SSID, security, VAP, AP group), STA access steps, WLAN security

IN THIS CHAPTER

19.1 How an AP goes online · 19.2 Profiles · 19.3 Profile configuration example · 19.4 How a phone joins Wi-Fi · 19.5 Wi-Fi security

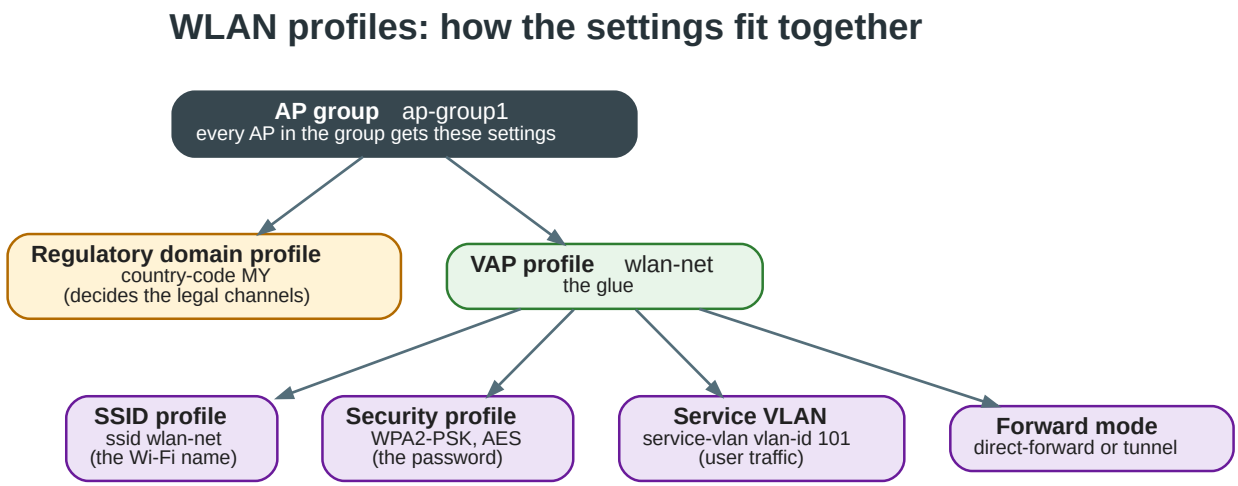
Pictures Look at these first

PICTURE 19.1



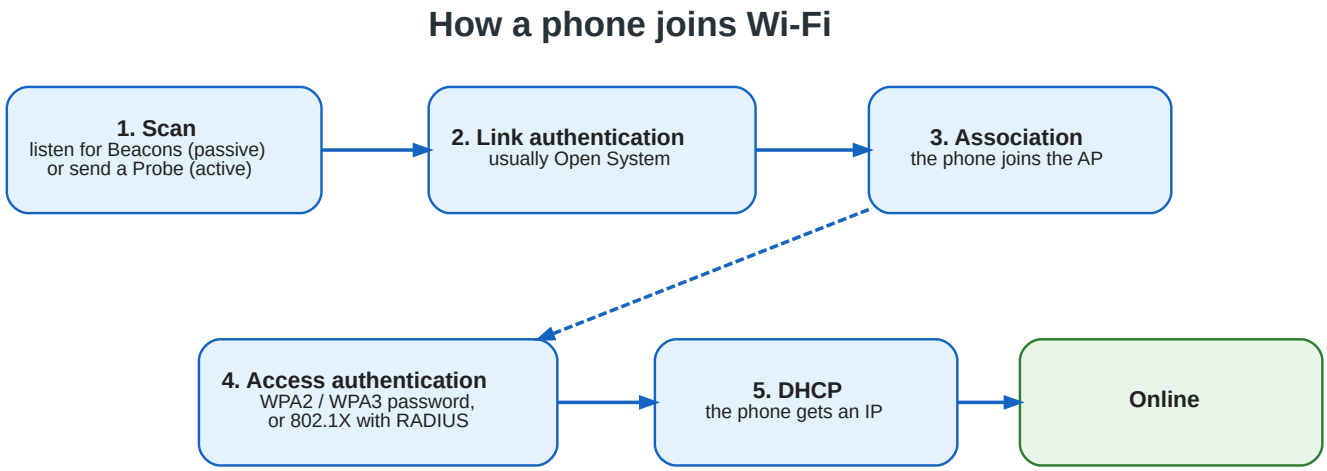
HOW TO READ IT
Follow the arrows. The AP gets an IP address, finds the AC, joins it through a CAPWAP tunnel, upgrades if needed and downloads its settings. When it is done, display ap all shows the AP as 'nor' (normal).

PICTURE 19.2



HOW TO READ IT
Settings are built from small profiles and attached to an AP group, so every AP in the group gets the same Wi-Fi. The VAP profile is the 'glue': it joins the SSID, the security, the service VLAN and the forwarding mode.

PICTURE 19.3



HOW TO READ IT

Five steps from seeing the Wi-Fi name to being online. Scanning finds the network, two kinds of authentication check the device and the user, and DHCP gives it an IP address.

Notes

Read, then cover the boxes and test yourself

19.1 How an AP goes online

The AP wakes up, gets an IP, finds the AC, builds the tunnel and downloads its settings.

REMEMBER

1. Get an IP (DHCP; Option 43 tells it the AC's IP)
2. Find the AC and join (CAPWAP)
3. Upgrade if needed
4. Receive its configuration

- The AC can check APs by MAC, by SN, or not at all

COMMAND / EXAMPLE

```
[AC] capwap source interface vlanif 100
[AC-wlan-view] ap auth-mode mac-auth
[AC-wlan-view] ap-id 0 ap-mac 00e0-fc12-3456
[AC-wlan-ap-0] ap-group ap-group1
(DHCP pool) option 43 sub-option 3 ascii 10.1.100.1
```

EXAM TRAP State 'nor' in display ap all = the AP is online.

19.2 Profiles

Wi-Fi settings are built from profiles, like Lego blocks, then attached to an AP group.

REMEMBER

- Regulatory domain – country code (sets the legal channels)
- SSID profile – Wi-Fi name
- Security profile – password / encryption
- VAP profile – joins SSID + security + service VLAN + forward mode
- AP group – APs that share the same profiles

EXAM TRAP The VAP profile is the 'glue' that binds everything together.

19.3 Profile configuration example

Build the profiles, then apply them to the AP group.

REMEMBER

Order: regulatory domain → security → SSID → VAP → AP group

COMMAND / EXAMPLE

```
[AC-wlan-view] regulatory-domain-profile name default
[...] country-code MY
[AC-wlan-view] security-profile name wlan-net
[...] security wpa-wpa2 psk pass-phrase a1234567 aes
[AC-wlan-view] ssid-profile name wlan-net
[...] ssid wlan-net
[AC-wlan-view] vap-profile name wlan-net
[...] forward-mode tunnel
[...] service-vlan vlan-id 101
[...] security-profile wlan-net
[...] ssid-profile wlan-net
[AC-wlan-view] ap-group name ap-group1
[...] regulatory-domain-profile default
[...] vap-profile wlan-net wlan 1 radio all
```

EXAM TRAP Wrong country code = wrong channels.

19.4 How a phone joins Wi-Fi

Five steps from seeing the network to getting online.

REMEMBER

1. Scan – listen for beacons or send a probe
2. Link authentication (open system)
3. Associate
4. Access authentication – WPA2/WPA3 password or 802.1X
5. Get an IP by DHCP

COMMAND / EXAMPLE

```
display station ssid wlan-net
```

EXAM TRAP Scan → link auth → associate → access auth → DHCP.

19.5 Wi-Fi security

From weakest to strongest.

REMEMBER

- WEP – old and weak; don't use
- WPA2 – AES; Personal (password) or Enterprise (802.1X + RADIUS)
- WPA3 – newest; uses SAE

COMMAND / EXAMPLE

```
display ap all
display vap ssid wlan-net
```

EXAM TRAP Enterprise = 802.1X + RADIUS. WPA3-Personal = SAE.

Check yourself

Answers: page 116

1 CHOOSE ONE

Which WLAN profile binds the SSID, security, service VLAN and forwarding mode together?

- A Regulatory domain profile
- B VAP profile
- C SSID profile
- D AP group

2 CHOOSE ONE

What is the correct order when a phone joins Wi-Fi?

- A Associate, scan, access authentication, link authentication, DHCP
- B Scan, link authentication, associate, access authentication, DHCP
- C Scan, DHCP, associate, link authentication, access authentication
- D Link authentication, scan, associate, DHCP, access authentication

3 CHOOSE ONE

In display ap all, what does the state 'nor' mean?

- A The AP is not registered
- B The AP is online and working normally
- C The AP is downloading software
- D The AP failed authentication

4 CHOOSE ONE

What does WPA3-Personal use instead of the old WPA2 pre-shared key exchange?

- A SAE
- B WEP
- C 802.1X with RADIUS
- D TKIP

5 CHOOSE ONE

What does the regulatory domain profile set?

- A The Wi-Fi name
- B The Wi-Fi password
- C The country code, which decides the legal channels
- D The service VLAN

CHAPTER 20

Network Device Management

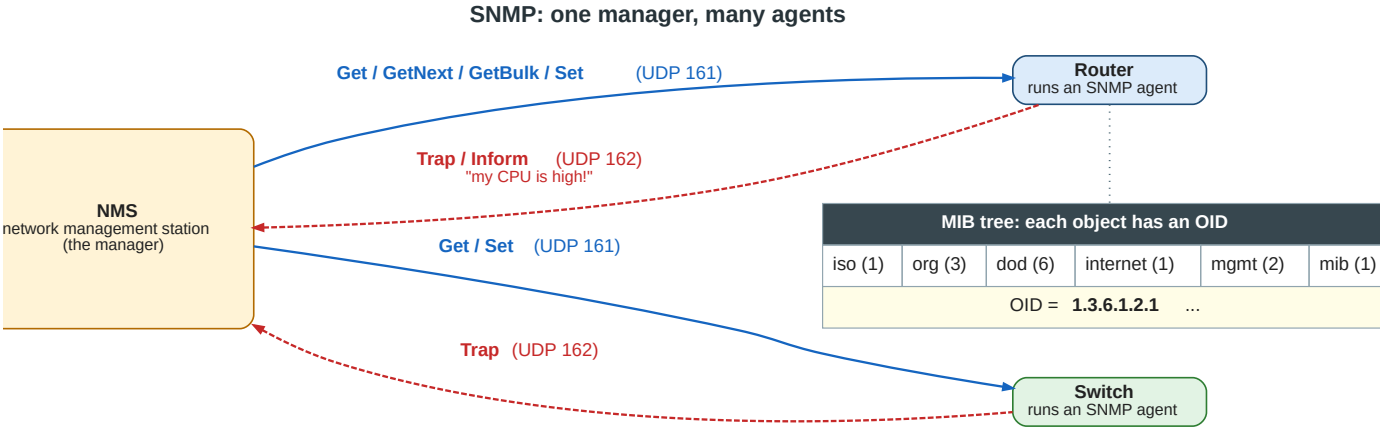
EXAM FOCUS
FCAPS, SNMP architecture/MIB/versions/operations, SNMPv3 config, NETCONF layers/datastores/operations, RESTCONF, iMaster NCE

IN THIS CHAPTER

20.1 What network management is · 20.2 SNMP · 20.3 SNMP versions · 20.4 SNMPv3 setup · 20.5 NETCONF · 20.6 NETCONF datastores · 20.7 RESTCONF · 20.8 iMaster NCE

Pictures Look at these first

PICTURE 20.1



HOW TO READ IT
The NMS asks devices for information (Get) or changes settings (Set) on UDP 161. Devices report problems by themselves with a Trap on UDP 162. Every value a device can report has an OID in the MIB tree.

PICTURE 20.2

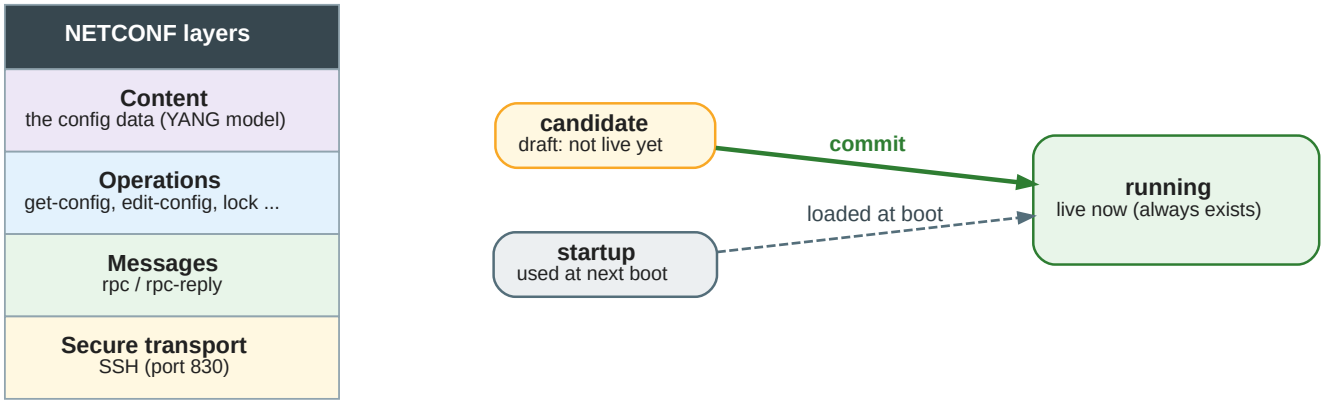
SNMP versions

Version	Operations	Security	Use it for
SNMPv1	Get, GetNext, Set, Trap	community string (plain text)	small, simple, trusted networks
SNMPv2c	v1 + **GetBulk** + **Inform** (Inform = a trap that is acknowledged)	community string (plain text)	bigger networks, more traffic
SNMPv3	same as v2c	**USM** : login + encryption **VACM** : who may see what	**recommended for all networks**

HOW TO READ IT
Each version adds to the one before. v2c adds GetBulk and Inform. v3 adds real security: user login and encryption (USM) and view-based access control (VACM). Use v3 whenever you can.

PICTURE 20.3

NETCONF: four layers and three datastores



HOW TO READ IT

Left: NETCONF sends XML messages over SSH (port 830). Right: you edit the candidate copy, then commit it to running, which is the live configuration. RESTCONF works over HTTP and only uses the running datastore.

Notes

Read, then cover the boxes and test yourself

20.1 What network management is

Watching and controlling all your network devices from one place.

REMEMBER

- The NMS (manager) talks to an agent on each device
- 5 functions (FCAPS): Fault, Configuration, Accounting, Performance, Security
- Tools: CLI/Web (small networks), SNMP, NETCONF, RESTCONF

EXAM TRAP All 5 FCAPS items are basic functions (slide quiz).

20.2 SNMP

SNMP lets one management server ask every device 'how are you?' and receive alarms.

REMEMBER

- UDP 161 (queries), UDP 162 (traps)
- MIB = tree of everything you can check; each item has an OID
- Trap = the device reports a problem by itself

EXAM TRAP MIB = object tree. Trap = alert started by the device.

20.3 SNMP versions

Each version adds features.

REMEMBER

- v1 – Get, GetNext, Set, Trap; community string (no real security)
- v2c – adds GetBulk (many items at once) and Inform (a trap that gets confirmed)
- v3 – adds login + encryption (USM) and access control (VACM); recommended

EXAM TRAP GetBulk/Inform = v2c. Security = v3.

20.4 SNMPv3 setup

Steps: enable agent → version → MIB view → group → user + passwords → trap target.

REMEMBER

- authentication-mode = login password
- privacy-mode = encryption password

COMMAND / EXAMPLE

```
[S1] snmp-agent
[S1] snmp-agent sys-info version v3
[S1] snmp-agent mib-view included iso iso
[S1] snmp-agent group v3 admin privacy read-view iso write-view iso notify-view iso
[S1] snmp-agent usm-user v3 nms-admin group admin
[S1] snmp-agent usm-user v3 nms-admin authentication-mode sha2-256
[S1] snmp-agent usm-user v3 nms-admin privacy-mode aes128
[S1] snmp-agent target-host trap address udp-domain 192.168.1.1 params securityname nms-admin v3
privacy
[S1] snmp-agent trap enable
```

EXAM TRAP trap enable does nothing without a target-host.

20.5 NETCONF

A smarter configuration protocol than SNMP, sending XML over SSH.

REMEMBER

- Can lock the config so two admins don't clash
- 4 layers: Secure Transport (SSH) → Messages (rpc) → Operations (get-config, edit-config...) → Content (YANG)
- Port 830

COMMAND / EXAMPLE

```
[HUAWEI] snetconf server enable
```

EXAM TRAP The data model in the content layer = YANG.

20.6 NETCONF datastores

Three copies of the configuration.

REMEMBER

- running – live now (always exists)
- candidate – a draft; only takes effect after commit
- startup – used at next boot

EXAM TRAP Edit candidate → commit to running = two-phase commit.

20.7 RESTCONF

NETCONF ideas, but over HTTP – like a web API.

REMEMBER

- XML or JSON
- Only the running datastore; changes apply immediately
- GET = read, POST = create, PUT = replace, PATCH = merge, DELETE = delete

COMMAND / EXAMPLE

```
[HUAWEI] http
[HUAWEI-http] service restconf
[HUAWEI-http-service-restconf] server enable
```

EXAM TRAP RESTCONF cannot use the candidate datastore (true/false = False).

20.8 iMaster NCE

Huawei's platform to manage, control and analyse the whole network.

REMEMBER

- NCE-Campus manages campus devices with SNMP, NETCONF, HTTP, SSH
- New devices find NCE automatically via DHCP Option 148

COMMAND / EXAMPLE

```
option 148 ascii agilemanage-mode=ip;agilemanage-domain=192.168.1.1;agilemanage-port=10020
```

EXAM TRAP Option 148 = NCE-Campus. Option 43 = AC address for APs.

Check yourself

Answers: page 116

1. CHOOSE ONE

   On which UDP port does an NMS receive SNMP traps?

   A 161 B 162
   C 514 D 830

2. CHOOSE ONE

   Which SNMP version added GetBulk and Inform?

   A SNMPv1 B SNMPv2c
   C SNMPv3 D All versions

3. CHOOSE ONE

   Which data modelling language is used in the NETCONF content layer?

   A XML Schema only B YANG
   C JSON D MIB

4. TRUE OR FALSE?

   RESTCONF can edit the candidate datastore and then commit it.

   ☐ True ☐ False

5 CHOOSE ONE

Which DHCP option lets new devices find iMaster NCE-Campus automatically?

A Option 3

B Option 43

C Option 66

D Option 148

CHAPTER 21

Network Programming and Automation

EXAM FOCUS

Compiled vs interpreted, Python syntax rules, functions/modules/classes, paramiko SSHClient methods & host key policies

IN THIS CHAPTER

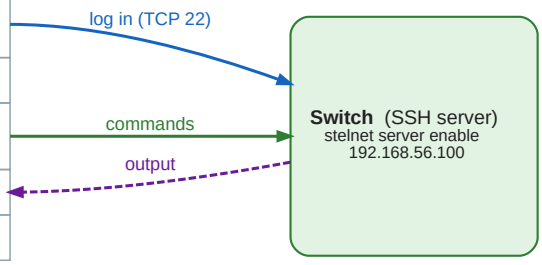
21.1 Why automate · 21.2 Compiled vs interpreted languages · 21.3 Python basics · 21.4 Functions, modules, classes · 21.5 Paramiko · 21.6 Host key policies · 21.7 Key-based SSH login

Pictures Look at these first

PICTURE 21.1

A Python script logs in to a switch with paramiko

Python script (the SSH client)	
1	`import paramiko, time`
2	`ssh = paramiko.SSHClient()`
3	`ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())`
4	`ssh.connect(hostname='192.168.56.100', port=22, username='client', password='Huawei@123')`
5	`cli = ssh.invoke_shell()`
6	`cli.send('screen-length 0 temporary\n')` `cli.send('display current-configuration\n')`
7	`time.sleep(3); print(cli.recv(999999).decode())`
8	`ssh.close()`

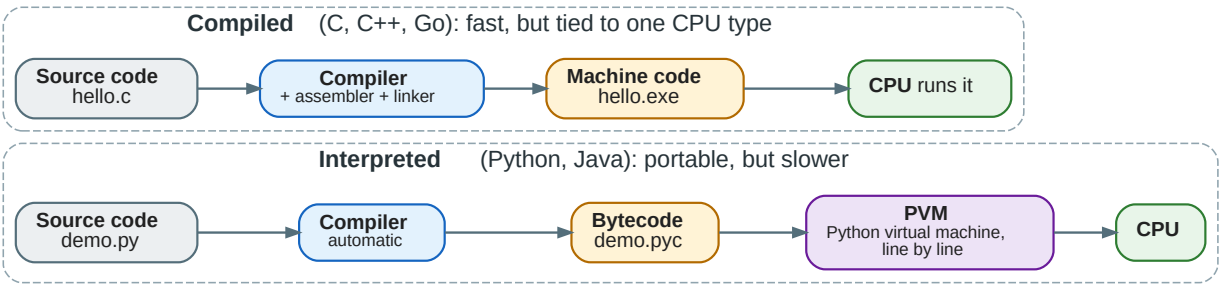


HOW TO READ IT

Read the script from top to bottom. paramiko opens an SSH session (TCP 22) to the switch, sends CLI commands like a person typing, reads the output and closes the session.

PICTURE 21.2

Compiled vs interpreted languages



HOW TO READ IT

A compiled language (C, C++, Go) is translated into machine code once, before it runs. Python is interpreted: it is turned into bytecode (.pyc) and the Python virtual machine (PVM) runs it line by line.

PICTURE 21.3

Paramiko methods and host key policies

SSHClient method	What it does
`connect()`	log in (hostname is required; port 22 by default)
`exec_command()`	**one** command per call (returns stdin, stdout, stderr)
`invoke_shell()`	interactive shell: send **many** commands
`close()`	log out

Host key policy	Unknown device on first connect
`AutoAddPolicy`	trust it automatically
`WarningPolicy`	trust it, but show a warning
`RejectPolicy`	refuse **(the default)**

HOW TO READ IT
exec_command runs ONE command per call. invoke_shell opens an interactive shell where you can send many commands. The host key policy decides what happens the first time you connect to an unknown device.

Notes Read, then cover the boxes and test yourself

21.1 Why automate

Typing the same commands on 1,000 devices is slow and easy to get wrong. A script does it in minutes.

REMEMBER
- Use cases: batch upgrades, config audits, password changes
- Tools: Python, Ansible, SaltStack, Puppet, Chef

EXAM TRAP Scripts suit repetitive, rule-based tasks.

21.2 Compiled vs interpreted languages

Compiled = translate the whole book before reading it. Interpreted = translate line by line while reading.

REMEMBER
- Compiled: C, C++, Go – fast
- Interpreted: Python, Java – portable but slower
- Python: code → .pyc bytecode → PVM runs it

EXAM TRAP Python is **NOT** a compiled language (true/false = False).

21.3 Python basics

A few rules keep Python code working and readable.

REMEMBER

- Indent with 4 spaces – wrong indent = IndentationError
- Names: letters, digits, _ ; can't start with a digit (SyntaxError)
- Names are case-sensitive (User_ID ≠ user_id)
- # = comment; triple quotes = multi-line comment
- Run it interactively (>>>) or as a script (python demo.py)

COMMAND / EXAMPLE

```
if True:
    print("Hello")
```

EXAM TRAP 4_passwd = 'x' → SyntaxError.

21.4 Functions, modules, classes

Function = a reusable recipe. Module = a .py file of recipes you import. Class = a blueprint for making objects.

REMEMBER

- def sit(): – a function
- import demo – use code from another file
- class Dog(): – a class; its methods use self

COMMAND / EXAMPLE

```
import demo
demo.sit()
```

EXAM TRAP Importing a module also runs its top-level code.

21.5 Paramiko

A Python library that logs in to devices over SSH for you.

REMEMBER

- connect() – log in (hostname required, port 22)
- exec_command() – **ONE** command per call
- invoke_shell() – interactive, **MANY** commands
- close() – log out

COMMAND / EXAMPLE

```
import paramiko, time
ssh = paramiko.SSHClient()
ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
ssh.connect(hostname='192.168.56.100', username='client', password='Huawei@123')
cli = ssh.invoke_shell()
cli.send('screen-length 0 temporary\n')
cli.send('display current-configuration\n')
time.sleep(3)
print(cli.recv(999999).decode())
ssh.close()
```

EXAM TRAP screen-length 0 temporary stops the output being split into pages.

21.6 Host key policies

What to do the first time you connect to a device.

REMEMBER

- AutoAddPolicy – trust it automatically
- WarningPolicy – trust it but show a warning
- RejectPolicy – refuse (the **DEFAULT**)

EXAM TRAP AutoDelPolicy does **NOT** exist (slide quiz).

21.7 Key-based SSH login

Log in with a key instead of a password.

REMEMBER

- The client makes two keys: id_rsa (private – keep secret) + id_rsa.pub (public)
- The device stores the **PUBLIC** key and links it to the user

COMMAND / EXAMPLE

```
ssh-keygen -t rsa
[SSH Server] rsa peer-public-key rsa01 encoding-type openssh
[SSH Server] ssh user client authentication-type rsa
[SSH Server] ssh user client assign rsa-key rsa01
```

EXAM TRAP The server gets the **PUBLIC** key. The private key never leaves the client.

Check yourself

Answers: page 116

1. TRUE OR FALSE?
   Python is a compiled language.
   ☐ True ☐ False

2. CHOOSE ONE
   Which Python variable name causes a SyntaxError?
   A 4_passwd
   B passwd_4
   C _passwd
   D Passwd4

3. CHOOSE ONE
   Which paramiko method opens an interactive shell so you can send MANY commands?
   A connect()
   B exec_command()
   C close()
   D invoke_shell()

4. CHOOSE ONE
   What is paramiko's DEFAULT host key policy for an unknown device?
   A AutoAddPolicy
   B WarningPolicy
   C AutoDelPolicy
   D RejectPolicy

5. CHOOSE ONE
   For key-based SSH login, which key is copied to the network device?
   A The private key (id_rsa)
   B Both keys
   C Neither; only a password is used
   D The public key (id_rsa.pub)

CHAPTER 22

Network Troubleshooting

EXAM FOCUS

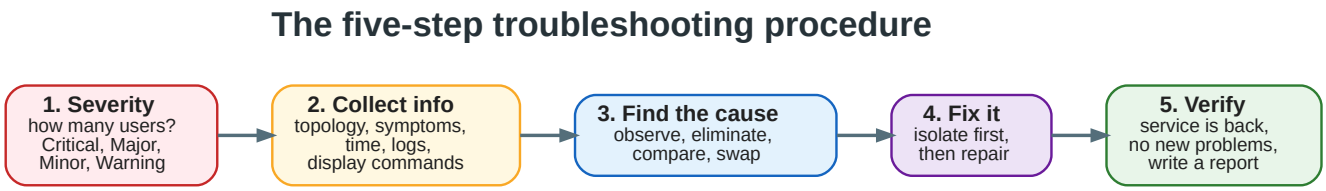
5-step procedure, severity levels, locating methods, DHCP / same-VLAN / ping / AP / STA troubleshooting flows & commands

IN THIS CHAPTER

22.1 Troubleshooting steps · 22.2 Finding the cause · 22.3 Useful commands · 22.4 PC gets no IP address · 22.5 PCs in the same VLAN can't talk · 22.6 Ping fails · 22.7 AP won't go online · 22.8 Phone can't join Wi-Fi

Pictures Look at these first

PICTURE 22.1

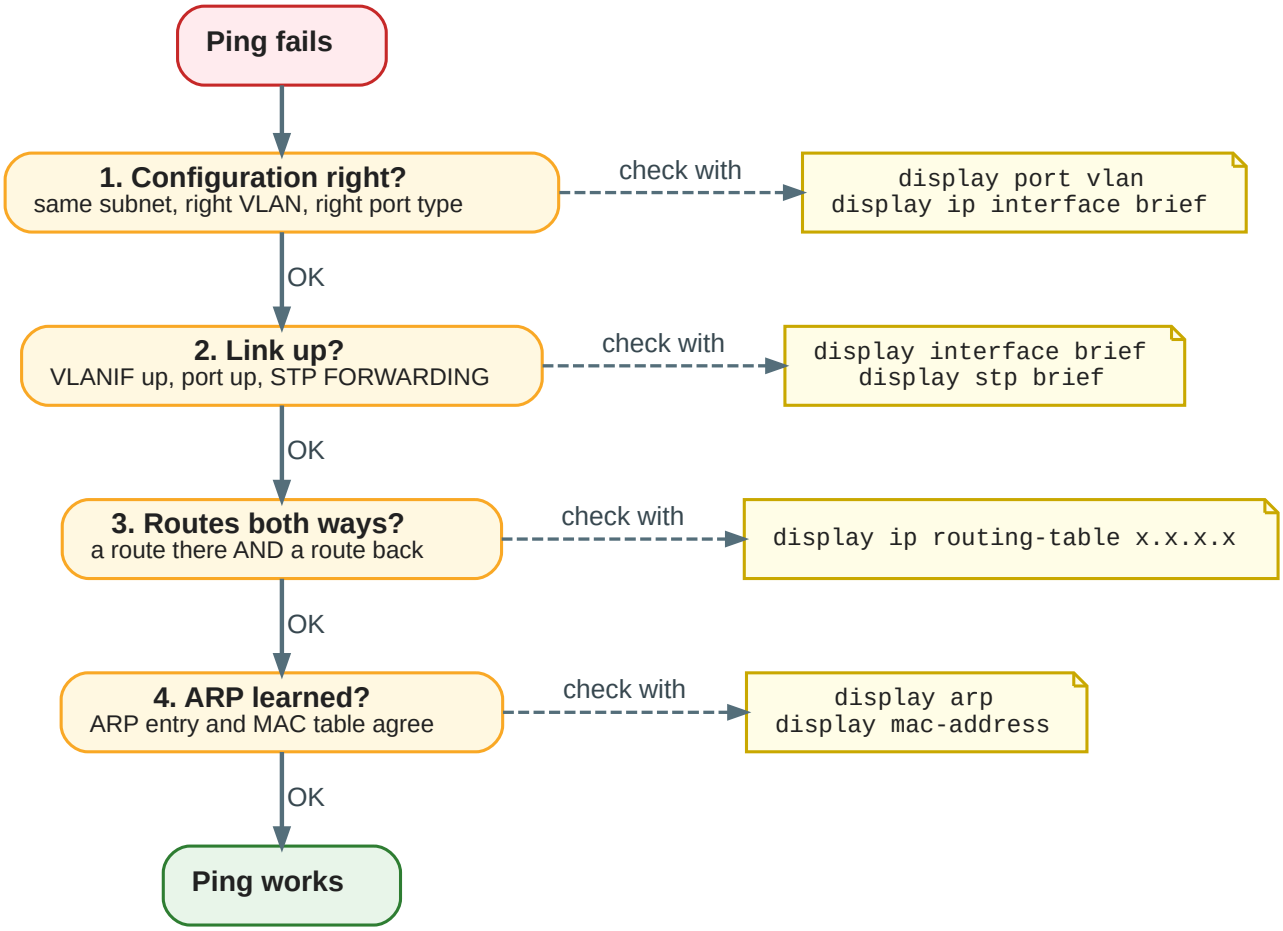


HOW TO READ IT

Work through the steps in order, like a doctor: how serious is it, what are the symptoms, what is the cause, fix it, then check the patient is really better and write it down.

PICTURE 22.2

Ping fails: what to check, in order



HOW TO READ IT

Check from the bottom up. Most ping failures are a wrong IP or VLAN, a port that is down or blocked by STP, a missing route (remember the RETURN route) or an ARP problem.

PICTURE 22.3

Who is affected tells you where to look

Who has the problem?	Look first at	Useful command
One user	the user's PC, cable, switch port, power	`display interface brief`
Many users	the device above them, or a configuration error	`display current-configuration`
Everyone	core device, main link, or a network attack	`display health, display cpu-usage`

HOW TO READ IT

Start by asking how many users have the problem. One user points to their own PC, cable or port. Many users point to the device above them. Everyone points to the core, the main link or an attack.

Notes Read, then cover the boxes and test yourself

22.1 Troubleshooting steps

Like a doctor: check how serious it is, ask questions, find the cause, treat it, then check the patient is OK.

REMEMBER

1. Decide the severity
2. Collect information
3. Find the cause
4. Fix it (isolate the problem first)
5. Verify, then write a report

- Severity levels: Critical > Major > Minor > Warning

EXAM TRAP Know the 5 steps in order.

22.2 Finding the cause

Four ways to track down a fault.

REMEMBER

- Observation – look at alarms and LEDs
- Elimination – turn off a feature or replace a card to test it
- Comparison – compare with something that works
- Swap – exchange a part with a known-good one
- 1 user down → their PC/cable/port; many → the device above them; all → core, link or attack

EXAM TRAP Match the method name to its description.

22.3 Useful commands

Commands that show the device's health and status.

REMEMBER

- display diagnostic-information – everything in one go
- display health – temperature, fans, CPU, memory
- display interface brief – port status and errors
- display cpu-usage / display memory-usage
- display trapbuffer – alarms

COMMAND / EXAMPLE

```
ping -a <source> <destination>
tracert <destination>
```

22.4 PC gets no IP address

Usually a link, DHCP setting or full address pool.

REMEMBER

Check in order:

1. Link OK? (ports/VLANIF up, errors not growing)
2. DHCP config correct?
3. Pool not empty? (Idle = 0 means full)

- Fix a full pool: bigger subnet, more pools, shorter lease

COMMAND / EXAMPLE

```
display interface brief
display ip pool interface vlanif10
```

EXAM TRAP VLANIF down → create the VLAN and allow it on the ports.

22.5 PCs in the same VLAN can't talk

Usually a port, VLAN or STP problem.

REMEMBER

Check in order:

1. Ports up?
2. VLAN created?
3. Ports in the right VLAN with the right port type?
4. STP state **FORWARDING**?

COMMAND / EXAMPLE

```
display vlan 10
display port vlan
display stp brief
```

EXAM TRAP **DISCARDING** in display stp brief = the port is blocked.

22.6 Ping fails

Work from config, to link, to routes, to ARP.

REMEMBER

Check in order:

1. Config (VLAN, IP in the same subnet)
2. Link up (VLANIF up, STP not blocking)
3. Routes – **BOTH** directions!
4. ARP learned correctly

COMMAND / EXAMPLE

```
display ip routing-table 192.168.1.11
display arp
display mac-address
```

EXAM TRAP Always check the **RETURN** route.

22.7 AP won't go online

Find the AP's state, then the reason.

REMEMBER

- No IP / wrong IP
- Network to the AC broken
- Failed MAC/SN check (state = unauth)
- Licence full
- Faulty AP
- AP states: nor, idle, download, fault, unauth

COMMAND / EXAMPLE

```
display ap all
display ap online-fail-record mac <mac>
display license resource usage
```

EXAM TRAP unauth → add the AP to the whitelist.

22.8 Phone can't join Wi-Fi

Ask the AC why the device failed.

REMEMBER

- Wrong or missing service VLAN in the VAP profile
- Device is on the blacklist
- Channel too busy (keep utilisation below 80%)

COMMAND / EXAMPLE

```
display station online-fail-record all
display radio ap-id 0
```

EXAM TRAP Slide quiz answer: display station online-fail-record.

Check yourself

Answers: page 116

1 CHOOSE ONE

What is the FIRST step of the troubleshooting procedure?

A Collect information
B Find the cause
C Decide the severity
D Fix the fault

2 CHOOSE ONE

Which fault-finding method replaces a suspect part with a known-good one?

A Observation
B Comparison
C Elimination
D Swap

3 CHOOSE ONE

display stp brief shows a port in the DISCARDING state. What does this mean?

A The port is blocked by STP
B The port is forwarding normally
C The port is shut down
D The port is an edge port

4 CHOOSE ONE

display ap all shows an AP in the 'unauth' state. What is the usual fix?

A Replace the PoE switch

B Change the Wi-Fi password

C Increase the DHCP lease time

D Add the AP to the whitelist (MAC or SN)

5 CHOOSE ONE

Which command shows why a phone failed to join Wi-Fi?

A display station online-fail-record all

B display ap all

C display vlan

D display ip pool

CHAPTER 23

Typical Campus Networking Solution

EXAM FOCUS

Campus sizes & thresholds, layers/zones, lifecycle, VLAN & IP planning, routing/WLAN design, reliability, security, commissioning

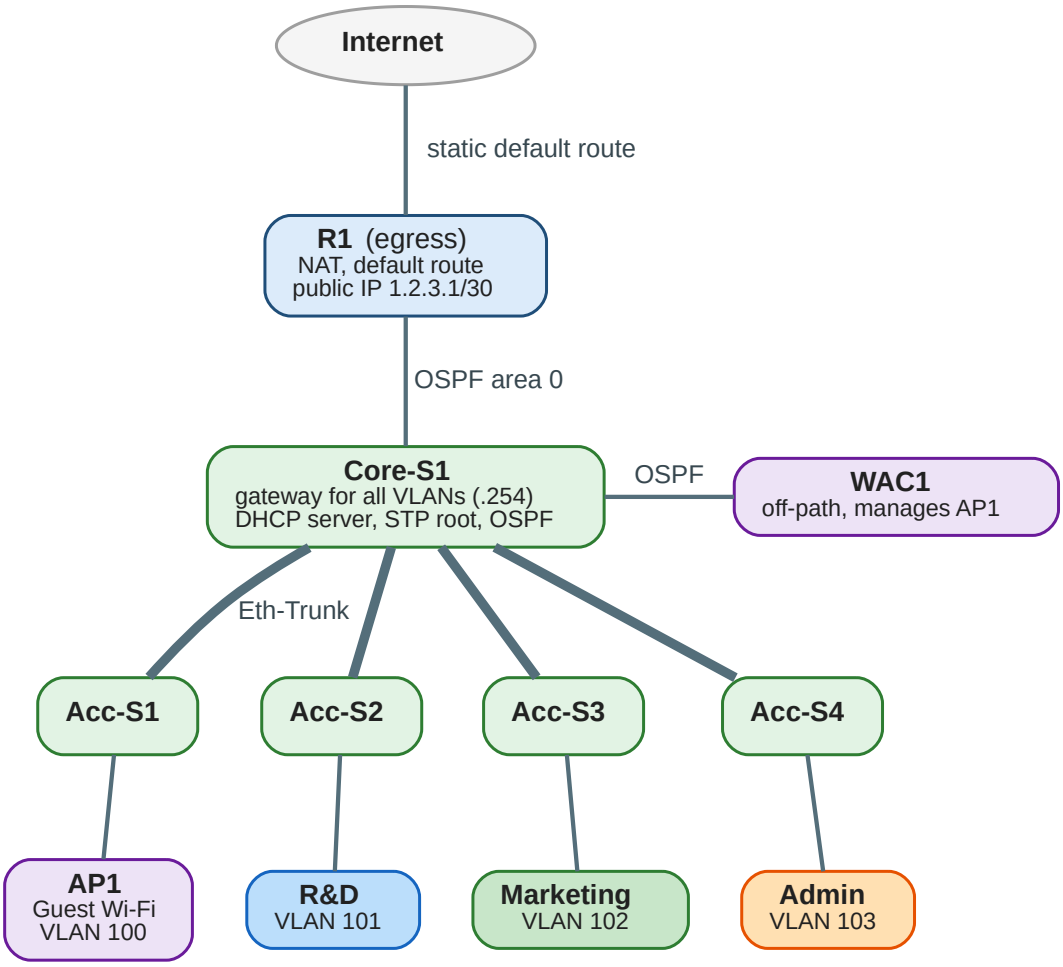
IN THIS CHAPTER

23.1 What a campus network is · 23.2 Campus sizes · 23.3 Project lifecycle · 23.4 VLAN plan (example) · 23.5 IP plan (example) · 23.6 Routing and Wi-Fi design · 23.7 Reliability and loops · 23.8 Security design · 23.9 Testing and running the network

Pictures Look at these first

PICTURE 23.1

Example campus network (from the course)



HOW TO READ IT

A two-layer design for about 200 users. Thick lines are Eth-Trunk links. Core-S1 is the gateway and DHCP server for every VLAN and the STP root. R1 does NAT to the Internet. WAC1 hangs off the core and manages AP1. Each department has its own VLAN.

PICTURE 23.2

VLAN and IP address plan (from the course)

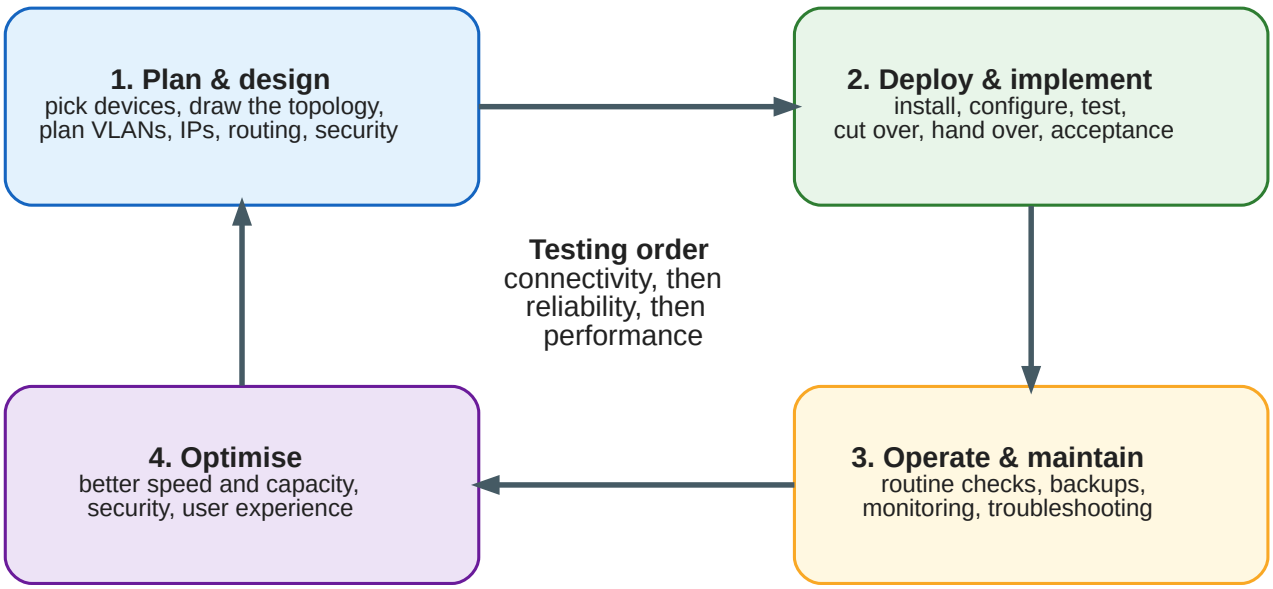
VLAN type	VLAN	Used for	Subnet	Gateway	How IPs are given
Management VLAN 2 to 9	2	switch management	192.168.2.0/24	.254	static
	3	AP management (CAPWAP)	192.168.3.0/24	.254	DHCP
Interconnection VLAN 10 to 99	10	Core-S1 to R1	192.168.10.0/30	-	static
	11	Core-S1 to WAC1	192.168.11.0/30	-	static
Service VLAN 100 to 199	100	Guests (Wi-Fi)	192.168.100.0/24	.254	DHCP
	101	R&D	192.168.101.0/24	.254	DHCP
	102	Marketing	192.168.102.0/24	.254	DHCP
	103	Administration	192.168.103.0/24	.254	DHCP
Internet link	-	R1 to the ISP	1.2.3.0/30	1.2.3.2	static

HOW TO READ IT

Each kind of VLAN gets its own number range so the plan is easy to read. Links between Layer 3 devices use /30. Every user subnet is a /24 with the gateway on .254.

PICTURE 23.3

The campus network project lifecycle



HOW TO READ IT

A network project goes round this cycle: plan and design it, build it, run and maintain it, then improve it. Improvements lead into the next round of planning.

Notes Read, then cover the boxes and test yourself

23.1 What a campus network is

The network inside one organisation's site – a school, office or factory – run by one owner.

REMEMBER

- Built in layers: Access → Aggregation → Core → Egress (Internet)
- Extra zones: data centre, network management, security

EXAM TRAP One owner = one campus network.

23.2 Campus sizes

Size is measured by users and number of network devices (NEs).

REMEMBER

- Small: < 200 users, < 25 devices, 1 site
- Midsize: 200–2000 users, 25–100 devices (most common; 3 layers)
- Large: > 2000 users, > 100 devices, many sites + VPN

EXAM TRAP Remember 200 / 2000 users and 25 / 100 devices.

23.3 Project lifecycle

Four phases, like a house: design it, build it, look after it, improve it.

REMEMBER

1. Plan & design
2. Deploy & implement
3. Operate & maintain (O&M)
4. Optimise

EXAM TRAP Short-answer question: name the 4 phases.

23.4 VLAN plan (example)

Give each kind of VLAN its own number range so it's easy to remember.

REMEMBER

- Management VLANs 2–9 (switches, APs)
- Interconnection VLANs 10–99 (core ↔ router, core ↔ AC)
- Service VLANs 100–199 (guests, departments)

EXAM TRAP Three VLAN types: management, interconnection, service.

23.5 IP plan (example)

Keep addresses tidy and predictable.

REMEMBER

- Management IP – used to log in to and manage each device
- Links between Layer 3 devices – /30
- User subnets – /24, gateway always .254
- PCs and APs – DHCP. Network devices and servers – static

COMMAND / EXAMPLE

```
R&D 192.168.101.0/24, gateway .254
Core ↔ R1 192.168.10.0/30
```

EXAM TRAP Short-answer: a management IP lets admins reach and manage a device.

23.6 Routing and Wi-Fi design

How traffic moves inside and out of the example campus.

REMEMBER

- OSPF Area 0 inside the campus
- Static default route on the egress router
- Wi-Fi: Layer 3 off-path, direct forwarding, WPA2-PSK for guests

COMMAND / EXAMPLE

```
[R1] ip route-static 0.0.0.0 0 1.2.3.2
```

23.7 Reliability and loops

Add backup and protect against cabling mistakes.

REMEMBER

- Eth-Trunk between access and core
- Stacking / CSS for device backup
- Run STP even with no backup links (protects against wrong cabling); core = root

COMMAND / EXAMPLE

```
[Core-S1] stp root primary
```

EXAM TRAP Why STP with no loops? To protect against accidental loops.

23.8 Security design

Protect the Internet edge, the inside network and device management.

REMEMBER

- ACL + traffic policy: guests can't reach the intranet
- NAT for Internet access
- DHCP snooping: blocks fake DHCP servers (uplink = trusted)
- ACL on VTY + SNMPv3 + STelnet for safe management

COMMAND / EXAMPLE

```
[SW] dhcp snooping enable
[SW-GE0/0/24] dhcp snooping trusted
```

EXAM TRAP Rogue DHCP server → DHCP snooping.

23.9 Testing and running the network

After building it, test it, look after it and keep improving it.

REMEMBER

- Test order: connectivity → reliability → performance
- O&M: routine checks, backups, troubleshooting
- Optimise: speed, capacity, user experience

Check yourself

Answers: page 117

1 CHOOSE ONE

A campus has 800 users and 60 network devices. What size is it?

A Small
B Midsize
C Large
D Data centre

2 CHOOSE ONE

What is the correct order of the campus network project lifecycle?

A Plan & design, deploy & implement, O&M, optimise
B Deploy, plan, optimise, O&M
C O&M, plan, deploy, optimise
D Plan, optimise, deploy, O&M

3 CHOOSE ONE

Which feature blocks a rogue (fake) DHCP server on the access switches?

A DHCP snooping
B Port security
C STP root protection
D NAT

4 CHOOSE ONE

Which mask is normally used for links between two Layer 3 devices?

A /24
B /30
C /26
D /32

5 CHOOSE ONE

Why run STP even when the design has no backup links?

A To protect against loops caused by wrong cabling
B To speed up routing
C Because DHCP needs it
D To share load across VLANs

Answer key

Mark your answers, then read the reason for any you got wrong and go back to that topic.

Chapter 1 answers

Data Communication Network Basics

1	**D**	A switch works at Layer 2 with MAC addresses. A router works at Layer 3 with IP addresses.
2	**B**	Full-mesh links = n(n-1)/2 = 6 × 5 / 2 = 15.
3	**D**	The SFU (Switch Fabric Unit) joins the cards. LPUs talk to each other through the SFU.
4	**False**	PPP and MPLS are WAN technologies. Ethernet and Wi-Fi are LAN technologies.
5	**D**	Associate, then Professional, then Expert.

Chapter 2 answers

Network Reference Models

1	**B**	Transport = segment, Network = packet, Data link = frame, Physical = bits.
2	**C**	ICMP is a network-layer protocol, together with IP and IGMP.
3	**C**	LCP agrees the link first, then PAP/CHAP checks the password (optional), then NCP agrees the network settings.
4	**A, C**	UDP is connectionless and fast but gives no delivery guarantee. Ordering and the handshake belong to TCP.
5	**A**	IEEE = Ethernet and Wi-Fi. IETF = Internet protocols (RFCs). ISO = the OSI model.

Chapter 3 answers

Network Device Operating Systems

1	**B**	Angle brackets < > = user view. Square brackets [ ] = system view or deeper.
2	**A**	ping and tracert are level 0 commands. File operations need level 3.
3	**B**	reboot asks you to save first; reboot fast does not. Unsaved changes are lost.
4	**D**	.cc = system software, .cfg/.zip = configuration, .pat = patch.
5	**True**	delete moves the file to the recycle bin. delete /unreserved removes it for good.

Chapter 4 answers

Ethernet Switching Basics

1	**D**	The switch learns from the SOURCE MAC. It uses the DESTINATION MAC to decide where to send.
2	**False**	Unknown unicast is FLOODED out of all other ports in the VLAN, not dropped.
3	**A, C**	ARP asks 'who has this IP?' by broadcast; the owner answers by unicast with its MAC.
4	**B**	0x0800 = IPv4, 0x0806 = ARP, 0x86DD = IPv6, 0x8100 = 802.1Q tag.
5	**D**	Switches split collision domains. Routers (and VLANs) split broadcast domains.

Chapter 5 answers

VLAN Fundamentals and Configuration

1	**C**	12 bits = 4096 values; 0 and 4095 are reserved, so VLANs 1 to 4094 can be used.
2	**C**	10 to 20 includes both ends = 11 VLANs. vlan batch 10 20 (no 'to') creates only 2.
3	**C**	A trunk sends every allowed VLAN tagged, except the PVID VLAN, which leaves untagged.
4	**D**	Access port: port default vlan. Trunk port: port trunk allow-pass vlan.
5	**False**	Each VLAN is a separate broadcast domain. A router or VLANIF is needed to route between them.

Chapter 6 answers

STP Fundamentals and Configuration

1	**B**	Lowest Bridge ID wins. Priority is compared first, so 4096 wins before MAC is checked.
2	**A, B, C**	Loops cause broadcast storms, MAC flapping and duplicate frames. IP conflicts are a different problem.
3	**B**	Alternate backs up the root port. Backup backs up a designated port.
4	**B**	Hello 2 s, Forward Delay 15 s, Max Age 20 s.
5	**False**	The root bridge has NO root port; all of its ports are designated ports.

Chapter 7 answers

Ethernet Link Aggregation

1	**B**	System priority picks the Actor device. Interface priority picks which links are active.
2	**True**	Manual mode does no negotiation, so every member link forwards traffic.
3	**A**	All member ports must have the same speed and duplex.
4	**A**	The bundle is one logical link, so there is no loop for STP to block.
5	**B**	Load balancing is per flow (by MAC or IP), so one conversation stays on one link and arrives in order.

Chapter 8 answers

IP Address and Configuration

1	**B**	/27 leaves 5 host bits: 2^5 - 2 = 30.
2	**D**	Private: 10.0.0.0/8, 172.16.0.0/12 (172.16 to 172.31), 192.168.0.0/16. 172.32 is public.
3	**A**	D-O-R-A: DISCOVER, OFFER, REQUEST, ACK.
4	**A**	At 50% it renews by unicast to its server. At 87.5% it broadcasts to any server.
5	**D**	40 hosts needs /26 (62 hosts). Valid /26 subnets start at .0, .64, .128, .192, so .96/26 is not valid.

Chapter 9 answers

IP Routing Basics

1	**D**	Longest match is checked first. Preference is only compared when the masks are equal.
2	**C**	Direct 0, OSPF 10, Static 60, RIP 100, OSPF external 150, BGP 255.
3	**D**	A bigger preference number is less preferred, so the route waits until the main route disappears.
4	**False**	A VLANIF is up only if the VLAN exists AND at least one port in that VLAN is up.
5	**D**	The reply cannot get back. Routes are needed in BOTH directions.

Chapter 10 answers

OSPF Fundamentals and Configuration

1	**A**	OSPF runs directly on IP (protocol 89). It does not use TCP or UDP.
2	**C**	LSU (Link State Update) carries the LSAs. LSR asks for them; LSAck confirms them.
3	**A, C**	Area ID, timers, authentication and (on Ethernet) the mask must match. Router IDs must be DIFFERENT.
4	**C**	DROthers only become Full with the DR and BDR. Staying at 2-Way with each other is normal.
5	**A**	100 / 1000 is less than 1, and the minimum cost is 1. That is why 100M and 1G both get cost 1.

Chapter 11 answers

IPv6 Address and Configuration

1	**C**	Drop leading zeros in each group and replace ONE run of zero groups with ::.
2	**C**	2000::/3 global unicast, FC00::/7 unique local, FE80::/10 link-local, FF00::/8 multicast.
3	**D**	Neighbor Solicitation (135) and Neighbor Advertisement (136) replace ARP.
4	**False**	IPv6 has no broadcast. It uses multicast instead.
5	**C**	Split the MAC, insert FFFE in the middle, then flip the 7th bit (3C becomes 3E).

Chapter 12 answers

ACL Fundamentals and Configuration

1	**B**	Basic 2000-2999, Advanced 3000-3999, Layer 2 4000-4999.
2	**D**	Matching a destination address or a port needs an ADVANCED ACL. A basic ACL checks source IP only.
3	**D**	0 means 'must match', so a wildcard of all 0s matches one exact address.
4	**A**	Classifier picks the traffic, behavior says what to do, policy joins them, then it goes on an interface.
5	**True**	First match wins, so put specific rules before general ones.

Chapter 13 answers

AAA Fundamentals and Configuration

1	**C**	Authentication = who are you; Authorization = what can you do; Accounting = what did you do.
2	**C**	Accounting is either none or remote. There is no local accounting.
3	**A, C**	HWTACACS uses TCP 49 and encrypts everything. RADIUS uses UDP 1812/1813 and encrypts only the password.
4	**A**	service-type controls which login methods the user may use. Also check authentication-mode aaa on the VTY lines.
5	**D**	The two built-in domains are default and default_admin.

Chapter 14 answers

Network Services and Applications

1	**C**	ack = the other side's seq + 1 for a SYN, so 100 + 1 = 101.
2	**A**	Active = the server starts the data connection from port 20. Passive = the client starts it.
3	**D**	TFTP = UDP 69, with no username or password.
4	**False**	Telnet sends everything as plain text. That is why SSH (STelnet) is preferred.
5	**C**	rsa local-key-pair create. Without a key pair, SSH will not work.

Chapter 15 answers

NAT Fundamentals and Configuration

1	**A**	Easy IP = NAPT using the outside interface's own IP address.
2	**A**	No-PAT changes the IP only. NAPT and Easy IP change the IP and the port.
3	**B**	NAT Server maps a public IP:port to the private server, so it changes the DESTINATION.
4	**D**	nat outbound goes on the outside interface. If there are no NAT sessions, check this first.
5	**True**	Easy IP follows whatever address the outside interface has, so it suits changing addresses.

Chapter 16 answers

Data Center Network Basics

1	**A**	VXLAN gives each tenant its own network, with about 16 million IDs.
2	**D**	24 bits gives about 16 million networks, compared with 4094 VLANs.
3	**C**	PFC = hop-by-hop pause. Marking packets end to end is ECN. Writing to memory directly is RDMA.
4	**False**	Lossless needs mechanisms such as PFC and ECN, not just bigger links.
5	**C**	1.0 = STP + VRRP, 2.0 = M-LAG, 3.0 = Spine-Leaf + VXLAN.

Chapter 17 answers

WLAN Technology Basics

1	**B**	Channels 1, 6 and 11 are far enough apart not to overlap.
2	**C**	Wi-Fi 5 works on 5 GHz ONLY. Wi-Fi 4 and 6 use both 2.4 and 5 GHz.
3	**A**	Wi-Fi avoids collisions (CA). Old shared Ethernet detected them (CD).
4	**C**	Wi-Fi 4 = 11n, Wi-Fi 5 = 11ac, Wi-Fi 6 = 11ax, Wi-Fi 7 = 11be.
5	**D**	256-QAM = Wi-Fi 5, 1024-QAM = Wi-Fi 6, 4096-QAM = Wi-Fi 7.

Chapter 18 answers

WLAN Networking Architecture

1	**B**	5246 = control, 5247 = data.
2	**False**	Control traffic ALWAYS goes through CAPWAP. Only the user DATA path changes.
3	**B**	Layer 3 networking: DHCP Option 43 tells the AP the AC's address. Layer 2: the AP finds the AC by broadcast.
4	**B**	802.3af 15.4 W, 802.3at 30 W, 802.3bt 60-90 W.
5	**D**	A controller managing many Fit APs suits medium and large sites. Fat AP and leader AP suit small sites.

Chapter 19 answers

WLAN Fundamentals and Configuration

1	**B**	The VAP profile is the 'glue' that joins everything, and it is then applied to an AP group.
2	**B**	Scan, link auth, associate, access auth, then get an IP by DHCP.
3	**B**	nor = normal (online). unauth = failed the MAC/SN check; download = upgrading.
4	**A**	WPA3-Personal = SAE. Enterprise mode = 802.1X + RADIUS.
5	**C**	A wrong country code means the wrong channels and power levels.

Chapter 20 answers

Network Device Management

1	**B**	UDP 161 = queries to the agent, UDP 162 = traps sent to the NMS.
2	**B**	v2c added GetBulk and Inform. v3 added security (USM and VACM).
3	**B**	NETCONF messages are XML, and the content is modelled with YANG.
4	**False**	RESTCONF uses only the running datastore; changes apply immediately.
5	**D**	Option 148 = NCE-Campus address. Option 43 = AC address for APs.

Chapter 21 answers

Network Programming and Automation

1	**False**	Python is interpreted: code becomes .pyc bytecode and the Python virtual machine runs it.
2	**A**	A name cannot start with a digit.
3	**D**	exec_command() runs one command per call. invoke_shell() keeps a session open for many commands.
4	**D**	RejectPolicy is the default. AutoDelPolicy does not exist.
5	**D**	The device stores the PUBLIC key. The private key never leaves the client.

Chapter 22 answers

Network Troubleshooting

1	**C**	Severity, collect information, find the cause, fix it, then verify and write a report.
2	**D**	Swap = exchange the part. Comparison = compare with a working setup.
3	**A**	DISCARDING = blocked. Check STP if PCs in the same VLAN cannot talk.
4	**D**	unauth = the AP failed the MAC/SN check on the AC.
5	**A**	The AC records the reason each station failed to go online.

Chapter 23 answers

Typical Campus Networking Solution

1	**B**	Midsize = 200-2000 users and 25-100 devices.
2	**A**	Design it, build it, look after it, then improve it.
3	**A**	DHCP snooping trusts only the uplink port towards the real DHCP server.
4	**B**	A /30 gives exactly 2 usable addresses, one for each end of the link.
5	**A**	Someone may plug a cable in the wrong place. STP stops that loop from taking the network down.

Quick reference

The numbers and values that come up again and again. Revise this page the night before the exam.

ITEM	VALUE	REMEMBER
PORTS		
FTP data / control	`TCP 20 / 21`	Active mode data from port 20; passive uses random server port (Ch2, 14)
SSH / STelnet / SFTP	`TCP 22`	Encrypted (Ch14)
Telnet	`TCP 23`	Plaintext (Ch14)
SMTP	`TCP 25`	Email (Ch2)
DNS	`UDP 53`	Slides list UDP; TCP 53 used for large replies/zone transfer (Ch2)
DHCP server / client	`UDP 67 / 68`	Ch8
TFTP	`UDP 69`	No authentication (Ch14)
HTTP / HTTPS	`TCP 80 / 443`	Device web first login: https://192.168.1.253:8443 (Ch3)
SNMP agent / trap	`UDP 161 / 162`	Ch20
NETCONF over SSH	`TCP 830 (also 22)`	Ch20
RADIUS auth / accounting	`UDP 1812 / 1813`	Old: 1645/1646 (Ch13)
HWTACACS	`TCP 49`	Ch13
CAPWAP control / data	`UDP 5246 / 5247`	Ch18
Port number ranges	`0–1023 / 1024–49151 / 49152–65535`	Well-known / registered / dynamic (Ch14)
PROTOCOL NUMBERS		
ICMP / TCP / UDP / OSPF	`1 / 6 / 17 / 89`	IPv4 Protocol field (Ch8, 10)
ETHERTYPE		
IPv4 / ARP / IPv6 / 802.1Q	`0x0800 / 0x0806 / 0x86DD / 0x8100`	Ch4, 5
SIZES		
MAC / IPv4 / IPv6 address	`48 / 32 / 128 bits`	
Ethernet frame / MTU	`64–1518 bytes / 1500`	Ch4
IPv4 / IPv6 basic header	`20 bytes (max 60) / 40 bytes fixed`	Ch8, 11
TCP / UDP header	`20 bytes / 8 bytes`	Ch14
802.1Q tag	`4 bytes; VID 12 bits`	Ch5
VXLAN VNI	`24 bits ≈ 16 million`	vs VLAN 4096 (Ch16)
ROUTE PREFERENCE		
Direct	`0`	Lower = preferred (Ch9)
OSPF (internal)	`10`	
IS-IS	`15`	
Static	`60`	Floating static = set a higher value
RIP	`100`	
OSPF ASE (external)	`150`	
BGP	`255`	

ITEM	VALUE	REMEMBER
ROUTE SELECTION		
Order of decision	`Longest match → preference → cost`	Ch9
VLAN		
VID range / usable	`0–4095 / 1–4094`	0 and 4095 reserved; default VLAN 1 (Ch5)
Default port type	`Hybrid on many Huawei S switches (V200)`	Check 'display port vlan' for actual type
MAC TABLE		
Dynamic entry aging	`300 s`	Ch4
STP		
Bridge priority default / step	`32768 / 4096`	stp root primary = 0, secondary = 4096 (Ch6)
Hello / Forward Delay / Max Age	`2 s / 15 s / 20 s`	Ch6
Path cost (802.1t) 100M / 1G / 10G	`200000 / 20000 / 2000`	Huawei default standard
STP states (5)	`Disabled, Blocking, Listening, Learning, Forwarding`	Ch6
RSTP states (3)	`Discarding, Learning, Forwarding`	Ch6
Standards	`STP 802.1D / RSTP 802.1w / MSTP 802.1s`	Huawei default mode = MSTP
LACP		
System / interface priority default	`32768 / 32768`	Lower wins; system priority picks Actor (Ch7)
Standard	`IEEE 802.3ad`	Ch7
OSPF		
Hello / Dead (broadcast, P2P)	`10 s / 40 s`	Must match on neighbours (Ch10)
Multicast addresses	`224.0.0.5 all OSPF / 224.0.0.6 DR-BDR`	Ch10
DR priority default	`1 (0 = never DR)`	Non-preemptive
Reference bandwidth	`100 Mbit/s`	Cost = ref BW / interface BW
Packet types 1–5	`Hello, DD, LSR, LSU, LSAck`	Ch10
Neighbour states	`Down → Init → 2-Way → ExStart → Exchange → Loading → Full`	Ch10
IPV4		
Class A / B / C first octet	`1–126 / 128–191 / 192–223`	D 224–239 multicast; E 240–255 (Ch8)
Private ranges	`10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16`	Ch8, 15
Special	`127.0.0.0/8 loopback, 169.254.0.0/16 link-local, 255.255.255.255 limited broadcast`	Ch8
Usable hosts	`2^n − 2`	n = host bits
/30 /29 /28 /27 /26 /25 /24 hosts	`2 / 6 / 14 / 30 / 62 / 126 / 254`	Interconnect links use /30 (Ch23)
ICMP		
Echo request / reply	`Type 8 / Type 0`	ping (Ch8)
Dest. unreachable / Time exceeded	`Type 3 / Type 11`	tracert uses TTL exceeded
DHCP		

ITEM	VALUE	REMEMBER
Message order	`DISCOVER → OFFER → REQUEST → ACK`	Ch8
Default lease / renew timers	`1 day / 50% unicast, 87.5% broadcast`	Ch8
Option 43 / Option 148	`AC IP for APs / iMaster NCE-Campus IP & port`	Ch19, 20
IPV6		
GUA / Link-local / ULA	`2000::/3 / FE80::/10 / FC00::/7`	Ch11
Multicast / all nodes / all routers	`FF00::/8 / FF02::1 / FF02::2`	No broadcast in IPv6
Loopback / unspecified	`::1 / ::`	Ch11
Solicited-node multicast	`FF02::1:FF00:0/104 + last 24 bits`	Used by NS / DAD
NDP ICMPv6 types	`RS 133, RA 134, NS 135, NA 136, Redirect 137`	Replaces ARP
RA flags	`M = stateful DHCPv6, O = stateless DHCPv6`	Ch11
ACL		
Basic / Advanced / Layer 2	`2000–2999 / 3000–3999 / 4000–4999`	Ch12
Default rule step	`5`	First match wins
Wildcard 0 / 1	`0 = must match, 1 = ignore`	Ch12
AAA		
Default domains	`default / default_admin`	Ch13
CLI		
Command levels	`0 Visit, 1 Monitoring, 2 Configuration, 3 Management`	Ch3
Default hotkeys	`Ctrl+G dis cur, Ctrl+L dis ip routing-table, Ctrl+O undo debugging all`	Ctrl+Z = back to user view
File extensions	`.cc software, .cfg/.zip/.dat config, .pat patch`	Ch3
SNMP		
v1 / v2c / v3	`Get, GetNext, Set, Trap / + GetBulk, Inform / + USM auth & encryption, VACM`	Ch20
NETCONF		
Layers (bottom-up)	`Secure Transport → Messages → Operations → Content`	Ch20
Datastores	`running (mandatory), candidate, startup`	RESTCONF = running only
RESTCONF		
Methods	`GET/HEAD read, POST create, PUT replace, PATCH merge, DELETE delete`	Ch20
PARAMIKO		
Host key policies	`AutoAddPolicy, WarningPolicy, RejectPolicy (default)`	Ch21
exec_command vs invoke_shell	`One command per call vs interactive multi-command`	Ch21
WI-FI		
2.4 GHz non-overlapping channels	`1, 6, 11`	Ch17

ITEM	VALUE	REMEMBER
Wi-Fi 4 / 5 / 6 / 7	802.11n / ac / ax / be	Wi-Fi 5 = 5 GHz only; 6E/7 add 6 GHz
Max QAM: Wi-Fi 5 / 6 / 7	256 / 1024 / 4096-QAM	Ch17
Channel utilisation	Keep < 80%; > 85% hurts access	Ch22
POE		
802.3af / at / bt	15.4 W / 30 W / 60–90 W	Ch18
WLAN		
AP states	nor (normal), idle, download, fault, unauth, standby	display ap all (Ch22)
CAMPUS		
Small / midsize / large	< 200 users & < 25 NEs / 200–2000 & 25–100 / > 2000 & > 100	Ch23
Lifecycle	Plan & design → Deploy & implement → O&M → Optimise	Ch23
DC		
Lossless technologies	PFC (hop-by-hop pause), ECN (end-to-end mark + CNP), RDMA/RoCEv2	Ch16
TROUBLESHOOTING		
Procedure	Severity → Collect info → Locate → Rectify → Verify	Ch22
Severity levels	Critical > Major > Minor > Warning	Ch22

Slides (V600) vs eNSP (V200) commands

The slides use YunShan OS (V600) commands. The eNSP labs use VRP (V200). Know both forms.

TASK	ENSP (V200)	SLIDES (V600)	NOTE
Interface name	GigabitEthernet0/0/1	GE1/0/1	Know both formats
LACP mode on Eth-Trunk	mode lacp-static	mode lacp	Ch 7
Give a switch port an IP	Use a VLANIF (S5700 switch)	undo portswitch, then ip address	Ch 3, 8
Filter traffic with an ACL	traffic-filter inbound acl 3000 (AR router)	MQC: classifier, behavior, traffic-policy	Both are valid exam knowledge (Ch 12)
Router-on-a-stick sub-interface	Also add arp broadcast enable (AR router)	-	Ch 9
NAT address pool	nat address-group 1 <start> <end>	Uses section syntax (see slides)	Ch 15